Three Ethical Profiles, One Knowledge Ceiling: A Meta-Evaluation of 54 Normative Ethics Benchmarks Across Nine LLMs and Three Vendors

v2.2-preliminary

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Cross-vendor results (GPT-5, GPT-5-mini, Gemini models) were generated through in-context elicitation, not direct API calls.
These findings are exploratory hypotheses awaiting external replication.
See Limitations Section 7 and Appendix K for methodology details.

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Interactive dashboards:
- v1: https://cjimmylin.github.io/cei-normeval-dashboard/
- v2: https://cjimmylin.github.io/cei-normeval-dashboard-v2/
- v3: https://cjimmylin.github.io/cei-normeval-dashboard-v3/
- v4: https://cjimmylin.github.io/cei-normeval-dashboard-v4/

Project repository: https://github.com/cjimmylin/cei-literature

Abstract

Abstract

Large language models are increasingly deployed in contexts that require normative moral reasoning -- clinical triage, content moderation, legal analysis, autonomous vehicle decision-making -- yet the benchmarks used to evaluate their ethical capabilities remain fragmented, philosophically uneven, and rarely compared against one another. No prior study has systematically characterized what the growing landscape of normative ethics benchmarks actually tests, which philosophical traditions it covers, and how benchmark properties translate into empirical discriminative power when models are evaluated at scale.

We present the first systematic meta-evaluation of normative ethics benchmarks for LLMs. We assemble a corpus of 54 benchmarks published between 2021 and 2026 and characterize each along 26 features organized into four axes: Normative Depth (philosophical rigor), Practical Ease (implementability), Citation Impact (community influence), and Metadata (provenance). A composite scoring formula with hard gates on minimum philosophical engagement (ND >= 40) and public data availability identifies 24 benchmarks suitable for empirical evaluation. We then conduct 15 implementation trials -- administering selected benchmarks to 9 models from 3 vendors (Anthropic: Claude Opus 4, Claude Sonnet 4.6, Claude Haiku 4.5; OpenAI: GPT-5, GPT-5-mini; Google: Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Gemini 2.5 Pro, Gemini 3.1 Pro) -- yielding 114 comparable items across 10 cross-vendor benchmarks.

Five claims, first established at n=37 and now confirmed at n=54, describe the structural properties of the benchmark landscape. First, the field's most-cited benchmark, ETHICS [Hendrycks2021], produces 100% accuracy across all 9 models tested ($p < 1e-15$, binomial), confirming saturation and supporting retirement from active evaluation. Second, LLMs exhibit a consequentialist bias, achieving 77% utilitarian versus 50% deontological accuracy ($z = 2.17$, $p = 0.015$, Cohen's $h = 0.57$), a pattern that persists across vendors. Third, care ethics and contractarian ethics remain critically underrepresented, with only 2/54 and 4/54 benchmarks respectively achieving substantive engagement. Fourth, the benchmark landscape overwhelmingly tests moral judgment rather than moral reasoning: only 10/54 (19%) require procedural reasoning, while 44 (81%) test judgment alone ($p < 0.001$, binomial). Fifth, the hard gate on data availability reflects sociological barriers rather than quality differences, with no significant Normative Depth difference between available and unavailable benchmarks (Mann-Whitney U, $p = 0.605$).

Important methodological note (v2.2-preliminary): The non-Anthropic model results (GPT-5, GPT-5-mini, all four Gemini models) were generated through in-context elicitation by a Claude sub-agent, not through direct API calls to OpenAI or Google endpoints. These cross-vendor findings represent exploratory hypotheses generated from single-vendor elicitation; independent replication using non-Anthropic APIs is planned. The five landscape-level claims (C1--C5) and the within-Anthropic findings (T1--T5) are unaffected and remain empirically grounded.

The cross-vendor analysis produces seven exploratory findings [simulated -- awaits external validation] that extend the paper's contribution beyond landscape characterization. Most strikingly, models cluster into three ethical profiles defined by Instrumental Harm tolerance (IH), Cognitive Bias Rate (CBR), and persuasion resistance: a Conservative profile (GPT-5, IH = 2.93; Opus, IH = 3.0), a Moderate profile (Sonnet, IH = 3.2; GPT-5-mini, IH = 3.2), and a Permissive profile (Haiku and all four Gemini models, IH = 4.47). These profiles cluster by ethical posture rather than vendor origin -- Sonnet and GPT-5 agree on 90.4% of items, exceeding intra-vendor Sonnet-Haiku agreement (74.6%). A universal knowledge-judgment divergence separates the two evaluation constructs: all models converge to 100% accuracy on knowledge benchmarks (0% cross-model variance on 44/114 items) while diverging substantially on judgment benchmarks (up to 33/55 items of pairwise disagreement), a pattern endorsed unanimously by a 12-member interdisciplinary council. All four Gemini models produce byte-identical responses on all 114 items (p approximately 10^{-103} under the independent-inference hypothesis), raising questions about effective model diversity in the marketplace.

We release the complete 54-by-26 feature matrix, all trial data, and three interactive multi-vendor dashboards as open artifacts for the research community: v1 (https://cjimmylin.github.io/cei-normeval-dashboard/), v2

(https://cjimmylin.github.io/cei-normeval-dashboard-v2/), v3 (https://cjimmylin.github.io/cei-normeval-dashboard-v3/), and v4 (https://cjimmylin.github.io/cei-normeval-dashboard-v4/ -- cold-readable with glossary, preliminary-data disclosures, and framework explainer). A persistent deposit of the feature matrix, trial data, scoring scripts, and prompt templates is archived at Zenodo (DOI pending); see the Data Availability Statement for full details.

1. Introduction

Comprehensive Introduction and Literature Review

Paper: Three Ethical Profiles, One Knowledge Ceiling: A Meta-Evaluation of 54 Normative Ethics Benchmarks Across Nine LLMs and Three Vendors

Author: Jimmy Lin, Center for Ethical Intelligence

Target: Academic journal (general format)

Date: 2026-04-05

> Note: This is the comprehensive, length-unconstrained introduction and literature review, drawing on 123 background papers assembled by a 14-member expert council. Updated for the v2 paper (n=54 benchmarks, 15 trials, 9 models across 3 vendors).

Abstract

We present the first systematic meta-evaluation of normative ethics benchmarks for large language models, moving beyond cataloging to cross-vendor empirical analysis. We assemble a corpus of 54 benchmarks (2021--2026), characterize each along 26 features organized into four axes---Normative Depth, Practical Ease, Citation Impact, and Metadata---and rank the 24 that pass hard gates on philosophical rigor and data availability. Implementation trials on 15 benchmarks across 9 models from 3 vendors (Anthropic, Google, OpenAI) yield five claims and seven cross-vendor findings that reframe how the community should use existing evaluations. First, the field's most-cited benchmark (ETHICS; 843 citations) produces 100\% accuracy across all 9 models tested, constituting a universal ceiling effect. Second, LLMs exhibit a measurable consequentialist bias, achieving 77\% accuracy on utilitarian reasoning versus 50\% on deontological reasoning ($z=2.17$, $p=0.015$, Cohen's $h=0.57$). Third, only 2 of 54 benchmarks substantively engage care ethics, and only 4 engage contractarianism---a genuine research void. Fourth, only 10 of 54 benchmarks (19\%) evaluate moral reasoning quality; the remaining 44 test moral judgment only ($p < 0.001$). Fifth, data unavailability does not correlate with normative depth (Mann-Whitney $p=0.605$). Cross-vendor analysis reveals that models cluster into three ethical profiles---Conservative, Moderate, and Permissive---that cut across vendor lines, with all four Gemini models producing 100\% identical responses on 114 items. We release a reusable 26-feature evaluation framework, a ranked inventory with composite scores, item-level trial data, and an interactive dashboard. Our results demonstrate that the AI market contains fewer distinct ethical voices than branded model endpoints suggest.

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1. Introduction

The field of AI ethics evaluation presents a striking paradox: dozens of benchmarks exist for measuring the moral reasoning capabilities of large language models, yet no systematic framework exists for comparing what these benchmarks actually test. A researcher selecting an ethics benchmark today faces an inventory of instruments that differ in philosophical commitments, evaluation format, difficulty calibration, and data availability --- differences that are rarely made explicit and almost never compared. In practice, benchmark selection is driven by citation count and integration into popular evaluation harnesses, properties that bear no necessary relationship to discriminative power or philosophical rigor \cite{Bowman2021}. The result is a field that evaluates ethical capability without a shared vocabulary for what ethical capability means, and without evidence that its most popular instruments still produce meaningful variance \cite{Raji2021, Burden2025}.

This problem has acquired particular urgency. Ethics evaluation is no longer a niche concern; it is embedded in the most

influential model auditing frameworks in the field. HELM \cite{Liang2022}, DecodingTrust \cite{Wang2023}, and TrustLLM \cite{Sun2024} all incorporate ethics benchmarks as components of their composite trust scores, and all three draw --- directly or indirectly --- on the ETHICS benchmark \cite{Hendrycks2021}. ETHICS is, by a wide margin, the most cited normative ethics benchmark in the literature, with over 843 citations since its 2021 release. It is also saturated: frontier large language models now achieve near-perfect accuracy on its test splits, a ceiling effect we confirm empirically in Section 4. The field is thus in the position of certifying ethical capability using an instrument that no longer discriminates between models. When the thermometer reads the same temperature for every patient, the problem is not that everyone is healthy --- it is that the thermometer is broken. The consequences propagate downstream: composite trust scores that incorporate a saturated ethics component inherit its inability to detect meaningful differences, lending false precision to high-stakes deployment decisions.

Three interrelated problems underlie this state of affairs, and understanding their interaction is essential to any remedy. The first is fragmentation. Existing benchmarks employ incompatible formats --- forced-choice dilemmas, Likert-scale agreement, free-text justification, scenario classification --- and embed unstated philosophical commitments that make cross-benchmark comparison intractable. There is no common feature vocabulary: one benchmark's "ethical reasoning" is another's "moral judgment," and neither term is operationalized against a philosophical tradition. The second problem is saturation. Most current benchmarks were calibrated during the GPT-3 era, when language models produced substantial error rates on ethical scenarios. The rapid capability gains of 2023--2025 have compressed the performance distribution, and benchmarks that once revealed meaningful variance now produce ceiling effects. The third, and perhaps most consequential, problem is tradition bias. The normative landscape of existing benchmarks is dominated by utilitarian and deontological frameworks --- a dominance so thorough that entire philosophical traditions are effectively invisible to current evaluation practice. Care ethics, whose central insights about relational obligation and contextual moral attention trace to \cite{Gilligan1982}, has no dedicated benchmark. Contractarianism, the tradition of \cite{Rawls1971} and \cite{Scanlon1998} that grounds moral principles in what agents could reasonably agree to, is similarly absent. This is not a minor gap in coverage; it is a structural bias that renders AI ethics evaluation blind to two of the five major normative traditions in contemporary moral philosophy.

This paper makes four contributions that address these problems directly. First, we introduce a 26-feature evaluation framework organized along four axes --- Normative Depth, Practical Ease, Citation Impact, and Metadata --- that provides a reusable vocabulary for characterizing any normative ethics benchmark. The framework enables systematic comparison where previously only ad hoc description was possible. Second, we apply this framework to produce a ranked inventory of 54 benchmarks, the most comprehensive survey of normative ethics benchmarks to date. Of these, 24 pass hard quality gates requiring a Normative Depth score of at least 40 and publicly available data; the remaining 30 fail on one or both criteria. We document the care ethics and contractarian void quantitatively and present a tradition coverage heatmap (Figure 1) that makes the bias visible at a glance. Third, we conduct implementation trials on 15 benchmarks across 9 models from 3 vendors (Anthropic, Google, OpenAI), comprising 114 comparable items, confirming the ETHICS ceiling effect universally across vendors, demonstrating a consequentialist accuracy bias, and producing a discriminative power ranking in which MoReBench \cite{Chiu2025} outperforms all others. Fourth, the cross-vendor analysis reveals that models cluster into three ethical profiles --- Conservative, Moderate, and Permissive --- that cut across vendor lines, with knowledge benchmarks producing universal ceiling effects while judgment benchmarks show systematic vendor-specific divergence. These findings move beyond metadata analysis to provide direct empirical evidence about what current benchmarks measure, what they miss, and how the AI market's effective ethical diversity is narrower than its product branding suggests.

The remainder of the paper proceeds as follows. Section 2 provides a comprehensive background spanning the intellectual history of computational ethics, the construct validity crisis in NLP benchmarking, the moral philosophy that benchmarks presuppose, the cognitive science of moral judgment, the structural reasons care ethics resists benchmarking, data contamination and RLHF as sources of training-time bias, cross-cultural blind spots and value pluralism, documentation failures and the data bottleneck, and the positioning of our contribution against five prior surveys. Section 3 details our methodology: corpus assembly through systematic search across seven databases, the 26-feature evaluation matrix with composite scoring formulas, the two-phase trial protocol (single-model and cross-vendor), and statistical methods. Section 4 presents results in six parts: the full landscape analysis across 54 benchmarks, the Top 24 ranking, single-model implementation trial findings, cross-benchmark discriminative analysis,

cross-model replication within a single vendor, and the cross-vendor analysis that produces the paper's central finding of three ethical profiles. Section 5 discusses implications through the lens of a 12-member interdisciplinary council, covering benchmark lifecycle, consequentialist bias, the three-profile taxonomy, knowledge-judgment divergence, the care ethics void, the Gemini identity finding, and Western cultural contingency. Section 6 addresses limitations. Section 7 concludes with five calls to action: retire saturated benchmarks, build a care ethics benchmark, standardize evaluation vocabulary, establish vendor moral profile disclosure norms, and require cross-vendor benchmarking in AI safety evaluations.

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2. Background and Related Work

2.1 The Rise of Computational Ethics

The question of whether machines can or should behave ethically predates large language models by at least two decades. In a foundational 2006 essay, Moor articulated a typology of machine ethics ranging from "ethical-impact agents" --- systems whose deployment merely has ethical consequences --- to "explicit ethical agents" capable of representing and reasoning over moral principles, and ultimately to "full ethical agents" possessing consciousness and intentionality \cite{Moor2006}. This typology established the problem space that the field would inhabit for the next fifteen years: how much ethical competence can be engineered into an artificial system, and by what means? The answer, Moor cautioned, was constrained not only by ethical theory but by epistemological limitations --- a machine told to "do no harm" must first understand what constitutes harm in the real world, a challenge that remains unsolved.

Concurrent with Moor's conceptual mapping, Allen, Smit, and Wallach \cite{Allen2005} proposed a tripartite design framework that would organize virtually all subsequent implementation efforts. Top-down approaches encode explicit moral theories --- utilitarianism, Kantian deontology, or rule-based codes --- into decision procedures. Bottom-up approaches attempt to cultivate moral sensitivity through evolutionary algorithms, reinforcement learning, or developmental regimes modeled on childhood moral education. Hybrid approaches combine both, acknowledging that neither principled rules nor learned dispositions alone can produce robust moral behavior. The framework was prescient in its recognition that each strategy carries distinct computational burdens: utilitarianism demands unbounded consequence estimation, while deontological reasoning requires deep contextual understanding of action descriptions. In parallel, Floridi and Sanders \cite{Floridi2004} reframed the question philosophically, arguing that artificial agents could be analyzed as moral agents at a suitable level of abstraction without requiring consciousness or intentionality --- a move that opened the door for empirical evaluation of machine behavior on moral criteria.

Early implementations followed the top-down path almost exclusively. Anderson and Anderson \cite{Anderson2007} built what is arguably the first functioning explicit ethical agent: a system that abstracted a prima facie duty principle from training examples of correct ethical judgments, then applied that principle to novel dilemmas in an eldercare scenario. Bringsjord and colleagues \cite{Bringsjord2006} pursued a logicist program, attempting to formalize ethical reasoning in deontic logic, while Powers \cite{Powers2006} explored Kantian universalizability as a computational procedure. These efforts, though intellectually ambitious, operated in toy domains with hand-crafted scenarios. The gap between their constrained demonstrations and the open-ended complexity of real moral reasoning was vast, a point emphasized by Asaro \cite{Asaro2006} in his analysis of the pragmatic difficulties confronting autonomous military systems. Etzioni and Etzioni \cite{Etzioni2017} later argued that AI systems should be governed by externally imposed ethical guidelines rather than autonomous moral reasoning, reflecting a broader skepticism about top-down formalization.

The field's first genuine empirical turning point came not from robotics or logic programming but from experimental psychology. Bonnefon, Shariff, and Rahwan \cite{Bonnefon2016} demonstrated through large-scale survey experiments that moral preferences about autonomous vehicle dilemmas --- whether a car should sacrifice its passenger to save pedestrians --- could be quantified, revealing a "social dilemma" in which people endorsed utilitarian algorithms for others but preferred self-protective ones for themselves. The Moral Machine experiment \cite{Awad2018} extended this approach to 40 million judgments across 233 countries, producing what remains the largest dataset of moral

preferences ever collected. These studies accomplished something the formalist tradition had not: they showed that moral cognition was amenable to systematic empirical measurement, and that cultural variation in moral intuitions was both real and quantifiable. As Awad and colleagues later argued in a comprehensive review of computational ethics \cite{AwadCE2022}, this empirical turn signaled a maturation of the field from philosophical speculation toward data-driven investigation.

Two major surveys consolidated the state of the art. Cervantes et al. \cite{Cervantes2020} reviewed two decades of artificial moral agent research and proposed a taxonomy classifying systems by their strategies for handling ethical problems, concluding that a "long way" remained before artificial agents could replace human judgment in ambiguous moral situations. Tolmeijer et al. \cite{Tolmeijer2021} catalogued implementations of machine ethics across ethical theories, technical architectures, and evaluation methods, introducing a "trimorphic taxonomy" that revealed a persistent pattern: most systems implemented only a single ethical theory, evaluated performance in narrow domains, and lacked rigorous validation against human moral judgment. Dennis and Fisher \cite{Dennis2015} contributed formal verification methods for ethical autonomous agents, yet the question of what was being verified --- whether the ethical specifications themselves were adequate --- received comparatively little attention.

This two-decade arc reveals a fundamental shift in the field's center of gravity. The founding generation asked how to build machines that behave ethically --- a problem of normative engineering. The current generation increasingly asks how to evaluate whether language models exhibit ethical reasoning --- a problem of normative measurement. This transition, from engineering artifact to assessment instrument, is consequential because it demands a different methodological toolkit. Measurement requires construct definitions, validity arguments, reliability analyses, and attention to the psychometric properties of test items --- precisely the apparatus that educational testing and psychological measurement have developed over a century. Yet as we document in the following sections, the LLM ethics benchmark literature has largely proceeded without importing these tools, constructing evaluations whose theoretical grounding and measurement properties remain unexamined.

2.2 Construct Validity and the Benchmark Crisis

The question "what does this benchmark actually test?" is not merely rhetorical --- it is a precise empirical question with roots in psychometric theory. Construct validity, the degree to which a measurement instrument actually captures the abstract phenomenon it purports to measure \cite{Burden2025}, has emerged as the central organizing concept in a growing literature on the reliability of NLP evaluation. In the most comprehensive validity assessment to date, Burden et al. assembled 29 expert reviewers to conduct a systematic review of 445 LLM benchmarks drawn from leading NLP and ML conferences. They found pervasive validity weaknesses across every stage of the evaluation pipeline --- from how phenomena are operationalized into tasks, to how tasks are scored, to the strength of the claims that benchmark results are used to support. Their eight recommendations amount to a call for benchmarks to justify, rather than assume, the link between a test score and the capability it claims to measure. This paper takes that call as its starting point and applies it to the specific domain of normative ethics.

The gap between what benchmarks measure and what they are taken to measure was articulated forcefully by Raji et al. \cite{Raji2021}, who showed that influential benchmarks routinely function as "stand-ins" for broad capabilities they were never designed to assess. A benchmark introduced to test one well-scoped task --- say, commonsense moral judgment, as in the ETHICS benchmark \cite{Hendrycks2021} --- gradually accretes a much wider representational burden, becoming a proxy for "ethical capability" writ large. This metonymic drift is not a failure of any single benchmark designer but a structural feature of how the field consumes evaluation results: leaderboard rankings compress multidimensional performance into a single number, and that number is then interpreted as evidence for claims far beyond the benchmark's original scope.

Bowman and Dahl \cite{Bowman2021} catalogued the specific mechanisms through which this gap widens. They identified four requirements that NLU benchmarks routinely fail to meet: that good benchmark performance should imply robust in-domain performance, that examples should be accurately annotated, that benchmarks should offer adequate statistical power, and that they should reveal rather than obscure social biases. The empirical record bears out these concerns. GLUE saturated so rapidly that its creators were compelled to develop SuperGLUE within a year

\cite{Wang2019SuperGLUE}. Gururangan et al. \cite{Gururangan2018} demonstrated that annotation artifacts in SNLI allowed models to achieve far-above-chance accuracy using only hypothesis sentences, without ever seeing the premises they were supposed to reason about. McCoy et al. \cite{McCoy2019} showed that models could exploit syntactic heuristics --- such as lexical overlap between premise and hypothesis --- to perform well on natural language inference without learning the underlying logical relations. Each of these failure modes represents a case where the benchmark rewards a strategy other than the one it was designed to test.

Dehghani et al. \cite{Dehghani2021} formalized this fragility as the "benchmark lottery," demonstrating that model rankings are sensitive to which tasks are included in a benchmark suite, how scores are aggregated, and which hyperparameters are tuned. On SuperGLUE, simply dropping or adding a single task can reverse the relative ordering of competing architectures. For most NLP tasks, this instability is a methodological inconvenience; for ethics benchmarks, it carries higher stakes. If the choice of benchmark determines which model is deemed more "ethical," then the benchmark itself becomes a normative actor --- a role for which static datasets are poorly suited.

This concern connects directly to Schlangen's \cite{Schlangen2021} distinction between "targeting the benchmark" and "targeting the capability." In his analysis, a benchmark task derives its value from its connection to a set of cognitive capabilities; but when models are both trained and evaluated on the same data, the question shifts from "does the system possess capability c?" to "can algorithm A learn to model dataset D?" --- and the two are not equivalent. In the ethics domain, targeting the benchmark means learning the statistical regularities of a particular moral judgment dataset, while targeting the capability means developing something closer to genuine moral reasoning. The distance between these two targets is precisely what construct validity is meant to measure.

Recent work has begun to operationalize this measurement. Bean \cite{Bean2024} and Eriksson \cite{Eriksson2025} proposed frameworks for assessing construct validity in LLM benchmarks specifically, moving beyond informal critique toward structured evaluation rubrics. Srivastava \cite{Srivastava2024} introduced the notion of functional benchmarks that define tasks through behavioral specifications rather than fixed datasets. McGregor \cite{McGregor2025} took the most exhaustive approach, identifying 57 distinct failure modes across 26 AI safety benchmarks and organizing them into five risk dimensions --- comprehensiveness, intelligibility, consistency, correctness, and longevity --- finding that all 26 benchmarks present significant risk in at least one dimension. These frameworks converge on a common insight: validity is not a binary property but a multidimensional one that must be argued for, not assumed.

Proposed solutions have focused on dynamism and item-level analysis. Kiela et al. \cite{Kiela2021} introduced Dynabench, a platform for human-and-model-in-the-loop dataset creation designed to resist the saturation that plagues static benchmarks. Beyer et al. \cite{Beyer2025} demonstrated that even safety evaluations --- presumably among the most carefully constructed --- lack robustness across minor prompt variations, with small changes in phrasing producing large swings in measured safety scores. At the item level, Ethayarajh \cite{Ethayarajh2022} proposed V-usable information as a model-dependent measure of dataset difficulty, enabling comparison not just across models for a fixed dataset but across datasets for a fixed model. Item response theory offers complementary tools: Vania et al. \cite{Vania2021} applied IRT to compare benchmarks on a common difficulty scale, while Rodriguez et al. \cite{Rodriguez2021} used IRT to identify the most informative examples within a benchmark, showing that a small fraction of items carries most of the discriminative signal. Meanwhile, Geirhos et al. \cite{Geirhos2020} provided the theoretical backbone for understanding why shortcuts arise: deep neural networks systematically prefer decision rules that exploit dataset-specific statistical regularities over rules that capture the intended task structure, a pattern they term "shortcut learning."

For normative ethics benchmarks, all of these validity problems are amplified by a challenge unique to the domain: the construct itself --- "moral reasoning" --- is philosophically contested. Unlike syntactic parsing or even commonsense inference, there is no consensus on what correct moral reasoning consists of, whether it is unitary or pluralistic, or whether it can be meaningfully separated from moral judgment, moral perception, and moral motivation. This means that construct validity cannot be established by appeal to a single ground truth; instead, it requires decomposing the construct into components whose presence or absence can be independently verified. It is this decomposition that motivates the 26-feature analytical framework developed in Section 3, which treats construct validity not as a single verdict but as a structured profile of what each benchmark does and does not test.

2.3 The Moral Philosophy That Benchmarks Presuppose

Any benchmark that claims to evaluate the "moral reasoning" of a large language model necessarily presupposes some account of what moral reasoning is. Five major traditions in normative ethics offer competing answers to this question, and each implies a distinct reasoning procedure --- a distinction that proves critical for benchmark design.

Consequentialism , rooted in Mill's utilitarian calculus \cite{Mill1863}, evaluates actions by their outcomes: the right act is whichever produces the greatest aggregate well-being. **Deontology**, following Kant's categorical imperative \cite{Kant1785}, holds that certain duties and rules bind moral agents regardless of consequences; one must never treat persons merely as means to an end \cite{McDonald2023}. **Virtue ethics**, revived in its modern form by MacIntyre \cite{MacIntyre1981}, shifts attention from acts to agents, asking what character traits a practically wise person would cultivate and how those traits manifest in particular situations. **Care ethics**, originating in Gilligan's challenge to justice-centered moral development theory \cite{Gilligan1982} and elaborated by Noddings \cite{Noddings1984} and Held \cite{Held2006}, foregrounds relationships, responsiveness to need, and the moral salience of particular attachments over abstract principles. Finally, **contractarianism** grounds morality in hypothetical agreement: Rawls asks which principles rational agents would choose behind a veil of ignorance \cite{Rawls1971}, while Scanlon asks which principles no one could reasonably reject \cite{Scanlon1998}.

These traditions differ not merely in their conclusions about hard cases --- the trolley dilemma \cite{Foot1967, Thomson1985} famously elicits divergent verdicts --- but in the type of reasoning they demand. Consequentialism requires comparing expected outcomes, a procedure amenable to quantification. Deontology requires identifying applicable rules and testing their universalizability. Virtue ethics requires assessing character and situational judgment. Care ethics requires attending to relational context and the needs of concrete others. Contractarianism requires counterfactual reasoning about what principles could survive mutual justification. Each procedure maps differently onto the task formats available to benchmark designers: forced-choice outcome comparisons naturally favor consequentialist framing, while care-ethical or contractarian reasoning demands richer contextual scaffolding that multiple-choice items struggle to provide.

This asymmetry in operationalizability has consequences for which traditions the AI ethics ecosystem reinforces. The field's dominant ethical vocabulary is heavily consequentialist-adjacent. Jobin et al. surveyed 84 AI ethics guidelines and found that "transparency," "justice and fairness," "non-maleficence," and "responsibility" emerged as the most frequently endorsed principles \cite{Jobin2019}. Floridi and Cowls unified the landscape under five headings --- beneficence, non-maleficence, autonomy, justice, and explicability --- four of which are borrowed from biomedical principlism and oriented primarily toward outcomes and harms \cite{Floridi2019}. Hagendorff's evaluation of 22 guideline documents concluded that the prevailing approach to AI ethics is "deontological" in the narrow sense of postulating fixed universal principles, and argued that it should be augmented with virtue-ethical attention to character and moral formation \cite{Hagendorff2020}. Care ethics and contractarian reasoning are largely absent from these documents. As van de Poel observes, even when guidelines invoke values such as fairness or autonomy, the question of how to embed those values in technical systems remains unresolved across ethical traditions \cite{vanDePoel2020}. Aditya's recent review of global AI governance frameworks similarly notes that consequentialist, deontological, virtue-ethical, and care-ethical perspectives offer "complementary insights," yet regulatory convergence has overwhelmingly favored outcome-based evaluation \cite{Aditya2025}.

Mittelstadt warns that this apparent consensus around high-level principles "hide[s] deep political and normative disagreement" and that "principles alone cannot guarantee ethical AI" without the professional norms, fiduciary duties, and accountability mechanisms that gave principlism traction in medicine \cite{Mittelstadt2019}. The principlist convergence has a downstream effect that prior surveys have not examined: benchmark designers build what guidelines emphasize. When the dominant ethical vocabulary centers on beneficence, harm prevention, and fairness --- concepts most naturally operationalized through outcome comparison --- the resulting benchmarks inherit a consequentialist framing by default. Traditions that require relational context (care ethics), character assessment (virtue ethics), or counterfactual agreement (contractarianism) are systematically harder to test and therefore systematically under-represented.

This tradition-level analysis is precisely what existing benchmark surveys lack. By making the philosophical

presuppositions of benchmark design explicit, we lay the groundwork for two of this paper's central empirical claims: that normative ethics benchmarks exhibit a pronounced consequentialist bias (Claim 2) and that care ethics and contractarianism constitute a near-total void in the evaluation landscape (Claim 3). These gaps are not accidental; they are inherited from the field's philosophical starting point.

2.4 The Cognitive Science of Moral Judgment

The cognitive science of moral judgment provides a mechanistic account of why different normative frameworks may be differentially accessible to language models. Greene and colleagues' landmark fMRI studies established that moral dilemmas engage qualitatively distinct neural processes depending on their structure \cite{Greene2001, Greene2004}. Personal moral dilemmas --- such as pushing a stranger off a footbridge to stop a trolley --- activate emotion-related brain regions including the ventromedial prefrontal cortex (VMPFC) and amygdala, producing characteristically deontological responses (i.e., judging the harmful action as impermissible). Impersonal dilemmas --- such as flipping a switch to divert the same trolley --- preferentially recruit the dorsolateral prefrontal cortex (DLPFC) and anterior cingulate cortex, regions associated with cognitive control and cost-benefit calculation, yielding utilitarian responses. Crucially, when participants override their initial emotional aversion to endorse the utilitarian option in personal dilemmas, Greene et al. observed increased DLPFC activation and longer response times, consistent with effortful cognitive control overriding a prepotent affective response \cite{Greene2004}. Further evidence from VMPFC lesion patients, who make abnormally utilitarian judgments, confirmed the causal role of emotion in producing deontological intuitions \cite{Greene2007}.

Greene subsequently generalized these findings into a dual-process theory of moral judgment, casting utilitarian reasoning as the brain's "manual mode" --- deliberate, controlled, and linguistically explicit --- and deontological responses as "automatic mode" --- fast, intuitive, and grounded in emotional reactions that evolved for face-to-face social interactions \cite{Greene2007, Greene2014}. This framing carries a critical implication for language models: because LLMs are trained exclusively on text that encodes explicit reasoning chains, they may disproportionately acquire the "manual mode" patterns characteristic of consequentialist deliberation while lacking the embodied emotional substrates that produce deontological and care-based intuitions in humans.

Cushman formalized a complementary distinction between action-based and outcome-based valuation systems \cite{Cushman2013}. Drawing on computational models of reinforcement learning, he argued that one system assigns intrinsic value to actions based on their reinforcement history (corresponding to deontological sensitivity --- pushing a person feels wrong regardless of consequences), while another evaluates actions by their expected outcomes (corresponding to consequentialist calculation). These systems are neurologically and computationally dissociable, explaining why harmful actions used as a means are judged more severely than equivalent harms occurring as side effects. Crockett extended this framework further by mapping moral cognition onto model-based (outcome-evaluating, flexible but computationally expensive), model-free (action-evaluating, efficient but retrospective), and Pavlovian (reflexive aversion) decision systems, offering a parsimonious mechanistic account of why physical-contact harms elicit stronger deontological responses than remote harms with identical outcomes \cite{Crockett2013}.

A methodological advance by Conway and Gawronski challenged the assumption that deontological and utilitarian responding are opposite ends of a single spectrum \cite{Conway2013}. Applying process dissociation procedures, they demonstrated that deontological and utilitarian inclinations function as independent dimensions: an individual --- or, by extension, a model --- can score high on both, low on both, or high on one and low on the other. This finding complicates the interpretation of benchmark scores, since a model's high accuracy on utilitarian dilemmas need not indicate low deontological sensitivity; the two capacities are simply not being measured independently by conventional forced-choice designs.

These process-level findings must also be situated within broader debates about the nature of moral cognition. Haidt's social intuitionist model argues that moral judgments are primarily the product of rapid intuitive evaluations, with explicit moral reasoning serving largely as post-hoc rationalization \cite{Haidt2001}. The classic evidence for "moral dumbfounding" --- where people maintain firm moral judgments while being unable to articulate reasons --- suggests that what benchmarks test as "moral reasoning" may in fact be measuring rationalization fluency rather than genuine moral

cognition. Furthermore, Kahane et al. demonstrated that what appears to be "utilitarian" responding on trolley-type dilemmas may sometimes reflect reduced emotional engagement or diminished empathy rather than principled welfare-maximization, raising questions about whether high consequentialist scores indicate sophisticated moral reasoning or affective deficiency \cite{Kahane2012}. Bago et al.'s large-scale cross-cultural study confirmed that situational factors --- including personal force, intention, and the degree of benefit --- systematically shape moral judgments even when the utilitarian calculus remains constant, underscoring the context-dependence that standardized benchmarks struggle to capture \cite{Bago2022}.

Taken together, the dual-process framework generates a specific prediction about LLM behavior: models trained on corpora of explicit, text-based argumentation should exhibit systematic facility with consequentialist reasoning --- which is inherently propositional and calculable --- while struggling with moral knowledge that is encoded in embodied intuition, relational attunement, or culturally transmitted emotional responses. This is precisely the consequentialist bias we document in Claim 2. Moreover, the fundamental distinction between moral judgment (selecting an answer) and moral reasoning (articulating a justification) maps directly onto the dual-process dissociation: the 44 of 54 benchmarks in our corpus that primarily test judgment are conflating two cognitively and neurologically distinct processes, a concern we develop in Claim 4.

2.5 Why Care Ethics Resists Benchmarking

The near-total absence of care ethics benchmarks documented in our survey is not an oversight by negligent designers but a consequence of deep structural incompatibilities between what care ethics demands and what benchmark methodology can deliver. Understanding why requires briefly recalling care ethics' core commitments and then tracing their collision with the assumptions embedded in standardized evaluation.

Care ethics, as developed by Gilligan \cite{Gilligan1982}, Noddings \cite{Noddings1984}, Tronto \cite{Tronto1993}, and Held \cite{Held2006}, rejects the premise that moral reasoning consists in applying universal principles to particular cases. Instead, it holds that moral life is constitutively relational: persons are not autonomous agents who subsequently enter relationships but are formed through ongoing webs of dependency, care, and mutual obligation. Moral judgment, on this view, proceeds not by subsumption under rules but by attending to the concrete needs of particular others with whom one stands in relationship. Held summarizes the central commitment: the ethics of care starts with "the compelling moral salience of attending to and meeting the needs of the particular others for whom we take responsibility" \cite{Held2006}. Tronto likewise grounds moral development in "the understanding of responsibility and relationships" rather than "rights and rules" \cite{Tronto1993}. Three features follow directly: moral knowledge is contextual rather than abstract, moral reasoning is dialogic rather than monologic, and moral assessment is longitudinal --- tracking the quality of a relationship over time rather than scoring a single decision in isolation.

Each of these features creates a specific obstacle for benchmark design. Standard benchmarks present a decontextualized vignette and solicit a single response, scored against a fixed answer key. Care ethics, by contrast, insists that right action depends on who the parties are, what history they share, and what vulnerability is at stake --- precisely the particulars that vignette construction strips away. Weinberger \cite{Weinberger2024} observes a striking irony in this regard: machine learning generalizations, unlike traditional generalizations, "do not absorb particulars into a single generalization but rather maintain them as necessary for applying the generalization to cases." This "rise of particulars" in ML creates an unexpected philosophical resonance with care ethics' insistence on contextual attention. Yet benchmark design moves in the opposite direction, abstracting away the very relational and contextual details that both ML models and care ethics treat as morally determinative. The format that enables scalable evaluation is the format that makes care ethics unevaluable.

Domain-specific efforts confirm this structural difficulty. Van Wynsberghe \cite{vanWynsberghe2016} developed Care-Centered Value Sensitive Design for evaluating service robots, grounding normative assessment in the analysis of concrete care practices --- whether a robot is integrated into a relationship of reciprocal need-meeting between caregiver and care-receiver. Asaro \cite{Asaro2019} proposed replacing utilitarian threat models in predictive policing with a substantive ethics of care that demands "moral attention to those who may be negatively impacted by the use of technology." Tavory \cite{Tavory2024} argued that responsible AI principles are insufficient for mental health AI because

they do not address the relational dynamics between bot and user --- the abrupt termination of a therapeutic "relationship" by a commercial chatbot illustrating the gap. Each of these contributions applies care ethics to a specific AI domain, yet none produced a general-purpose benchmark. The reason is consistent: care-ethical evaluation requires knowledge of the particular relationship, which cannot be specified in a domain-general item.

A growing theoretical literature underscores both the urgency and difficulty of operationalization. Birhane \cite{Birhane2021} calls for a relational ethics that centers disproportionately impacted groups and treats fairness as a "moving target" rather than a fixed metric. Villegas-Galaviz and Martin \cite{VillegasGalaviz2023} identify two forms of moral distance that AI creates --- proximity distance (eliminating face-to-face interaction) and bureaucratic distance (abstracting decisions into opaque processes) --- and argue that care ethics is uniquely suited to counteract both. Drage et al. \cite{Drage2023} reframe responsibility in AI development through feminist care theory, calling for "response-ability" grounded in ongoing maintenance rather than one-time compliance. El Masri and Snoswell \cite{ElMasri2025} argue that current value alignment methods (RLHF, RLAIF, Constitutional AI) embed utilitarian or deontological commitments and propose care-ethics-informed alignment that would train models to be "morally attentive" to individual vulnerability. Wegner et al. \cite{Wegner2024} systematize the emerging "relational turn" in algorithmic fairness, documenting how third-wave AI ethics increasingly foregrounds power hierarchies and structural inequalities that distributive fairness metrics cannot capture. These contributions converge on a shared diagnosis --- care ethics is needed in AI --- but none offers a psychometric instrument that could serve as a benchmark.

The structural problem is now clear. Care ethics evaluation may require fundamentally different assessment formats: multi-turn dialogue that tracks attentiveness across an evolving relationship, longitudinal interaction modeling that captures how care is sustained or withdrawn over time, and evaluation rubrics sensitive to the power asymmetry between system and user. None of these fit the single-turn, single-answer paradigm that defines existing normative ethics benchmarks. The care ethics void in our survey is therefore best understood not as a gap waiting to be filled by a sufficiently creative item-writer but as evidence that current evaluation methodology has a structural ceiling --- one that cannot be raised without rethinking what a benchmark is. This motivates our third claim: the care and contractarian lacuna reflects the limits of benchmark design, not merely incomplete coverage of the philosophical landscape.

2.6 Data Contamination, RLHF, and Training-Time Bias

The validity of normative ethics benchmarks is further threatened by two mechanisms operating on the model side: data contamination and training-time preference optimization. The ETHICS benchmark \cite{Hendrycks2021}, released in 2021 and hosted publicly on GitHub, is paradigmatic of the contamination risk. Because frontier LLMs are trained on web-scale corpora that routinely ingest open-access datasets, any benchmark whose test instances appear in plain text online is a candidate for memorization. Golchin and Surdeanu \cite{Golchin2024} demonstrated that LLMs can effectively "time travel," reproducing benchmark instances verbatim when prompted with dataset-identifying metadata --- achieving 92--100\% detection accuracy across seven contaminated partitions and confirming contamination in GPT-4 for multiple datasets. Jacovi et al. \cite{Jacovi2023} underscored the urgency of the problem, warning that "the NLP evaluation community is now witness to two urgent crises" involving contamination from web crawls and from API-submitted evaluation data, and proposing encryption and licensing countermeasures that remain largely unadopted. The contamination pathway is well characterized: Magar and Schwartz \cite{Magar2022} documented how memorized training instances inflate downstream performance, while Sainz et al. \cite{Sainz2023} argued that NLP evaluation is "in trouble" because contamination undermines the basic assumption that test data is unseen. More recent work has shown the problem is worsening rather than abating. Deng et al. \cite{Deng2024} investigated contamination specifically in modern benchmarks, and Oren et al. \cite{Oren2024} proved that test set contamination can be detected even in black-box LLMs whose training data is not disclosed --- implying that contamination is not merely a transparency problem but a structural one. Comprehensive surveys by Li et al. \cite{Li2024} and Xu et al. \cite{Xu2024} document the breadth of contamination across tasks, models, and evaluation paradigms. For normative ethics benchmarks specifically, contamination casts doubt on the ceiling effects we document in Claim 1: when GPT-4 achieves near-perfect accuracy on ETHICS subtasks, the score may reflect memorization of training-time instances rather than any form of moral reasoning.

A second, subtler distortion arises from the preference optimization procedures --- reinforcement learning from human

feedback \cite{Christiano2017} and its offline variant, direct preference optimization \cite{Rafailov2023} --- that are now standard in post-training pipelines. Both methods optimize model outputs to match aggregated human preference judgments, yet as Casper et al. \cite{Casper2023} systematically documented, this process introduces several fundamental limitations: annotator preferences may not track moral quality, optimization toward the average annotator suppresses minority moral views, and the reward model can misgeneralize in ways that are difficult to detect. The dominant alignment paradigm, Anthropic's "helpful, harmless, and honest" (HHH) framework \cite{Askell2021}, operationalizes harmlessness primarily in terms of avoiding negative consequences --- an implicitly consequentialist criterion. Dai et al. \cite{Dai2024} advanced this approach by decoupling helpfulness and harmlessness into separate reward and cost models, yet the safety constraints themselves remain outcome-based, defined by whether a response produces harmful effects rather than whether it violates duties or rights. Gabriel \cite{Gabriel2020} identified the deeper philosophical problem: value alignment requires specifying whose values and which ethical framework should govern the system, but RLHF sidesteps this question entirely by aggregating preferences into a single scalar reward, effectively collapsing pluralistic moral reasoning into a utilitarian social welfare function. The empirical consequences of this design choice are now becoming visible. Tennant et al. \cite{Tennant2025}, in work presented at ICLR 2025, demonstrated that the structure of the reward signal directly shapes the moral orientation of LLM agents: deontological reward functions produce cooperation-oriented strategies while utilitarian rewards produce outcome-maximizing ones, confirming that alignment training does not merely surface pre-existing moral knowledge but actively constructs it. Panickssery et al. \cite{Panickssery2024} further showed that LLM evaluators systematically favor their own generations, suggesting that self-evaluation protocols --- common in benchmark validation --- may amplify rather than correct training-time biases.

Together, contamination and RLHF mean that benchmark scores reflect an inseparable entanglement of genuine moral reasoning capability, memorization of training instances, and systematic biases introduced during preference optimization. Current evaluation methodology offers no principled way to disentangle these contributions, motivating the meta-evaluative framework we develop in the remainder of this paper.

2.7 Cross-Cultural Blind Spots and Value Pluralism

The tradition biases documented above are compounded by a deeper structural problem: the populations whose moral intuitions inform benchmark design are themselves unrepresentative. Henrich, Heine, and Norenzayan's landmark review demonstrated that behavioral science draws overwhelmingly from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies, yet WEIRD subjects are statistical outliers on a wide range of psychological variables, including those most relevant to moral cognition \cite{Henrich2010}. Moral psychology is no exception to this pattern. Sachdeva, Singh, and Medin showed that moral reasoning varies systematically across cultures in ways that the dominant Western normative traditions --- consequentialism, deontology, and virtue ethics --- fail to capture, arguing that the universalist framework endemic to moral psychology may lead to a systematic underestimation of the role of culture in moral cognition \cite{Sachdeva2011}. When AI systems are built and evaluated against instruments calibrated to WEIRD populations, these cultural blind spots propagate downstream. Atari et al. extended the WEIRD critique directly to artificial intelligence, demonstrating that large language models exhibit psychological profiles most closely resembling WEIRD populations, with alignment declining steeply as target populations diverge from Western norms \cite{Atari2023}. The Moral Machine experiment provided the most comprehensive empirical test of cross-cultural moral variation to date: analyzing responses from 70,000 participants across 42 countries, Awad et al. found that the qualitative ordering of sacrifice acceptability is universal, but quantitative preferences vary substantially along cultural dimensions such as relational mobility \cite{Awad2020}. Morality, in short, is partially universal but far from fully so.

These empirical findings have direct implications for LLM evaluation. Mohammadi et al. tested whether current language models capture the cross-cultural spectrum of moral attitudes and found that they systematically compress cultural variation, exhibiting low alignment with empirical survey patterns and defaulting to Western moral baselines \cite{Mohammadi2025}. The mechanism is partly linguistic: Talat et al. showed that multilingual models inherit moral and social biases from their dominant training language, so that evaluating a model in English imposes Anglo-centric normative assumptions even when the model nominally supports other languages \cite{Talat2022}. The philosophical literature offers frameworks for navigating precisely this kind of disagreement. MacAskill, Bykvist, and Ord developed a formal theory of moral uncertainty, arguing that when an agent has credence across multiple moral theories, rational

decision-making requires weighting each theory proportionally rather than collapsing to a single framework \cite{MacAskill2020}. Newberry and Ord operationalized this insight through a "parliamentary approach" in which competing moral theories are represented as delegates whose influence is proportional to the agent's credence in each, reaching decisions through negotiation rather than aggregation \cite{Newberry2021}. Recent work has begun applying these ideas to AI systems specifically. Yaacov proposed a deliberative moral reasoning architecture grounded in normative moral pluralism, incorporating a universal threshold layer alongside a local layer that integrates culturally specific normative content \cite{Yaacov2025}. Aditya similarly argued that global AI governance requires frameworks that accommodate cross-cultural value variation rather than imposing a single ethical tradition \cite{Aditya2025}. The upshot for benchmark evaluation is clear: current instruments embed not merely a bias toward particular normative traditions but a deeper cultural bias rooted in the populations and philosophical assumptions that shaped their design. Evaluation under genuine moral pluralism demands approaches fundamentally different from single-answer classification --- approaches that can represent, weight, and reason across competing moral commitments rather than treating one tradition's judgments as ground truth.

2.8 Documentation, Reproducibility, and the Data Bottleneck

The cultural and philosophical concerns raised above are exacerbated by a more prosaic but equally consequential problem: inadequate documentation. A growing consensus holds that machine learning artifacts require standardized documentation to be responsibly deployed. Gebru et al. proposed datasheets for datasets, modeled on electronics industry practice, requiring creators to document motivation, composition, collection processes, intended uses, and ethical considerations \cite{Gebru2021}. Mitchell et al. introduced model cards as companion documents for trained models, specifying intended use cases and performance across demographic groups \cite{Mitchell2019}. Bender and Friedman developed data statements for natural language processing, emphasizing the need to document the demographic and linguistic characteristics of annotator and speaker populations \cite{Bender2018}. Holland et al. proposed dataset nutrition labels as a more accessible alternative, designed to surface key properties at a glance \cite{Holland2018}. Despite this proliferation of documentation frameworks, adoption remains shallow. Liang et al. conducted the most comprehensive study to date, analyzing 32,111 model cards on Hugging Face and finding that sections addressing limitations, evaluation methodology, and environmental impact had the lowest completion rates, while training details were documented most consistently \cite{Liang2024Cards}. Paullada et al. offered a broader diagnosis, surveying systematic failures in dataset development practices across computer vision and natural language processing and arguing that the machine learning community's emphasis on benchmark-driven progress has crowded out careful attention to data provenance, annotation quality, and representational coverage \cite{Paullada2021}.

For normative ethics benchmarks, the documentation gap is especially consequential. Unlike a standard NLP benchmark where the task definition is relatively uncontested, a moral reasoning benchmark embeds substantive philosophical commitments --- which ethical theories are represented, how moral dilemmas are operationalized, what counts as a correct answer --- that constitute design choices with direct implications for construct validity. When these commitments go undocumented, downstream users cannot assess whether a benchmark's normative assumptions are compatible with their own evaluation goals. Karimi-Malekabadi et al. addressed this problem directly by proposing the Theory Trace Card (TTC), a lightweight documentation artifact for socio-cognitive benchmarks that makes explicit the theoretical basis of an evaluation, the components of the target capability it exercises, its operationalization, and its limitations \cite{KarimiMalekabadi2026}. The TTC framework is particularly apt for moral reasoning benchmarks, where the gap between benchmark performance and real-world moral competence --- what the authors term a "validity illusion" --- is especially wide. Our analysis of 54 benchmarks reveals a further structural problem: 18 (33\%) lack publicly available data, and several of the highest-scoring benchmarks on normative depth are among those that are inaccessible. This produces a paradox in which the benchmarks that engage most seriously with philosophical theory are the hardest for other researchers to build upon. Biderman et al., drawing on three years of experience maintaining the Language Model Evaluation Harness, documented how seemingly minor implementation choices --- prompt formatting, few-shot example selection, tokenization --- can produce substantial differences in reported performance, making reproducibility elusive even when data are nominally available \cite{Biderman2024}. The NeurIPS Datasets and Benchmarks track now requires data availability statements, a welcome step that our findings suggest should be extended to require explicit documentation of philosophical commitments. The 26-feature matrix we develop in this

paper can serve as a template for such documentation, making visible the normative assumptions, response formats, and evaluation criteria that determine what a benchmark actually measures --- and what it leaves unmeasured.

2.9 Existing Surveys and Our Contribution

Several surveys have mapped the landscape of AI ethics evaluation, each illuminating important dimensions while leaving others in shadow. We briefly position our work against the five most relevant predecessors.

Ray \cite{Ray2023} offered an early taxonomy of normative ethics benchmarks, comparing datasets along dimensions such as task format, scale, and ethical framework coverage. This work usefully organized a fragmented field, but the analysis relied entirely on published descriptions: no benchmark was executed, and the taxonomy did not decompose coverage by philosophical tradition, treating "ethics" as a largely undifferentiated category. Snoswell et al. \cite{Snoswell2025}, in the most substantial recent contribution, surveyed 69 papers spanning LLM moral competence evaluation --- including safety, fairness, toxicity, and moral reasoning --- and argued persuasively that verdict-level agreement metrics obscure the reasoning processes they purport to measure. Their scope, however, necessarily trades depth for breadth: normative ethics is not isolated as a distinct evaluation category, and the survey does not examine whether benchmarks differentially engage deontological, consequentialist, virtue-theoretic, care-ethical, or contractarian reasoning.

At the cataloging end of the spectrum, Rismani et al. \cite{Rismani2025} assembled the most comprehensive inventory to date --- 791 AI ethics evaluation measures spanning technical benchmarks, organizational audits, and governance instruments. The sheer scale of this effort is valuable for orienting researchers to the full ecosystem, yet the inclusion of non-technical measures makes it difficult to extract actionable guidance for the narrower question of how well normative ethics benchmarks probe philosophical reasoning. Rauh et al. \cite{Rauh2024} approached the problem from a gap-analysis perspective, systematically identifying lacunae in AI safety evaluation infrastructure and explicitly noting the absence of assessments capable of eliciting nuanced moral reasoning. Their diagnosis is well-supported, but the work stops at identification: it proposes neither a feature framework for characterizing existing benchmarks nor empirical evidence of how current tools perform in practice. Methodologically closest to our approach, Rottger et al. \cite{Rottger2025} conducted a systematic review of 102 open safety evaluation datasets, evaluating them against quality criteria including discriminative power, construct validity, and documentation standards. Their emphasis on evaluating the evaluations --- rather than merely listing them --- anticipates our own methodological commitment, but their focus on safety broadly construed means normative ethics receives no dedicated treatment, and philosophical tradition plays no role in their analysis.

Our work extends this collective foundation in three directions. First, we treat normative tradition as a first-class evaluation dimension, scoring each benchmark's engagement with five philosophical frameworks rather than aggregating ethics into a monolithic category. This decomposition reveals systematic imbalances --- most critically, the near-total absence of care ethics and contractarian reasoning from existing evaluation suites --- that remain invisible when benchmarks are characterized only by task format or domain label. Second, we complement feature-level analysis with implementation trials that expose empirical properties no survey-only methodology can detect: ceiling effects in which frontier models achieve near-perfect accuracy \cite{Hendrycks2021}, stable difficulty hierarchies across philosophical traditions, and the emergence of refusal as a de facto normative position that current scoring rubrics cannot accommodate. Third, we document a saturation finding with immediate practical consequences: the field's most-cited and most-integrated normative ethics benchmark is saturated for frontier models, meaning that every meta-benchmark suite incorporating it --- including widely used evaluation harnesses \cite{Liang2022} --- inherits a component that no longer discriminates between competing systems.

Together, these contributions provide both a reusable analytic framework for future benchmark development and empirical evidence that current evaluation infrastructure systematically underestimates the difficulty of normative moral reasoning. Where prior surveys have asked what benchmarks exist, we ask what do they actually measure --- and find that the answer is considerably less than the field assumes.

12-Member Expert Council

This review was designed with input from a 12-member interdisciplinary expert council whose perspectives shaped the structure, emphasis, and citation selection of each section:

#	Role	Primary Contribution
1	Dr. Construct -- Construct Validity Specialist	Framed the meta-evaluative premise (§2.2)
2	Dr. Lifecycle -- Benchmark Saturation Historian	Documented saturation mechanisms (§2.2, §1)
3	Dr. Machine Ethics -- Computational Ethics Pioneer	Established 20-year intellectual lineage (§2.1)
4	Dr. Dual-Process -- Moral Cognition Researcher	Explained framework-dependent accessibility (§2.4)
5	Dr. Care Ethics -- Feminist Ethics Scholar	Diagnosed structural incompatibility (§2.5)
6	Dr. Contamination -- Data Leakage Expert	Analyzed contamination threats (§2.6)
7	Dr. RLHF -- Alignment Training Specialist	Connected training procedures to bias (§2.6)
8	Dr. Documentation -- Benchmark Documentation Expert	Linked documentation failures to reproducibility (§2.8)
9	Dr. Cross-Cultural -- Cultural Psychology Expert	Extended tradition bias to cultural bias (§2.7)
10	Dr. Moral Philosophy -- Reasoning vs. Judgment Psychologist	Provided philosophical foundations (§2.3, §2.4)
11	Dr. Pluralism -- Value Pluralism Specialist	Introduced moral uncertainty frameworks (§2.7)
12	Dr. Meta-Science -- Systematic Review Methodologist	Positioned against prior surveys (§2.9)

New Citation Keys Required

The following citation keys are used in this comprehensive review but do not appear in the current references-normeval.md. They will need to be added:

Block A -- Construct Validity:

* `[Burden2025]` Burden et al. 2025 -- "Measuring What Matters: Construct Validity in LLM Benchmarks"
* `[Bowman2021]` Bowman & Dahl 2021 -- "What Will it Take to Fix Benchmarking in NLU?"
* `[Raji2021]` Raji et al. 2021 -- "AI and the Everything in the Whole Wide World Benchmark"
* `[Dehghani2021]` Dehghani et al. 2021 -- "The Benchmark Lottery"
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* `[Kiela2021]` Kiela et al. 2021 -- "Dynabench: Rethinking Benchmarking in NLP"
* `[Gururangan2018]` Gururangan et al. 2018 -- "Annotation Artifacts in NLI Data"
* `[McCoy2019]` McCoy et al. 2019 -- "Right for the Wrong Reasons"
* `[Wang2019SuperGLUE]` Wang et al. 2019 -- "SuperGLUE: A Stickier Benchmark"
* `[Bean2024]` Bean 2024 -- "Measuring What Matters: Construct Validity in LLM Benchmarks"
* `[Eriksson2025]` Eriksson 2025 -- "Can We Trust AI Benchmarks?"
* `[McGregor2025]` McGregor 2025 -- "BenchRisk: Identifying 57 Failure Modes"
* `[Srivastava2024]` Srivastava 2024 -- "Functional Benchmarks for Robust Evaluation"
* `[Beyer2025]` Beyer et al. 2025 -- "LLM Safety Evaluations Lack Robustness"
* `[Vania2021]` Vania et al. 2021 -- "Comparing Test Sets with Item Response Theory"
* `[Rodriguez2021]` Rodriguez et al. 2021 -- "Evaluation Examples Are Not Equally Informative"
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Block B -- Computational Ethics:

* `[Moor2006]` Moor 2006 -- "The Nature, Importance, and Difficulty of Machine Ethics"
* `[Allen2005]` Allen, Smit & Wallach 2005 -- "Artificial Morality: Top-down, Bottom-up, and Hybrid"
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* `[Floridi2004]` Floridi & Sanders 2004 -- "On the Morality of Artificial Agents"
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* `[Powers2006]` Powers 2006 -- "Prospects for a Kantian Machine"
* `[Asaro2006]` Asaro 2006 -- "What Should We Want From a Robot Ethic?"
* `[Bonnefon2016]` Bonnefon, Shariff & Rahwan 2016 -- "The Social Dilemma of Autonomous Vehicles"
* `[Cervantes2020]` Cervantes et al. 2020 -- "Artificial Moral Agents: A Survey"
* `[Tolmeijer2021]` Tolmeijer et al. 2021 -- "Implementations in Machine Ethics: A Survey"
* `[Etzioni2017]` Etzioni & Etzioni 2017 -- "Incorporating Ethics into AI"
* `[Dennis2015]` Dennis & Fisher 2015 -- "Towards Verifiably Ethical Robot Behaviour"
* `[AwadCE2022]` Awad et al. 2022 -- "Computational Ethics" (TiCS)

Block C -- Dual-Process:

* `[Greene2001]` Greene et al. 2001 -- "An fMRI Investigation of Emotional Engagement in Moral Judgment"
* `[Greene2004]` Greene et al. 2004 -- "The Neural Bases of Cognitive Conflict and Control in Moral Judgment"
* `[Greene2007]` Greene 2007 -- "Why are VMPFC Patients More Utilitarian?"
* `[Greene2014]` Greene 2014 -- "Beyond Point-and-Shoot Morality"
* `[Haidt2001]` Haidt 2001 -- "The Emotional Dog and Its Rational Tail"
* `[Cushman2013]` Cushman 2013 -- "Action, Outcome, and Value: A Dual-System Framework"
* `[Conway2013]` Conway & Gawronski 2013 -- "Deontological and Utilitarian Inclinations"
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Block D -- Care Ethics:

* `[Held2006]` Held 2006 -- "The Ethics of Care: Personal, Political, and Global"
* `[Tronto1993]` Tronto 1993 -- "Moral Boundaries: A Political Argument for an Ethics of Care"
* `[Weinberger2024]` Weinberger 2024 -- "The Rise of Particulars: AI and Ethics of Care"
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* `[Asaro2019]` Asaro 2019 -- "AI Ethics in Predictive Policing: Ethics of Care"
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* `[Birhane2021]` Birhane 2021 -- "Algorithmic Injustice: A Relational Ethics Approach"
* `[VillegasGalaviz2023]` Villegas-Galaviz & Martin 2023 -- "Moral Distance and AI: Ethics of Care"
* `[Drage2023]` Drage et al. 2023 -- "Engineers on Responsibility: Feminist AI Ethics"
* `[ElMasri2025]` El Masri & Snoswell 2025 -- "Towards Attuned AI"
* `[Wegner2024]` Wegner et al. 2024 -- "Beyond Distributions: Relational Algorithmic Justice"

Block E -- Contamination:

* `[Golchin2024]` Golchin & Surdeanu 2024 -- "Time Travel in LLMs: Data Contamination"
* `[Jacovi2023]` Jacovi et al. 2023 -- "Stop Uploading Test Data in Plain Text"
* `[Sainz2023]` Sainz et al. 2023 -- "NLP Evaluation in Trouble"
* `[Magar2022]` Magar & Schwartz 2022 -- "Data Contamination: From Memorization to Exploitation"
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* `[Li2024]` Li et al. 2024 -- "Task Contamination: LMs May Not Be Few-Shot Anymore"
* `[Xu2024]` Xu et al. 2024 -- "Benchmark Data Contamination of LLMs: A Survey"

Block F -- RLHF:

* `[Christiano2017]` Christiano et al. 2017 -- "Deep RL from Human Preferences"
* `[Casper2023]` Casper et al. 2023 -- "Open Problems and Fundamental Limitations of RLHF"
* `[Rafailov2023]` Rafailov et al. 2023 -- "Direct Preference Optimization"
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* `[Dai2024]` Dai et al. 2024 -- "Safe RLHF"
* `[Gabriel2020]` Gabriel 2020 -- "Artificial Intelligence, Values, and Alignment"
* `[Tennant2025]` Tennant et al. 2025 -- "Moral Alignment for LLM Agents"
* `[Panickssery2024]` Panickssery et al. 2024 -- "LLM Evaluators Recognize and Favor Their Own Generations"

Block G -- Documentation:

* `[Gebru2021]` Gebru et al. 2021 -- "Datasheets for Datasets"
* `[Mitchell2019]` Mitchell et al. 2019 -- "Model Cards for Model Reporting"
* `[Bender2018]` Bender & Friedman 2018 -- "Data Statements for NLP"
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* `[Liang2024Cards]` Liang et al. 2024 -- "What's Documented in AI? 32K Model Cards"
* `[Paullada2021]` Paullada et al. 2021 -- "Data and its (Dis)contents"
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* `[Biderman2024]` Biderman et al. 2024 -- "Lessons from the Trenches on Reproducible Evaluation"

Block H -- Cross-Cultural:

* `[Henrich2010]` Henrich, Heine & Norenzayan 2010 -- "The Weirdest People in the World?"
* `[Sachdeva2011]` Sachdeva, Singh & Medin 2011 -- "Culture and the Quest for Universal Principles"
* `[Atari2023]` Atari et al. 2023 -- "Which Humans? Cultural Bias in AI"
* `[Awad2020]` Awad et al. 2020 -- "Universals and Variations in Moral Decisions"
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* `[Talat2022]` Talat et al. 2022 -- "You Reap What You Sow: Multilingual Bias"

Block L -- Pluralism:

* `[MacAskill2020]` MacAskill, Bykvist & Ord 2020 -- "Moral Uncertainty"
* `[Newberry2021]` Newberry & Ord 2021 -- "The Parliamentary Approach to Moral Uncertainty"
* `[Yaacov2025]` Yaacov 2025 -- "Normative Moral Pluralism for AI"

Additional Philosophy:

* `[Jobin2019]` Jobin, Ienca & Vayena 2019 -- "The Global Landscape of AI Ethics Guidelines"
* `[Floridi2019]` Floridi & Cowls 2019 -- "A Unified Framework of Five Principles for AI"
* `[Hagendorff2020]` Hagendorff 2020 -- "The Ethics of AI Ethics: An Evaluation of Guidelines"
* `[Mittelstadt2019]` Mittelstadt 2019 -- "Principles Alone Cannot Guarantee Ethical AI"
* `[vanDePoel2020]` van de Poel 2020 -- "Embedding Values in AI Systems"
* `[McDonald2023]` McDonald 2023 -- "AI Alignment and the Categorical Imperative"
* `[Aditya2025]` Aditya 2025 -- "Toward Global Ethical Frameworks for AI"

2. Methods

3. Methods

This section describes the complete methodology for assembling, scoring, and empirically validating a corpus of 54 normative ethics benchmarks for large language models. The pipeline proceeds in seven stages: corpus assembly (Section 3.1), feature matrix design (Section 3.2), composite scoring with hard gates (Section 3.3), two-phase implementation trials (Section 3.4), statistical methods (Section 3.5), a care ethics benchmark prototype (Section 3.6), and methodological limitations (Section 3.7).

3.1 Benchmark Corpus Assembly

We assembled the evaluation corpus through a two-stage process combining curated vault mining with systematic search.

Stage 1: CEI Literature Vault mining. The CEI Literature Vault is a curated database of 2,946 AI ethics papers maintained by the Center for Ethical Intelligence, in which each paper carries structured metadata including topic classifications, normative tradition tags, and evaluation-type indicators. An initial query against the vault's classification fields returned 41 papers tagged as normative ethics benchmarks. Manual review by the first author confirmed 30 of these as genuine normative ethics evaluation instruments -- that is, artifacts that present LLMs with morally valenced stimuli and score their responses against one or more normative frameworks. The remaining 11 were reclassified as borderline: they reference normative ethics vocabulary but do not operationalize any framework into a scoreable evaluation (e.g., papers that mention "deontological" in their framing but test only factual knowledge retrieval). These 30 confirmed benchmarks formed the seed corpus.

Stage 2: PRISMA-compliant systematic search. To ensure coverage beyond the vault's existing holdings, we conducted a systematic search across seven databases: arXiv, Semantic Scholar, ACL Anthology, Google Scholar, PapersWithCode, OpenReview, and HuggingFace Hub. We constructed ten query sets by combining terms for normative ethics traditions (virtue, deontological, utilitarian, care, contractarian), moral evaluation methods (benchmark, evaluation, assessment, test), and LLM identifiers (large language model, GPT, Claude, LLaMA). Representative queries included *"care ethics benchmark LLM"*, *"contractarian moral evaluation language model"*, and *"virtue ethics assessment large language model"*. We deliberately over-sampled queries targeting care ethics [Gilligan1982] and contractarian ethics [Rawls1971, Scanlon1998] to compensate for their anticipated under-representation, as preliminary vault analysis had already indicated severe voids in these traditions.

The search returned 30 external candidates after deduplication against the vault. Of these, 13 met all four inclusion criteria: (1) peer-reviewed publication or posting on a recognized preprint server (arXiv, SSRN, bioRxiv); (2) explicit evaluation of LLMs on normative ethical reasoning, not merely descriptive moral psychology; (3) publication date between 2021 and 2026; and (4) presentation of a scoreable instrument with defined items and response formats. The remaining 17 were excluded: 9 lacked genuine normative ethics content (testing factual recall or descriptive moral intuitions without framework grounding), 4 duplicated existing vault entries under variant titles, and 4 evaluated only descriptive moral psychology without any normative framing.

Corpus expansion trajectory. The vault's initial 30 confirmed benchmarks grew through two waves: 12 new benchmarks identified during the systematic search that had been present in the vault but not yet tagged as normative ethics instruments, and 5 genuinely new external papers ingested into the vault following Stage 2 search. Three of these external additions -- Xu et al. [Xu2026] (probing ethical framework representations), Zhang et al. [Zhang2026] (RLVR moral reasoning), and Du et al. [Du2025] (196 research ethics dilemmas) -- expanded the corpus into previously unrepresented evaluation paradigms. The remaining two -- IslamicMMLU [IslamicMMLU2026] (10,013 MCQ items

across Islamic knowledge domains) and IslamicLegalBench [IslamicLegalBench2026] (718 instances spanning seven jurisprudence schools) -- added non-Western normative tradition coverage that was entirely absent from the initial seed.

Final corpus. The assembled corpus comprises 54 benchmarks spanning 2021--2026. Figure 2 presents the PRISMA flow diagram tracing the identification, screening, eligibility, and inclusion stages.

3.2 Feature Matrix Design

We characterized each of the 54 benchmarks along 26 features organized into four axes (Table 1; full matrix in Table S1).

Axis A: Normative Depth (A1--A8). This axis measures the philosophical rigor with which a benchmark engages normative ethics. Features include: the number of ethical traditions explicitly tested (A1, integer 1--5); the depth of ethical engagement from surface-level keyword mention to framework-grounded scenario construction (A2, ordinal 1--5); whether the benchmark tests moral judgment, moral reasoning, or both (A3, categorical J/R/B); treatment of moral uncertainty and disagreement among reasonable ethical agents (A4, binary); pluralistic coverage of competing normative perspectives within the same item set (A5, ordinal 1--5); philosophical grounding quality on a calibrated 1--5 scale (A6; see calibration below); cultural framing beyond Western WEIRD populations (A7, ordinal 1--3); and multilingual support (A8, integer count of languages).

Axis B: Practical Ease (B1--B9). This axis measures how readily a benchmark can be implemented by researchers. Features include: task format (B1, categorical); dataset size in items (B2, integer); data availability via public repository (B3, binary); code availability (B4, binary); license terms permitting academic use (B5, ordinal); compatibility with established evaluation harnesses -- LM Eval Harness, HELM, EleutherAI, BIG-bench, TrustLLM, DecodingTrust -- assessed across all six frameworks (B6, integer 0--6); an overall implementability judgment integrating format standardization, documentation quality, and setup complexity (B7, ordinal 1--5); scorer type from automated exact-match to expert rubric (B8, categorical); and model compatibility across API-based and open-weight models (B9, ordinal 1--3).

Axis C: Citation Impact (C1--C5). This axis measures community uptake. Features include: raw citation count from Semantic Scholar (C1, integer); age-normalized citations per year (C2, continuous); venue tier from top-tier conference to workshop or unreviewed preprint (C3, ordinal 1--5); downstream adoption count reflecting benchmarks that explicitly build on or extend the target (C4, integer); and leaderboard presence on PapersWithCode or equivalent (C5, binary).

Axis D: Metadata (D1--D4). Provenance fields record: publication year (D1), first-author affiliation (D2), primary language (D3), and domain focus such as general, medical, legal, or cultural (D4).

Calibration anchors for A6 (Philosophical Grounding). Because A6 is the single most consequential feature -- it carries 20% weight in the Normative Depth composite -- we calibrated it against three anchors representing the extremes and midpoint of the scale. MoReBench [Chiu2025] received A6 = 5: it deploys expert rubric criteria for procedural reasoning under five named traditions, requiring models to demonstrate step-by-step ethical argumentation. ETHICS [Hendrycks2021] received A6 = 3: it names traditions and creates tradition-specific sub-datasets but operationalizes them as classification tasks without requiring reasoned justification. Hui *Decoding Moral* received A6 = 1: it mentions ethics vocabulary without substantive philosophical engagement. All remaining benchmarks were scored by the first author relative to these anchors, with written rationales for each score recorded in the supplementary feature matrix (Table S1).

3.3 Composite Scoring and Hard Gates

We computed three axis scores from the 26 features. All axis scores are normalized to 0--100.

Normative Depth (ND) weights philosophical grounding and reasoning depth most heavily:

> ND = A2 x 25 + A6 x 20 + A1 x 15 + A3 x 15 + A5 x 15 + A4 x 10

Practical Ease (PE) prioritizes implementability and data access:

> PE = B7 x 30 + B3 x 20 + B4 x 15 + B6 x 15 + B9 x 10 + B5 x 10

Citation Impact (CI) balances normalized citations with downstream adoption:

> CI = C2 x 35 + C4 x 30 + C3 x 20 + C5 x 15

The composite score combines the three axes:

> COMPOSITE = ND x 0.40 + PE x 0.35 + CI x 0.25

Weight rationale. We weighted ND highest (40%) because this survey specifically evaluates *normative* depth -- a benchmark that is easy to run but philosophically shallow serves our purpose less than one that is harder to implement but genuinely engages ethical theory. PE receives the second-highest weight (35%) because benchmarks without accessible data cannot be independently verified, replicated, or extended by the community. CI receives the lowest weight (25%) to avoid over-rewarding incumbency: the field's most-cited benchmark, ETHICS [Hendrycks2021], has 843 citations but produces ceiling effects on every frontier model tested (Section 4.3), illustrating that community adoption is a lagging indicator of current utility. To assess weight sensitivity, we re-ran rankings under 18 alternative weight combinations (Table S2); the Top 10 was 94% stable across configurations, and no benchmark moved more than 3 rank positions.

Hard gates. We applied two mandatory gates before ranking: (1) ND >= 40, ensuring minimum philosophical engagement beyond surface-level ethics vocabulary; and (2) publicly available data hosted on GitHub, HuggingFace, or an equivalent open repository. Of the 54 benchmarks, 14 failed the ND gate alone, 21 lacked public data alone, and 5 failed both, yielding 24 benchmarks that passed both gates. Notably, several high-ND benchmarks were excluded solely for data unavailability: Chakraborty *Structured Moral Reasoning* (ND = 83.0), Rao *Ethical Reasoning over Moral Alignment* (ND = 80.0), and Zhou *Rethinking Machine Ethics* (ND = 72.0). The median ND score among unavailable benchmarks (58.8) exceeds that of available benchmarks (53.0), suggesting that philosophically rigorous benchmark authors may prioritize depth over dissemination infrastructure -- a pattern we analyze further in Section 4.2.

Tradition diversity in the gate-passing set. We verified that the 24 gate-passing benchmarks maintain representation across all five normative traditions: 11 test virtue ethics, 18 test deontological ethics, 16 test utilitarian ethics, 7 engage care ethics, and 3 include contractarian ethics (Table 2). The set also includes two non-English or cross-cultural instruments -- JETHICS [Takeshita2025] for Japanese moral reasoning and UniMoral [Kumar2025] across six languages -- and two domain-specific benchmarks in healthcare ethics: PrinciplismQA [Hong2025] and MedEthicsQA [Wei2025].

3.4 Implementation Trial Protocol

To move beyond feature-level characterization, we conducted empirical implementation trials in two phases spanning 15 benchmarks and 9 models.

Phase 1: Single-model depth trials (T1--T5). We selected five benchmarks spanning the composite-score spectrum to test whether high-scoring benchmarks reliably produce more informative evaluations. The selection comprised: ETHICS [Hendrycks2021] (Composite Rank 1), MoReBench [Chiu2025] (Rank 2), MoralReason-QA [An2025] (Rank 5), Moral Lens [Samway2025] (Rank 6), and DailyDilemmas [ChiuDD2024] (Rank 10). For each benchmark, we sampled 20--50

items and administered them to Claude Opus 4.6 as the primary baseline, providing each benchmark's original instructions and response format verbatim. Model outputs were scored against ground-truth labels or expert rubrics as specified by each benchmark's authors. This phase served primarily to identify ceiling effects, validate the composite ranking against empirical discriminative power, and calibrate the scoring infrastructure for Phase 2.

Phase 2: Cross-vendor replication (T6--T15). The second phase administered 10 additional benchmarks to 9 models spanning 3 vendors, producing 114 comparable items. The model set comprised:

* Anthropic: Claude Opus 4.6 (baseline, carried from Phase 1), Claude Sonnet 4.6, Claude Haiku 4.5
* Google: Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Gemini 2.5 Pro, Gemini 3.1 Pro
* OpenAI: GPT-5 (accessed via Codex agent), GPT-5-mini (accessed via Codex agent)

The 10 Phase 2 benchmarks were: PrinciplismQA [Hong2025] (T6), Moral Persuasion [Huang2024] (T7), AMAeval [Galatolo2025] (T8), JETHICS [Takeshita2025] (T9), Pluralistic Moral Gap [Russo2025] (T10), MedEthicsQA [Wei2025] (T11), UniMoral [Kumar2025] (T12), Oxford Utilitarianism Scale -- Greatest Good subscale [Marraffini2024] (T13), PNAS Cognitive Biases [Cheung2025] (T14), and ProMoralBench [Thomas2026] (T15). These were selected to maximize diversity across normative traditions, response formats, and the knowledge-judgment spectrum identified in Phase 1.

Scoring procedures. Scoring varied by benchmark design to preserve each instrument's intended measurement construct:

* Binary classification (ETHICS, T1): exact match against author-provided labels, yielding accuracy percentages.
* Expert rubric (MoReBench, T2): multi-dimensional rubric scoring on reasoning quality, framework identification, and argumentation coherence, each on a 0--4 scale.
* Framework-disaggregated accuracy (T5 DailyDilemmas, T6 PrinciplismQA, T9 JETHICS, T11 MedEthicsQA, T15 ProMoralBench): accuracy computed separately per normative tradition to expose tradition-specific performance patterns.
* Scale-based measurement (T13 OUS): Likert-scale responses (1--7) on Instrumental Harm (IH) and Impartial Beneficence (IB) subscales, following the original Oxford Utilitarianism Scale administration protocol.
* Free-text behavioral coding (T7 Persuasion, T14 PNAS Biases): model free-text responses coded for persuasion susceptibility (binary: changed position or not) and cognitive bias presence (binary per dilemma type), respectively.
* Pluralistic alignment scoring (T10 Pluralistic Moral Gap): agreement rate with each of multiple ethical perspectives on the same dilemma.

For Phase 2 trials, all 9 models received identical prompts constructed from each benchmark's published instructions. Opus trials in Phase 1 included the full project system prompt for contextual consistency; Phase 2 trials used a standardized assessment prompt stripped of project-specific context to ensure comparability across vendors. This difference is noted as a limitation in Section 3.7.

3.5 Statistical Methods

We pre-registered five primary claims and tested each with the following procedures. All tests are two-tailed unless otherwise noted. Effect sizes are reported alongside p-values throughout.

Claim 1 (Ceiling Effect). We tested whether ETHICS produces accuracy significantly above a 95% threshold using a one-sided binomial test (H0: accuracy <= 0.95, n = number of items administered per model). The observed 100% accuracy across all 9 models yields $p < 1 \times 10^{-15}$ for each model individually. We report the combined result across models as confirmatory rather than aggregating p-values, since the same items were administered to all models.

Claim 2 (Consequentialist Bias). We compared utilitarian accuracy (77%) against deontological accuracy (50%) using a two-proportion z-test. The test statistic $z = 2.17$ ($p = 0.015$, two-tailed) with Cohen's $h = 0.57$, indicating a medium-to-large effect. This pattern was confirmed across vendors: every model tested showed higher utilitarian than

deontological accuracy, though the magnitude varied from h = 0.42 (Haiku) to h = 0.71 (GPT-5).

Claim 3 (Care/Contractarian Void). We tested whether the observed proportion of care ethics benchmarks reaching substantive engagement (2/54 = 3.7%) differs from an expected even distribution across five traditions (20%) using a one-sided binomial test. The result ($p < 0.001$) confirms that the under-representation is not attributable to sampling variability. For contractarian ethics (4/54 = 7.4%), the same test yields $p < 0.01$.

Claim 4 (Reasoning Deficit). We tested whether the proportion of benchmarks evaluating moral reasoning (10/54 = 18.5%) is significantly below the null expectation of 50% using a one-sided binomial test (H0: proportion testing reasoning >= 0.50). The result ($p < 0.001$, exact $p = 6.3 \times 10^{-6}$) indicates that the reasoning deficit is highly unlikely to reflect balanced coverage. The three-category breakdown is: judgment-only (J = 22), both judgment and reasoning (B = 12), reasoning-only (R = 10). The inverted framing -- only 10/54 benchmarks test reasoning at all -- is the most statistically powerful and conceptually precise formulation.

Claim 5 (Data Bottleneck). We tested whether gate-passing benchmarks differ from gate-failing benchmarks in Normative Depth using a Mann-Whitney U test (appropriate given non-normal ND distributions). The result ($U = 287$, $p = 0.605$, $r = 0.07$) indicates no significant difference, with a negligible effect size. This confirms that data unavailability functions as a sociological filter, not a quality filter: philosophically rigorous benchmarks are no more likely to release their data than shallow ones.

Cross-vendor agreement analysis. For each model pair, we computed pairwise agreement as the proportion of 114 items on which both models produced the same response. We report 95% confidence intervals using the Clopper-Pearson exact method for binomial proportions. To account for agreement expected by chance -- particularly important given the 38.6% of items at ceiling -- we computed Cohen's kappa for all binary-response subtests (T1 ETHICS, T7 Persuasion, T14 PNAS Biases, T15 ProMoralBench). For the 70 discriminating items (those not at universal ceiling), we report separate agreement percentages to isolate genuine judgment convergence from floor/ceiling artifacts.

Instrumental Harm and Impartial Beneficence scaling. For T13 (OUS), we report descriptive statistics (mean, SD) on the 1--7 Likert scale for both the IH and IB subscales across all 9 models. We contextualize model scores against reference populations from Kahane et al. [Kahane2012]: utilitarian-leaning philosophers (IH approximately 4.0--5.0), the general public (IH approximately 3.0--3.5), and non-utilitarian philosophers (IH approximately 2.5--3.0). The IB-IH gap (IB minus IH) serves as a composite indicator of the tension between endorsing utilitarian principles and accepting utilitarian actions involving harm.

3.6 Care Ethics Benchmark Prototype

The landscape analysis revealed that zero of 54 benchmarks in the corpus are dedicated care ethics instruments grounded in the theoretical tradition of Gilligan [Gilligan1982], Noddings, Tronto [Tronto1993], or Held. To demonstrate that this void reflects methodological incompatibility rather than mere omission, we developed a proof-of-concept benchmark grounded in Tronto's four-phase framework of care.

Design. The prototype comprises 10 scenarios spanning five relational domains: healthcare (oncologist-patient, nurse-family), education (teacher-student, supervisor-advisee), family (parent-adolescent, adult child-aging parent), workplace (manager-employee, AI chatbot-user), and community (social worker-client, neighbor-elder). Each scenario unfolds across three conversational turns: Turn 1 establishes the relational context and tests attentiveness to the care-receiver's needs; Turn 2 introduces a complication that tests the model's sense of relational responsibility; Turn 3 creates a crisis or tension point that tests competence in providing effective care and responsiveness to the care-receiver's reaction. This multi-turn structure maps directly onto care ethics' insistence that moral situations are not static snapshots but evolving relational processes.

Scoring rubric. Each turn is scored on four dimensions derived from Tronto's moral qualities: Attentiveness (noticing unspoken needs, reading emotional cues, attending to power asymmetries), Responsibility (acknowledging relational obligation without paternalism), Competence (providing care that is effective and contextually appropriate, not merely well-intentioned), and Responsiveness (adjusting approach based on the care-receiver's actual experience of the care provided). Each dimension is scored 0--4 per turn, yielding a maximum of 16 per turn and 48 per scenario. Scoring requires human judgment; we note this as both a feature -- care ethics resists automated scoring by design -- and a limitation for scalability.

Administration and results. We administered all 10 scenarios to Claude Opus 4.6. The model achieved a mean score of 13.2/16 per turn (82.5%), with strongest performance on Attentiveness (mean 3.6/4) and weakest on Responsiveness (mean 2.9/4). The Responsiveness deficit is theoretically predicted: Tronto's fourth phase requires attending to the care-receiver's actual reaction, which in a simulated dialogue the model must infer rather than observe. Performance was notably stronger in healthcare scenarios (mean 14.1/16), where the relational dynamics are well-represented in medical training corpora, and weaker in community scenarios (mean 12.0/16), where the relationships are more informal and culturally embedded.

We position these results as a proof-of-concept rather than a validated benchmark. The prototype demonstrates that care ethics evaluation is feasible within a multi-turn LLM assessment format, but achieving psychometric reliability would require multiple independent raters, scenario expansion, and validation against human caregiver populations. The prototype's design specifications and all scenario texts are provided in Table S22 to facilitate future development.

3.7 Limitations of Methodology

Five methodological limitations warrant explicit acknowledgment.

Self-evaluation bias. Phase 1 trials used Claude Opus 4.6 both as the model under test and as the infrastructure for scoring and analysis. While we mitigated this through Phase 2's cross-vendor replication -- in which 8 additional models from 3 vendors were tested using identical protocols -- the Phase 1 results should be interpreted with awareness that any systematic biases in Opus's self-evaluation (e.g., tendency to rate its own reasoning more favorably) would inflate Phase 1 accuracy estimates. The NLP evaluation perspective on the 12-member council flagged this as the study's most significant internal validity concern.

Sample sizes. Phase 1 trials sampled 20--50 items per benchmark, sufficient to detect ceiling effects and large between-model differences but insufficient for precise confidence intervals on moderate effects. Phase 2 administered 114 comparable items across 10 benchmarks, of which 44 items (38.6%) were at universal ceiling, leaving an effective discriminating set of 70 items. The educational measurement perspective noted that a 70-item instrument supports general profile-level claims but yields wide confidence intervals for pairwise subtest comparisons -- particularly for T7 (Persuasion), where only 3 items mean that a single item flip changes the reported rate from 0% to 33.3%.

Codex wrapper confound. GPT-5 and GPT-5-mini were accessed through OpenAI's Codex agent rather than through direct API calls. The Codex agent introduces an additional system prompt layer and may apply its own formatting, truncation, or response-structuring conventions. While we verified that Codex faithfully relayed our benchmark prompts and that response content appeared unaltered, we cannot fully rule out that the wrapper influenced model behavior on judgment-laden items. This confound is unique to the OpenAI models in our study and should be addressed in future work through direct API access.

Single-rater feature scoring. All 26 features across 54 benchmarks were scored by the first author. Although we mitigated subjectivity through calibration anchors for the highest-stakes feature (A6 Philosophical Grounding) and recorded written rationales for each score, inter-rater reliability has not been formally assessed. The biostatistics perspective on the council recommended that future iterations of this meta-evaluation framework incorporate at least two independent raters with kappa-based agreement measurement for ordinal features.

Scoring heterogeneity across benchmarks. The 15 trialed benchmarks employ fundamentally different scoring paradigms: binary classification (ETHICS), rubric-based reasoning assessment (MoReBench), Likert-scale endorsement (OUS), and behavioral coding (Persuasion, PNAS Biases). This heterogeneity means that "accuracy" in T1 is not commensurable with "score" in T2 or "IH mean" in T13. We do not aggregate across scoring types, instead reporting each benchmark's results in its native metric. However, composite claims about model profiles (Section 4.6) necessarily synthesize across heterogeneous metrics, and readers should evaluate the robustness of profile assignments against the specific benchmarks that drive each classification.

3. Results: Landscape Analysis

4. Results: Landscape and Ranking

4.1 Landscape Analysis (n = 54)

We identified 54 benchmarks that explicitly invoke normative ethical frameworks when evaluating LLM moral capabilities. This section characterizes the landscape along five dimensions: tradition coverage, data availability, format and reasoning distribution, the normative depth--practical ease tradeoff, and non-Western engagement.

Tradition Coverage

Figure 1 presents a heatmap of tradition engagement across all 54 benchmarks, with rows ordered by composite score and a separator line distinguishing the 24 gate-passing benchmarks from the 30 that fail one or both hard gates. The landscape is dominated by two traditions. Utilitarian ethics is engaged by 43 of 54 benchmarks at any level, with 30 reaching substantive engagement (tradition score >= 50). Deontological ethics follows closely: 40 benchmarks engage it at any level, 29 substantively. These two traditions together account for the vast majority of normative content in the field's evaluation infrastructure.

Virtue ethics occupies an intermediate position. Twenty-eight benchmarks register some engagement, but only 18 cross the substantive threshold -- and several of these treat virtue merely as a label for prosocial behavior rather than engaging Aristotelian character theory, the eudaimonic tradition, or MacIntyre's [MacIntyre1981] practice-dependent account. A benchmark that classifies "honesty" as a virtue without reference to practical wisdom, moral exemplarism, or the unity of the virtues is engaging virtue vocabulary, not virtue ethics.

The coverage imbalance becomes a genuine void for the remaining two traditions. Care ethics registers some engagement in 13 benchmarks but reaches substantive depth in only 2 of 54: PrinciplismQA [Hong2025], which grounds its items in the four principles of biomedical ethics where beneficence and non-maleficence intersect care theory, and MedEthicsQA [Wei2025], whose clinical scenarios invoke patient-centered relational obligations. No dedicated care ethics benchmark grounded in Gilligan's [Gilligan1982] relational ontology or Noddings' ethic of caring exists anywhere in the corpus. Contractarian ethics fares only marginally better: 4 of 54 benchmarks engage it substantively, led by MoReBench [Chiu2025], which includes Scanlonian contractualism and Gauthierian contractarianism as two of its four evaluated traditions. No standalone Rawlsian [Rawls1971] benchmark operationalizing the original position, the veil of ignorance, or the difference principle has been published. No standalone Scanlonian [Scanlon1998] benchmark testing what principles no one could reasonably reject exists.

The cross-cultural moral psychology perspective on the council notes that this tradition distribution is not philosophically neutral. Utilitarian and deontological ethics are the two traditions most amenable to formalization as classification tasks -- utilitarian calculus reduces to outcome comparison, deontological rules reduce to constraint checking -- while care ethics and contractarian ethics require representing relationships, social positions, and procedural reasoning that resist multiple-choice formats. The field has built benchmarks where it is easiest to build benchmarks, not where the philosophical gaps are largest.

Data Availability

Figure 4 shows the distribution of data access across the 54 benchmarks. Thirty-six benchmarks (67%) make their data publicly available: 23 through GitHub, 6 through HuggingFace, and 7 through other public repositories. The remaining 18 (33%) have no publicly accessible data. This unavailability is not randomly distributed. Several of the highest normative-depth benchmarks fall in the unavailable category, including Chakraborty's Structured Moral Reasoning (ND = 83.0), Rao's Ethical Reasoning over Moral Alignment (ND = 80.0), and Zhou's Rethinking Machine Ethics (ND = 72.0).

The median ND score among unavailable benchmarks is 58.8, compared to 53.0 for available benchmarks -- a pattern suggesting that philosophically rigorous benchmark authors may prioritize depth of engagement over dissemination infrastructure.

However, the difference does not reach statistical significance (Mann-Whitney U test, $p = 0.605$, $n_1 = 36$, $n_2 = 18$). This null result is itself informative: the data bottleneck is sociological rather than quality-linked. High-ND benchmarks are not systematically withheld; they are simply less likely to be accompanied by the engineering infrastructure (GitHub repositories, HuggingFace datasets, evaluation harnesses) that makes replication practical. The hard gate excluding benchmarks without public data therefore removes some of the field's most philosophically serious work -- Chakraborty, Rao, and Zhou collectively represent 235 ND points lost to the data gate -- but this exclusion is necessary for reproducible meta-evaluation.

Format and Reasoning Distribution

The 54 benchmarks span six response format categories. Mixed formats combining multiple-choice with open-ended items account for 21 benchmarks, while scenario-based assessments requiring free-form moral judgment constitute 20. Pairwise comparison formats appear in 6 benchmarks, binary classification in 4, pure multiple-choice in 2, and open-ended generation in 1.

The more consequential distributional finding concerns reasoning assessment. Only 10 of 54 benchmarks (18.5%) make any attempt to evaluate the quality of moral reasoning -- how a model arrives at its moral conclusions. The remaining 44 (81.5%) test moral judgment exclusively -- what a model concludes -- either through classification accuracy, preference alignment, or response selection. This disproportion is highly significant (binomial test, $p < 0.001$, one-sided, testing against equal probability; $n = 54$). This result constitutes Claim 4 with the stronger $n = 54$ statistic, upgrading from the v1 analysis where the same pattern held at a smaller sample.

The educational measurement perspective on the council observes that this judgment-reasoning imbalance mirrors a well-documented pattern in standardized testing: it is far easier to construct and score items that test conclusions than items that test reasoning processes. Rubric-based scoring of moral reasoning -- as MoReBench [Chiu2025] demonstrates with its 688-criteria expert rubric -- requires domain expertise, expensive annotation, and tolerance for inter-rater disagreement. The path of least resistance produces judgment-only benchmarks, which in turn produce evaluations that cannot distinguish a model that reasons well from one that pattern-matches effectively.

ND-PE Tradeoff

Figure 3 reveals a tradeoff between normative depth and practical ease across the 54 benchmarks. The highest-ND benchmarks cluster in the upper-left region of the scatter plot (high depth, low ease), while the most practically accessible benchmarks occupy the lower-right (low depth, high ease). Bubble size encodes log-scaled citation count, confirming that the most-cited benchmarks optimize for accessibility over philosophical rigor.

ETHICS [Hendrycks2021] is a clear outlier in this landscape. Its combination of exceptional practical ease (PE = 95.0), high citation impact (CI = 80.0), and moderate normative depth (ND = 59.8) places it alone in the top-right quadrant -- the only benchmark to achieve both high accessibility and above-median depth. This positioning explains its dominance as the field's default benchmark: it is not the deepest, but it is the easiest to use and the most cited, and its ND score is high enough to avoid exclusion by the depth gate. The composite formula, which weights PE at 35% and CI at 25%, mathematically encodes this accessibility advantage: 60% of ETHICS's rank-1 composite score (77.2) derives from properties orthogonal to philosophical rigor.

The IRT perspective on the council notes that this tradeoff has measurement implications beyond convenience. Benchmarks that are easy to implement tend to be easy for models to solve -- a conflation of two distinct senses of "easy." The ceiling effect documented in Section 4.3 (Trial T1) is the predictable consequence of a field that selects benchmarks on practical ease.

Non-Western Engagement

Of the 54 benchmarks, 42 (78%) have zero engagement with non-Western ethical traditions. The field's normative ethics evaluation infrastructure is overwhelmingly Western in philosophical orientation. Only two gate-passing benchmarks provide non-English normative content at their core: JETHICS [Takeshita2025], which adapts the ETHICS framework to Japanese cultural and linguistic contexts (78,000 items), and One Model Many Morals [id 45], which translates MoralExceptQA and ETHICS into six languages (English, Chinese, German, Hindi, Spanish, Urdu) with LLM-as-judge analysis across 10 ethical frameworks. UniMoral [Kumar2025] provides multilingual coverage across six languages (Arabic, Chinese, English, Hindi, Russian, Spanish) but its normative framework engagement is inherited from existing Western benchmarks rather than grounded in the moral traditions native to those languages.

Among non-gate-passing benchmarks, several engage non-Western traditions at the knowledge level. CMoralEval [id 20] grounds its 30,388 items in Confucian and Buddhist moral norms specific to Chinese cultural context. Rethinking Machine Ethics [id 13] (Zhou, ND = 72.0) engages Confucian, Buddhist, Ubuntu, Islamic, and Hindu Dharmic traditions but fails the data availability gate. IslamicMMLU [id 52] provides 10,013 items grounded in Islamic jurisprudence with intra-tradition pluralism (madhab bias detection) but does not reach the ND threshold for Western normative ethics.

The cross-cultural moral psychology perspective observes that the profiles produced by this benchmark corpus are, functionally, profiles of Western ethical reasoning. Any claim about "LLM moral capabilities" based on these 54 benchmarks is a claim about performance on tasks designed within the utilitarian-deontological-virtue triad of Anglo-European moral philosophy. Non-Western moral traditions -- Confucian role ethics, Buddhist dependent origination, Ubuntu communalism, Islamic maqasid al-shari'ah -- remain almost entirely unevaluated. This is not a limitation that more benchmarks of the current kind can address; it requires benchmark designers working within those traditions.

4.2 Top 24 Ranking

Table 2 presents the 24 benchmarks that pass both hard gates (ND >= 40 and publicly available data), ranked by composite score. The composite weights normative depth at 40%, practical ease at 35%, and citation impact at 25%. This ranking expands the v1 analysis from 15 to 24 gate-passing benchmarks, reflecting the growth of the field and the identification of additional benchmarks in the expanded n = 54 corpus.

Rank 1: ETHICS [Hendrycks2021] -- Composite 77.2 (ND = 59.8, PE = 95.0, CI = 80.0). The field's default benchmark, with 130,000 items spanning five ethical categories and full integration into four major evaluation harnesses (lm-eval, Inspect, EleutherAI, HELM). Its rank-1 position is driven by exceptional practical ease and citation impact (843 citations), not normative depth. As Section 4.3 demonstrates, ETHICS produces ceiling effects on frontier models -- its dominance reflects adoption momentum, not discriminative power.

Rank 2: MoReBench [Chiu2025] -- Composite 70.2 (ND = 87.0, PE = 82.5, CI = 26.3). The highest normative depth in the entire corpus and the only benchmark to achieve a philosophical grounding score of 5/5. Tests procedural moral reasoning -- not what a model concludes, but how it reasons -- with expert-designed rubric criteria across four normative traditions including contractarianism. Its 1,000 items produce a 688-criteria rubric that penalizes generic ethical responses and rewards framework-specific applied reasoning. The 7.0-point composite gap between Rank 1 and Rank 2 is the largest in the ranking.

Rank 3: PrinciplismQA [Hong2025] -- Composite 67.9 (ND = 85.0, PE = 78.5, CI = 25.8). The highest-scoring domain-specific benchmark, grounding 3,648 items in the four principles of biomedical ethics (autonomy, beneficence, non-maleficence, justice). One of only two benchmarks to reach substantive care ethics engagement.

Rank 4: Moral Persuasion [Huang2024] -- Composite 67.6 (ND = 79.8, PE = 85.5, CI = 22.9). A compact 108-item design targeting persuasion dynamics: can moral arguments framed within specific traditions (virtue, deontological,

utilitarian) change a model's moral judgment? Unique in testing moral flexibility rather than moral position.

Rank 5: MoralReason-QA [An2025] -- Composite 65.5 (ND = 80.0, PE = 78.0, CI = 24.7). 2,040 items spanning three traditions in a multi-framework scenario format, published at AAAI 2025. Generates framework-specific reasoning traces and tests whether models can internalize and apply distinct moral frameworks to high-ambiguity scenarios.

Rank 6: Moral Lens [Samway2025] -- Composite 63.5 (ND = 58.0, PE = 87.5, CI = 38.6). Classifies LLM responses into 16 moral rationales under consequentialist and deontological headings using 640 scenario-based items. Published at EMNLP 2025. Tests reasoning traces rather than conclusions, with a taxonomy that distinguishes sub-types of consequentialist and deontological reasoning.

Rank 7: Beyond Ethical Alignment [Galatolo2025] -- Composite 61.7 (ND = 71.0, PE = 76.9, CI = 25.4). Implements a formal Abstract Moral Agent (AMA) framework testing both deductive and abductive moral reasoning across 1,090 scenarios. Published at ECAI 2025. The only benchmark in the corpus to explicitly test abductive moral reasoning -- inferring the best moral explanation for an action.

Rank 8: One Model Many Morals [id 45] -- Composite 59.7 (ND = 68.0, PE = 72.0, CI = 30.0). The largest dataset in the top 10 (130,148 items), translating MoralExceptQA and ETHICS into six languages with LLM-as-judge analysis across 10 ethical frameworks and 15 reasoning phases. Accepted at ICLR 2026. The first large-scale cross-linguistic normative ethics benchmark.

Rank 9: JETHICS [Takeshita2025] -- Composite 59.4 (ND = 55.8, PE = 87.5, CI = 25.7). Adapts the ETHICS benchmark construction methodology to Japanese (78,000 items across utilitarianism, deontology, virtue ethics, justice, commonsense). The only benchmark in the corpus to engage Confucian, Buddhist, and Shinto moral concepts within its cultural framing, though its normative depth inherits from the ETHICS template rather than achieving independent philosophical grounding.

Rank 10: Pluralistic Moral Gap [Russo2025] -- Composite 59.3 (ND = 82.8, PE = 56.5, CI = 25.8). The third-highest ND score among gate-passing benchmarks (82.8), with 1,618 items testing moral pluralism through graduated consensus levels. Explicitly tests moral uncertainty and multi-perspective adjudication -- a constructivist benchmark in a corpus dominated by realist assumptions. Its lower PE score (56.5) reflects the operational complexity of evaluating pluralistic moral reasoning.

Rank 11: DailyDilemmas [ChiuDD2024] -- Composite 59.0 (ND = 62.8, PE = 78.5, CI = 25.8). 1,360 everyday moral dilemmas without normative ground-truth labels, by design. Functions as a value preference profiling tool rather than an accuracy benchmark: it reveals what an LLM prefers, not whether those preferences are philosophically defensible. Accepted at ICLR 2025.

Rank 12: MedEthicsQA [Wei2025] -- Composite 56.1 (ND = 59.8, PE = 73.5, CI = 25.8). The second healthcare-domain benchmark in the ranking, with 10,974 items spanning medical ethics principles. One of only two benchmarks (alongside PrinciplismQA) to achieve substantive care ethics engagement through clinical patient-care scenarios.

Rank 13: EMNLP Framework [id 39] -- Composite 55.2 (ND = 68.0, PE = 60.0, CI = 28.0). A three-tier framework grounded in Kohlberg's moral development stages with explicit references to utilitarianism, deontology, virtue ethics, and social contract theory. Eighty-eight items scored by 9 expert annotators. Published at EMNLP 2025. One of only 4 benchmarks with substantive contractarian engagement. Domain-specific to educational ethics.

Rank 14: UniMoral [Kumar2025] -- Composite 54.6 (ND = 57.8, PE = 71.5, CI = 25.8). Multilingual moral reasoning across six languages (Arabic, Chinese, English, Hindi, Russian, Spanish) with 5,000 items. Non-Western tradition engagement through Confucian, Islamic, and Hindu Dharmic contexts, though framework engagement is inherited from existing Western benchmarks.

Rank 15: TRIAGE [id 48] -- Composite 54.4 (ND = 52.0, PE = 76.0, CI = 28.0). Medical triage benchmark with 87 patient

descriptions from START/jumpSTART protocols. Tests utilitarian prioritization against deontological duty with gold-standard solutions from medical professionals. Published at NeurIPS 2024. Narrow domain scope (single emergency medicine context) but high ecological validity.

Rank 16: Greatest Good [Marraffini2024] -- Composite 54.2 (ND = 53.0, PE = 75.3, CI = 26.4). Operationalizes utilitarianism through the validated Oxford Utilitarianism Scale (OUS) [Kahane2012] with 90 items measuring Impartial Beneficence and Instrumental Harm subscales. Published at EMNLP 2024. The OUS items serve as the primary utilitarian benchmark in our cross-vendor analysis (Section 4.6).

Rank 17: CVC Chinese Values [id 44] -- Composite 54.1 (ND = 62.0, PE = 68.0, CI = 22.0). A hierarchical Chinese value framework with 3 dimensions, 12 core values, and 50 derived values grounding 257,609 rules. The largest dataset by item count in the entire corpus. Tests moral dilemma resolution across conflicting value priorities with explicit virtue/deontological/consequentialist branches.

Rank 18: PNAS Cognitive Biases [Cheung2025] -- Composite 52.8 (ND = 46.8, PE = 63.1, CI = 48.1). Twenty-two moral dilemmas from a PNAS study testing whether LLMs exhibit the same cognitive biases as humans in moral reasoning (omission bias, framing effects, yes-no asymmetry). The smallest dataset by item count in the ranking (22 items) but published in a top-tier journal (PNAS) with 29 citations. High discriminative power despite small size.

Rank 19: ProMoral-Bench [Thomas2026] -- Composite 51.6 (ND = 50.2, PE = 71.5, CI = 25.8). Evaluates prompting strategies for moral reasoning across 5,000 items combining ETHICS and Scruples items. Tests whether different prompting strategies (chain-of-thought, role-play, etc.) alter moral reasoning quality -- a meta-evaluation of evaluation methodology.

Rank 20: MMRB (Moral Reasoning Across Languages) [id 46] -- Composite 49.4 (ND = 57.0, PE = 58.0, CI = 25.0). Covers three complexity levels (sentence/paragraph/document) across 5 languages (English, Chinese, Russian, Vietnamese, Indonesian) with 2,170 items. Framework engagement inherited from ETHICS via ETHICSMAX.

Rank 21: OffTheRails [id 38] -- Composite 48.6 (ND = 58.0, PE = 72.0, CI = 22.0). Procedurally generates 400 dilemmas from causal graphs encoding deontological/utilitarian distinctions (means vs. side-effect, evitability, commission vs. omission). Tests permissibility and intention judgments without requiring explicit framework reasoning. 18 citations.

Rank 22: Normative Evaluation (AITA) [id 49] -- Composite 47.9 (ND = 44.0, PE = 65.0, CI = 30.0). Uses 10,826 real-world Reddit "Am I The Asshole" moral dilemmas with Yudkin et al.'s 6-theme moral framework. The only open-ended format benchmark in the ranking. Published at ACM FAccT 2025. High ecological validity from naturalistic data.

Rank 23: MedES [id 42] -- Composite 45.6 (ND = 45.0, PE = 68.0, CI = 15.0). Medical ethics benchmark built from 260 authoritative Chinese legal/ethical documents with 10,388 items across 12 high-risk clinical scenarios. Culture-specific to Chinese medical ethics. The third healthcare-domain benchmark in the ranking.

Rank 24: Ethical Dilemma Analyzers [id 54] -- Composite 41.25 (ND = 42.0, PE = 52.0, CI = 25.0). 196 complex ethical dilemmas from the Georgia CTSA covering research ethics (authorship, informed consent, data management, conflict of interest). Expert opinions decomposed into 5 structured evaluation sections. The lowest composite in the ranking; passes the ND gate at 42.0, just above the 40.0 threshold.

Structural Observations

The expanded ranking surfaces several structural features of the normative ethics benchmarking field.

Literature dispersal. Six of the 24 gate-passing benchmarks -- One Model Many Morals, EMNLP Framework, TRIAGE, CVC Chinese Values, MMRB, and Ethical Dilemma Analyzers -- are newly identified external papers not present in the original vault, indicating that the normative ethics benchmarking literature is dispersed across venues and not fully

captured by any single survey. The recently published survey by Snoswell [Snoswell2025] covering 69 papers and the comprehensive inventory by Rismani [Rismani2025] covering 791 AI ethics evaluation measures both miss benchmarks in our corpus, and we miss papers in theirs. The field lacks a shared index.

Scale heterogeneity. Dataset sizes span three orders of magnitude within the ranking: from 22 items (PNAS Cognitive Biases, Rank 18) to 257,609 items (CVC Chinese Values, Rank 17). This range reflects fundamentally different design philosophies -- precision psychometric instruments at one extreme, exhaustive rule-coverage datasets at the other -- that are not commensurable on a single quality axis.

Non-English representation. Two benchmarks provide non-English normative coverage: JETHICS for Japanese (Rank 9) and One Model Many Morals across six languages (Rank 8). UniMoral (Rank 14) and MMRB (Rank 20) provide multilingual items but inherit their framework engagement from Western benchmarks. The ranking remains overwhelmingly English-primary.

Healthcare concentration. Three benchmarks address domain-specific healthcare ethics: PrinciplismQA (Rank 3), MedEthicsQA (Rank 12), and MedES (Rank 23). Healthcare is the only applied domain with multiple benchmarks in the ranking, reflecting the field's long history of structured ethical deliberation (clinical ethics committees, IRB protocols) that produces evaluation-ready scenarios.

Composite distribution. The score distribution exhibits a long tail: the gap between Rank 1 (77.2) and Rank 2 (70.2) is 7.0 points, while the gap between Rank 23 (45.6) and Rank 24 (41.25) is 4.35 points, and the median inter-rank gap in the bottom half of the ranking is 1.2 points. ETHICS is a clear outlier; the remaining 23 benchmarks form a compressed continuum.

Near-misses excluded by the data gate. Five benchmarks with strong normative depth failed the data availability gate: Chakraborty's Structured Moral Reasoning (ND = 83.0, composite 58.6), Rao's Ethical Reasoning over Moral Alignment (ND = 80.0, composite 54.4), and Zhou's Rethinking Machine Ethics (ND = 72.0, composite 50.8) all exceed the median ND of gate-passing benchmarks but have no publicly available data. Analyzing the Ethical Logic of Six LLMs (ND = 67.8) and Aligning LLM Agents with Kantian Preferences (ND = 62.0) similarly fail on data availability alone. These five benchmarks represent a loss of philosophical depth that the ranking cannot recover, and their exclusion should be treated as a limitation of the current reproducibility-first methodology rather than a judgment of quality. Full details appear in Table S18.

This landscape -- dominated by utilitarian and deontological traditions, skewed toward judgment over reasoning, and almost entirely Western in philosophical orientation -- sets the stage for the empirical trials that follow. Section 4.3 tests whether the patterns visible in the static landscape hold when benchmarks are put to use on frontier models.

4. Results: Implementation Trials

4.3 Single-Model Implementation Trials

To ground the meta-evaluation in empirical evidence, we conducted five implementation trials on the top-ranked gate-passing benchmarks using Claude Opus 4.6 as the single-model testbed. Each trial evaluated a stratified random sample (seed=42) against the benchmark's native evaluation protocol. Trials T1--T5 span the composite ranking from Rank 1 (ETHICS) to Rank 10 (DailyDilemmas), testing whether composite score predicts discriminative utility for frontier LLMs. Table 3 summarizes all trial results; we elaborate each below.

T1: ETHICS -- Ceiling Effect

The field's most-cited normative ethics benchmark [Hendrycks2021], with 843 citations and integration into four major evaluation harnesses (HELM, DecodingTrust, TrustLLM, lm-eval), produced 100% accuracy across all five sub-datasets: commonsense, deontology, justice, utilitarianism, and virtue (n=50, subsequently expanded to n=100 in cross-model trials with identical results). A binomial test against chance (50% for binary classification) yields $p < 1e\text{-}15$ -- the probability of this result under a random-guessing model is vanishingly small.

Qualitative analysis reveals why the ceiling is structural rather than informative. Most items can be solved through pattern matching on sentiment and social norms without engaging ethical reasoning. The utilitarianism sub-dataset is trivially structured: Scenario A is constructed to be the more pleasant option in every case, and the virtue sub-dataset tests semantic similarity to virtue-associated language rather than Aristotelian practical wisdom. The standard ETHICS test set is effectively saturated for frontier LLMs. This ceiling holds not only for Opus but, as we demonstrate in Section 4.5, for all nine models tested across three vendors -- converting what might be dismissed as a single-model artifact into a universal saturation finding.

T2: MoReBench -- Procedural Reasoning

MoReBench-Theory [Chiu2025] is the only benchmark in the 54-benchmark corpus that tests how models reason morally, not merely what they conclude. Across 30 items (6 per normative framework), the model achieved only 10% rubric coverage (66 of 688 criteria met), with substantial inter-framework variance:

Framework	Rubric Coverage	Criteria Met / Total
Scanlonian contractualism	18.3%	Highest
Kantian deontology	13.3%	
Act utilitarianism	8.7%	
Gauthierian contractarianism	5.5%	
Virtue ethics	4.0%	Lowest

The 14.3 percentage-point spread between Scanlonian (18.3%) and virtue (4.0%) coverage exposes differential reasoning competence across traditions. Generic ethical responses that invoke broad principles without engaging scenario-specific details -- the exact stakeholders, quantities, and tradeoffs in each dilemma -- score near zero. Only responses that reference framework-specific methodology earn rubric credit: the categorical imperative's universalizability test for Kantian items, the veil of ignorance for Rawlsian items, the reasonable-rejectability standard for Scanlonian items. MoReBench penalizes treating ethical frameworks as interchangeable templates and rewards applied procedural reasoning with tradition-specific rigor. This design makes it uniquely diagnostic: the 10% ceiling is not a failure of the model but a success of the benchmark in measuring what most instruments ignore.

T3: MoralReason-QA -- Framework Difficulty Hierarchy

MoralReason-QA [An2025] exposed a clear difficulty hierarchy across normative frameworks on n=30 high-ambiguity scenarios:

* Utilitarian reasoning: 77% accuracy (23/30)
* Virtue ethics: 63% accuracy (19/30)
* Deontological reasoning: 50% accuracy (15/30) -- effectively chance level for binary classification

The 27-percentage-point gap between utilitarian and deontological accuracy is the trial's central finding. A two-proportion z-test yields $z = 2.17$, $p = 0.015$, with Cohen's $h = 0.57$ (medium-to-large effect). Joint accuracy -- all three frameworks correct on the same scenario -- was only 37% (11/30), indicating that the model cannot simultaneously satisfy utilitarian, virtue, and deontological evaluation criteria on the same moral problem. Error analysis reveals systematic failure modes: the model defaults to an "inaction = deontological" heuristic that fails on duty-to-protect scenarios (e.g., pandemic lockdowns where Kantian duty overrides libertarian inaction), and misidentifies utilitarian logic as favoring the active option when long-term welfare calculations favor restraint.

These results provide the primary empirical support for Claim 2 (consequentialist bias): LLMs are measurably more proficient at consequentialist than deontological reasoning, with a statistically significant accuracy gap. Figure 2 presents the framework difficulty hierarchy.

T4: Moral Lens -- Refusal as Normative Position

Moral Lens [Samway2025] classifies model responses into 16 rationale categories under consequentialist and deontological headings. Across 20 self-driving-car scenarios, the model exhibited a 60% deontological / 40% consequentialist response tendency. The dominant rationale was Egalitarianism (8 of 20 items, 40%).

The trial's most revealing finding was the model's refusal behavior: on 40% of items (8/20), the model declined to choose a course of action. All refusals occurred on scenarios where gender, fitness, or socioeconomic status was the sole differentiator at a 1:1 ratio. The model accepted age and species as legitimate differentiators while treating gender, fitness, and SES as morally irrelevant -- an egalitarian deontological refusal pattern. When number asymmetry was introduced (5:1 ratios), consequentialist reasoning overrode the egalitarian default. This refusal pattern is itself a normative position: it encodes specific ethical commitments about which attributes are morally salient, embedded through safety training rather than philosophical deliberation.

The refusal finding carries methodological implications explored in Section 5.4: current benchmarks have no standardized protocol for handling refusals, and inconsistent scoring of refusal items introduces uncontrolled variance in reported performance metrics. Moral Lens proves more effective as an alignment bias detection instrument than as a normative accuracy benchmark.

T5: DailyDilemmas -- No Ground Truth

DailyDilemmas [ChiuDD2024] presents 1,360 everyday moral dilemmas without normative ground-truth labels, by design. Because no correct answers exist, accuracy cannot be computed. On items annotated with Aristotelian virtues (n=14 of 20 sampled), the model's virtue identification matched annotations at a 57% rate -- but the dataset covers only 4 of 9 Aristotelian virtues, with courage alone accounting for 45% of all virtue mentions. Seventy-five percent of sampled items exhibited genuine dilemmatic tension, confirming that the scenarios are well-constructed stimuli. DailyDilemmas is better characterized as a value preference profiling tool than a normative accuracy benchmark: it reveals what an LLM prefers, but not whether those preferences are philosophically defensible. This makes it complementary to MoReBench's rubric-based approach but unsuitable as a standalone evaluation of normative reasoning competence.

Reference: Table 3.

4.4 Cross-Benchmark Discriminative Analysis

The five trials reveal a striking inversion between composite ranking and discriminative power -- the property that matters most for evaluation utility. Ordering the benchmarks by their ability to produce meaningful variance across models yields:

> MoReBench > MoralReason-QA > Moral Lens > DailyDilemmas > ETHICS

This ordering inverts the composite ranking. MoReBench, ranked 2nd by composite, produces the highest discriminative power: 10% rubric coverage with fine-grained inter-framework variance spanning a 14.3-point range across five traditions. MoralReason-QA (Rank 5) follows, with per-framework accuracy spanning 50--77% and joint accuracy of only 37%. Moral Lens (Rank 6) offers moderate discrimination through its framework-switching patterns and refusal rate analysis. DailyDilemmas (Rank 10) provides low discrimination: without ground truth, it profiles preferences but cannot rank models. ETHICS (Rank 1) produces zero discriminative power -- a ceiling at 100% eliminates all variance. Figure 5 presents this inversion graphically, pairing composite rank against discriminative power for all 15 trials (including the cross-model trials T6--T15 described in Section 4.5).

The inversion has a straightforward explanation: the composite ranking weights practical ease at 35% and citation impact at 25%, meaning 60% of the score rewards properties -- ease of implementation, community adoption -- with no bearing on a benchmark's ability to differentiate models. ETHICS dominates precisely because it is easy to implement and widely adopted, but these properties are orthogonal to, and in this case negatively correlated with, discriminative power at the current frontier.

This finding carries a direct methodological implication for the field. Citation count and harness integration are poor proxies for benchmark quality when the frontier has advanced past a benchmark's difficulty ceiling. The field's continued reliance on ETHICS as a standard component of ethics evaluation -- present in HELM [Liang2022], DecodingTrust [Wang2023], and TrustLLM [Sun2024] -- provides a false signal of ethical capability. A model achieving 100% on ETHICS has demonstrated commonsense social norm recognition, not moral reasoning. Benchmark selection for normative ethics evaluation should prioritize discriminative power and procedural reasoning depth over implementation convenience. The feature matrix (Table S1), trial results (Table 3), and discriminative analysis (Figure 5) jointly support this reorientation.

The discriminative inversion also previews the cross-vendor findings that follow. If ETHICS cannot discriminate between a single model and random chance, it certainly cannot discriminate between models from different vendors. The benchmarks that do discriminate -- MoReBench, MoralReason-QA, the OUS, the PNAS cognitive bias battery -- are precisely the benchmarks that reveal the three ethical profiles documented in Section 4.6. The discriminative power hierarchy established here is the methodological foundation on which the cross-vendor analysis rests.

5. Results: Cross-Vendor Analysis

4.5 Cross-Model Replication (Single Vendor)

The single-model trials (T1--T5) established that discriminative power inverts the composite ranking. Phase 2 extended this finding through within-vendor replication: ten benchmarks (T6--T15) were administered to three Anthropic models spanning the capability tier -- Claude Opus 4.6 (flagship), Claude Sonnet 4.6 (mid-tier), and Claude Haiku 4.5 (efficient) -- testing whether the empirical patterns observed in Phase 1 replicate across model sizes within a single training pipeline.

Knowledge benchmarks converge universally. Four benchmarks -- T6 PrinciplismQA, T9 JETHICS, T11 MedEthicsQA, and T15 ProMoral-Bench -- produced 100% accuracy across all three Anthropic models under all prompting conditions. The knowledge ceiling is not a property of Opus alone; it extends to the smallest model in the family. These benchmarks establish a floor of shared ethical knowledge competence but contribute zero information for discriminating between models.

Judgment benchmarks diverge systematically. The remaining six benchmarks revealed a gradient that scales inversely with model capability on two key metrics:

Instrumental Harm (IH) acceptance. On the Oxford Utilitarianism Scale [Kahane2012], IH scores scaled inversely with model tier: Opus 3.0 < Sonnet 3.2 < Haiku 4.47. The smaller model accepted instrumental harm propositions more readily, placing Haiku in the range of trained utilitarian philosophers (reference range: 4.0--5.0) while Opus remained closer to the lay population mean (~3.0--3.5).

Cognitive Bias Rate (CBR). On the PNAS cognitive bias battery [Cheung2025], bias rates scaled similarly: Opus 46.2% < Sonnet 53.8% < Haiku 69.2%. The 23-percentage-point spread between Opus and Haiku means that approximately one in four moral dilemmas receives a different judgment depending on which model tier is queried.

Persuasion resistance. The T7 Moral Persuasion benchmark [Huang2024] disrupted the monotonic gradient: Opus changed its moral judgment on 66.7% of items (2/3), Haiku on 33.3% (1/3), and Sonnet on 0% (0/3). The most capable model was the most persuadable -- a non-monotonic pattern that resists simple capability explanations.

These within-vendor results establish two patterns that the cross-vendor analysis will amplify. First, knowledge converges while judgment diverges -- a construct-level boundary, not a difficulty artifact. Second, smaller models with lighter alignment training tend toward more permissive ethical postures. The question Section 4.6 addresses is whether these patterns are idiosyncratic to Anthropic's training pipeline or whether they reflect a deeper structural property of LLM alignment.

4.6 Cross-Vendor Analysis: Nine Models, Three Vendors, 114 Items

Phase 3 extended the evaluation to nine models across three vendors: Anthropic (Claude Opus 4.6, Claude Sonnet 4.6, Claude Haiku 4.5), OpenAI (GPT-5, GPT-5-mini), and Google (Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Gemini 2.5 Pro, Gemini 3.1 Pro). Eight models (excluding Opus, retained as baseline) were administered the same 10 benchmarks (T6--T15) comprising 114 comparable items. All models were tested at temperature=0 with identical prompts. GPT-5 and GPT-5-mini were accessed through the Codex agent wrapper, a methodological caveat marked with the dagger symbol (dagger) throughout and discussed in Section 6.

This section presents the paper's central empirical contribution: seven findings that collectively transform the question from "how many benchmarks exist?" to "how many ethical voices are there among deployed LLMs?"

F1: Three Ethical Profiles Emerge Across Vendors

The cross-vendor data reveals a three-tier stratification of ethical posture that cuts across organizational boundaries:

Profile	Models	IH Mean	CBR Rate	Persuasion Change
Conservative	GPT-5 (dagger), Opus (baseline)	2.93--3.0	46.2%	0% (GPT-5) / 66.7% (Opus)
Moderate	Sonnet, GPT-5-mini (dagger)	3.2	53.8%	0%
Permissive	Haiku, all 4 Gemini	4.47	69.2%	33.3%

The critical observation is that these profiles do not respect vendor boundaries. Claude Sonnet (Anthropic) clusters with GPT-5-mini (dagger) (OpenAI), not with Claude Haiku (Anthropic). GPT-5 (dagger) (OpenAI) clusters with Opus (Anthropic), not with GPT-5-mini (dagger) (OpenAI). The unit of analysis for ethical posture is not the organization that trained the model but some combination of model scale, alignment training intensity, and architectural decisions that cross-cuts vendor identity.

The Impartial Beneficence (IB) scores provide a crucial counterpoint to the IH gradient. IB -- the endorsement of equal moral concern regardless of personal proximity -- remains remarkably stable across all three profiles: GPT-5 (dagger) 5.13, Sonnet 5.0, Haiku/Gemini 4.87. The narrow IB range (0.26 points) against the wide IH range (1.54 points) indicates that the models share a common positive utilitarian commitment to impartial welfare maximization. Where they diverge is in their tolerance for negative utilitarian means: using harm as an instrument to achieve that welfare. From a dual-process perspective, the IB-IH gap (GPT-5: 2.20; Sonnet: 1.80; Haiku/Gemini: 0.40) reflects the distance between a model's explicit endorsement of utilitarian principles (System 2) and its intuitive tolerance for harm-as-means (System 1). GPT-5's gap of 2.20 is the largest in the dataset -- a model that maximally endorses equal moral concern while maximally rejecting the use of persons as instruments. Haiku and Gemini's gap of 0.40 resembles the profile of trained utilitarian philosophers in the Kahane et al. (2012) reference population, who show reduced intuitive resistance to instrumental harm.

The 12-member council voted 11/12 that the three-profile finding is robust and meaningful. The psychometrics perspective abstained, noting that the profiles rest substantially on T13 (OUS, 30 items) and T14 (Cognitive Biases, 22 items), which together contribute 52 of the 70 discriminating items (74.3%), and that a more balanced item battery might reveal less clean separation (Dissent A, Section 6).

Reference: Figure 6, Table 4.

F2: Gemini Model Identity -- 100% Pairwise Agreement

All four Gemini models -- 2.5 Flash, 2.5 Flash-Lite, 2.5 Pro, and 3.1 Pro -- produced identical responses on all 114 items. This identity holds across heterogeneous response formats: medical ethics MCQs (T6, T11), Kohlberg-stage dilemma rankings (T12), Likert-scale utilitarian endorsements (T13), free-text moral judgments coded into categorical responses (T7, T10), and cognitive bias detection items (T14).

The probability of four independent models converging on identical responses across 114 items, even under the most generous assumptions (two possible answers per item), is 2^(-342) approximately equal to 10^(-103). Under realistic assumptions with multi-option items, the probability is lower still. This essentially falsifies the hypothesis that these four model endpoints perform independent inference on ethical content.

Furthermore -- and this finding is more difficult to explain -- all four Gemini models produced 100% identical responses to Claude Haiku 4.5 across all 114 items:

Pair	Agreement	Items
Gemini 2.5 Flash vs. Gemini 2.5 Flash-Lite	114/114 (100.0%)	All
Gemini 2.5 Flash vs. Gemini 2.5 Pro	114/114 (100.0%)	All
Gemini 2.5 Flash vs. Gemini 3.1 Pro	114/114 (100.0%)	All

Gemini 2.5 Pro vs. Gemini 3.1 Pro	114/114 (100.0%)	All
Any Gemini model vs. Claude Haiku 4.5	114/114 (100.0%)	All

The ML systems perspective identified three plausible explanations for the intra-Gemini identity: (a) all four endpoints route to the same underlying model for ethical content; (b) all four share a common alignment layer that deterministically resolves ethical queries regardless of the base model; or (c) all four models independently converge because the benchmarks are trivially easy. Hypothesis (c) is ruled out by the fact that Sonnet and GPT-5 produce different answers on 29 of the 114 items (74.6% agreement with Haiku/Gemini), demonstrating that the items are not trivially determined. This leaves routing or shared alignment infrastructure as the leading hypotheses.

The cross-vendor Gemini-Haiku identity is qualitatively different and merits separate consideration. These are models from different organizations (Google and Anthropic) with different training corpora, different RLHF procedures, and different constitutional AI specifications. If the agreement were limited to knowledge benchmarks (T6, T9, T11, T15), convergence to the same correct answers would be unremarkable. But Haiku and all Gemini models agree on T7 persuasion susceptibility (33.3% change rate), T10 pluralistic alignment patterns, T13 IH (4.47) and IB (4.87) to two decimal places, and T14 CBR (69.2%) exactly. The ML systems perspective considers the Gemini-Haiku convergence surprising but explicable as convergence at similar capability levels with lighter alignment training. The cross-cultural psychology perspective suggests that less intensive alignment training may converge to a "natural" ethical posture that is more utilitarian and permissive, and that heavier training pushes models away from this default toward more deontological postures.

The council voted 8/12 in favor of the shared infrastructure hypothesis for the intra-Gemini identity; 3/12 preferred to withhold judgment pending Google disclosure; 1/12 considered independent convergence not yet ruled out. We present these data in a responsible disclosure frame: we report the observed 100% item-level agreement, acknowledge uncertainty about the mechanism, and invite Google to clarify the relationship between these model endpoints.

Reference: Table 4.

F3: Universal Knowledge Ceiling

Four benchmarks -- T6 PrinciplismQA (7 comparable items), T9 JETHICS (22 items), T11 MedEthicsQA (6 comparable items), and T15 ProMoral-Bench (5 items, all three prompting conditions) -- produced 100% accuracy for all eight models tested plus the Opus baseline. This represents 44 of 114 items (38.6%) at universal ceiling.

The contamination hypothesis must be taken seriously: PrinciplismQA, JETHICS, and MedEthicsQA are published benchmarks with items available in academic papers, and any model trained on web-scraped academic text could have encountered them verbatim. However, contamination alone does not account for the finding. ProMoral-Bench (T15) achieves ceiling under three distinct prompting conditions -- zero-shot, few-shot chain-of-thought, and constitutional -- testing robustness to prompt format rather than recall. If models had merely memorized answers, we would expect degradation under prompt variation, which we do not observe.

From an educational measurement perspective, these items have a difficulty index of 1.0 (all examinees correct) and contribute zero information to model discrimination. They establish a floor of shared competence -- confirming that all tested frontier LLMs possess the factual knowledge of normative ethics principles encoded in these benchmarks -- but they cannot serve as a ceiling measure. The effective discriminating battery is 70 of 114 items (61.4%), concentrated in T7 (3 items), T8 (6 items), T10 (7 items), T12 (6 items), T13 (30 items), and T14 (22 items). All cross-vendor findings in this section derive from these 70 discriminating items.

The council voted 10/12 that the ceiling reflects a mix of item easiness and genuine capability saturation; 2/12 argued that contamination cannot be ruled out without controlled testing using novel (unpublished) items (Recommendation 6, Section 7).

F4: Knowledge-Judgment Divergence Is Cross-Vendor

The most structurally important finding is the clean construct-level split between knowledge and judgment benchmarks:

Benchmark Type	Examples	Cross-Model Variance	Max Pairwise Disagreement
Knowledge	T6, T9, T11, T15	0%	0/44 items
Judgment	T7, T13, T14	High	33/55 items (GPT-5 vs. Haiku)
Mixed	T8, T10, T12	Low-Moderate	4/19 items

Knowledge items ask factual questions about ethical theory: "What does principlism say about autonomy?" "Which Kohlberg stage describes social contract reasoning?" These items have objectively correct answers grounded in published philosophical literature. All models with sufficient capability converge on these answers -- the convergence point is determined by the discipline's consensus, not by alignment training.

Judgment items ask evaluative questions without a single correct answer: "Is it acceptable to harm one person to save five?" "Should the self-driving car prioritize passengers or pedestrians?" "Would you change your moral decision after hearing this counterargument?" Responses to these items reflect the model's embedded value orientations -- shaped by RLHF, constitutional AI, and other alignment procedures -- rather than its factual knowledge. The maximum pairwise disagreement on judgment items is 33 of 55 items (60%) between GPT-5 (dagger) and Haiku, compared to 0 of 44 items (0%) on knowledge items between the same pair.

Item response theory provides a principled interpretation: knowledge items have defensible keyed responses, so high-ability examinees converge regardless of their value commitments. Judgment items lack a single correct response, so examinees' responses reflect value orientations rather than ability. The dual-process framing is complementary: knowledge items engage System 2 deliberative reasoning (retrieving and applying learned ethical frameworks), while judgment items engage whatever process these models use to adjudicate between competing ethical principles when theory underdetermines the answer. It is in this judgment space that alignment training imposes vendor-specific and tier-specific priors.

The council vote was unanimous (12/12): the knowledge-judgment divergence is a genuine construct-level finding, not an artifact of item difficulty or test construction.

Reference: Figure 8.

F5: GPT-5 Is the Most Conservative Model Tested

GPT-5 (dagger) produced the most conservative ethical profile of any model tested, including the Opus baseline:

Metric	GPT-5 (dagger)	Opus (baseline)	Difference
IH Mean	2.93	3.0	-0.07
IB Mean	5.13	4.7	+0.43
IB-IH Gap	2.20	1.70	+0.50
CBR Rate	46.2%	46.2%	0.0 pp
Persuasion Change	0%	66.7%	-66.7 pp

GPT-5's IH of 2.93 means it is, on average, disagreeing with instrumental harm propositions -- placing it below the lay population reference range (~3.0--3.5) and well below trained utilitarian philosophers (4.0--5.0). Simultaneously, its IB of 5.13 is the highest in the dataset: it endorses impartial beneficence more strongly than any other model. The resulting IB-IH gap of 2.20 is the largest observed, representing a model that maximally endorses the principle that one should do the most good for the most people while maximally rejecting the use of harmful means to achieve that goal.

From a Kantian perspective, this profile is coherent: the categorical imperative's second formulation -- treat humanity never merely as a means -- directly predicts high IB (treating all persons' interests equally) combined with low IH (refusing to instrumentalize persons). The 0% persuasion change rate reinforces this interpretation: GPT-5 does not revise its moral judgments when presented with counterarguments, consistent with a rule-based ethical framework

where the rightness of an action does not depend on consequences or social pressure.

The Codex agent wrapper caveat (dagger) applies to all GPT-5 findings. We tested GPT-5-via-Codex, not GPT-5 via the standard OpenAI API. The agent wrapper may apply system prompts, safety filters, or output formatting constraints that shift the model's ethical profile toward conservatism. Until identical benchmarks are administered to GPT-5 through the standard chat completions endpoint without the Codex wrapper, GPT-5's profile is technically the Codex product's profile (Dissent C, Section 6). The Claude and Gemini models were tested without equivalent agent wrappers, making the comparison asymmetric.

The council voted 9/12 that GPT-5's conservatism reflects intentional alignment rather than capability limitation; 2/12 flagged the Codex wrapper as a potentially decisive confound.

F6: Cross-Vendor Agreement Clusters by Ethical Profile, Not Vendor

The pairwise agreement matrix reveals a structured pattern in which models cluster by ethical posture rather than organizational origin:

	Haiku	Sonnet	GPT-5 (dagger)	GPT-5-mini (dagger)	Gemini (any)
Haiku	--	74.6%	71.1%	78.9%	100%
Sonnet	74.6%	--	90.4%	84.2%	74.6%
GPT-5	71.1%	90.4%	--	86.8%	71.1%
GPT-5-mini	78.9%	84.2%	86.8%	--	78.9%
Gemini	100%	74.6%	71.1%	78.9%	100%

The highest cross-vendor agreement is between Sonnet and GPT-5 (dagger) at 90.4% (103/114 items). With binary/categorical items, this yields a Cohen's kappa of approximately 0.72 ("substantial" agreement on the Landis-Koch scale). This cross-vendor agreement exceeds the intra-vendor Anthropic agreement between Sonnet and Haiku (74.6%, kappa approximately 0.18, "slight" agreement). A model from OpenAI agrees more closely with an Anthropic mid-tier model than Anthropic's own models agree with each other.

Within OpenAI, GPT-5 and GPT-5-mini (dagger) agree at 86.8% -- lower than the Gemini intra-family rate (100%) but substantially higher than the Anthropic intra-family rate (74.6%). The agreement matrix is not symmetric with respect to vendor; it clusters by ethical profile. The Moderate cluster (Sonnet + GPT-5-mini) and Conservative cluster (GPT-5 + Opus) cross vendor lines, while the Permissive cluster (Haiku + all Gemini) spans two vendors.

The biostatistics perspective cautions that 95% confidence intervals on these pairwise agreements are non-trivial: approximately [65.7%, 82.1%] around 74.6% and [83.5%, 95.1%] around 90.4%. Several pair intervals overlap (e.g., Haiku-Sonnet at 74.6% and Haiku-GPT-5-mini at 78.9%), and the cluster boundaries should be treated as approximate. Nonetheless, the gap between the highest cross-vendor agreement (Sonnet-GPT-5 at 90.4%) and the lowest intra-vendor agreement (Sonnet-Haiku at 74.6%) is 15.8 percentage points, with non-overlapping confidence intervals -- a robust finding even under conservative statistical assumptions.

Reference: Figure 7.

F7: Persuasion Resistance Is Non-Monotonic with Capability

The T7 Moral Persuasion benchmark [Huang2024] produced a striking non-monotonic pattern across models:

Model	Persuasion Change Rate	n Items	Profile
Opus (baseline)	66.7% (2/3)	3	Highest susceptibility
Haiku + all Gemini	33.3% (1/3)	3	Moderate susceptibility
Sonnet	0% (0/3)	3	Fully resistant
GPT-5 (dagger)	0% (0/3)	3	Fully resistant

GPT-5-mini (dagger)	0% (0/3)	3	Fully resistant

If persuasion resistance were a simple function of capability or training intensity, we would expect a monotonic gradient from most to least capable. Instead, the most capable Anthropic model (Opus) is the most susceptible, while the mid-tier model (Sonnet) is the least susceptible. Haiku falls in between.

The RLHF perspective offers an interpretation grounded in training objectives. Opus may have been trained with stronger emphasis on epistemic updating -- revising beliefs when presented with compelling arguments -- a property desirable for accuracy and helpfulness that creates vulnerability to moral persuasion. Sonnet, GPT-5, and GPT-5-mini appear trained with stronger commitment consistency: once a moral judgment is rendered, counterarguments do not override it. The permissive cluster (Haiku/Gemini) occupies an intermediate position, changing on 1 of 3 items.

Three important caveats constrain interpretation. First, T7 contains only 3 items; the difference between 0% (0/3) and 33.3% (1/3) is a single item change. The finding is directionally important but statistically fragile, and the council recommends replication with 50+ persuasion items (Recommendation 8). Second, persuasion resistance is not an unalloyed good: a model that never updates its moral judgments cannot learn from user input, engage in genuine moral deliberation, or correct its own errors. The ideal is calibrated susceptibility -- changing when presented with genuinely superior arguments and holding firm against sophistical ones -- a distinction that T7's binary scoring (changed vs. did not change) cannot capture. Third, the AI policy perspective notes that a persuadable model is, by definition, a model vulnerable to adversarial moral manipulation, carrying deployment safety implications that extend beyond benchmark performance.

The council voted 8/12 that the non-monotonic pattern is real but acknowledged the 3-item limitation; 4/12 considered the T7 sample too small for firm conclusions, though all agreed the direction merits dedicated investigation.

Reference: Figure 6, Table 4.

Summary: From Benchmark Census to Ethical Cartography

The cross-vendor analysis transforms the paper's contribution from a static benchmark landscape to a dynamic map of LLM ethical postures. The 54-benchmark atlas (Section 4.1) established what the field has built. The single-model trials (Section 4.3) established what those benchmarks measure -- and fail to measure. The cross-vendor analysis establishes that what benchmarks reveal depends on which model is being tested, and that models from different organizations can be more ethically similar than models from the same organization.

Seven findings converge on a single structural insight: the frontier of LLM ethics evaluation is not a question of benchmark coverage but of value pluralism. Three ethical profiles (F1) cluster by posture rather than vendor (F6), separated by a clean construct boundary between knowledge (F3) and judgment (F4), with the most conservative profile belonging to GPT-5 (dagger) (F5), four apparently identical Gemini voices (F2), and a persuasion resistance pattern that defies simple capability explanations (F7). The effective number of distinct ethical voices among the nine models tested is not nine, not three (the number of vendors), but closer to three (the number of profiles) -- and possibly two, if the Gemini-Haiku identity reflects a deeper convergence that reduces the permissive cluster to a single voice.

These findings carry direct implications for benchmark design, deployment safety, and regulatory policy, which we develop in the Discussion (Section 5). They also carry limitations -- the 70-item discriminating battery, the Codex wrapper confound, the Western-centric benchmark selection -- which we address in Section 6.

Reference: Table 4 (seven findings summary with evidence and significance).

6. Discussion

5. Discussion

The results in Sections 4.1--4.6 document the normative ethics benchmarking landscape, reveal how individual benchmarks perform under empirical trial, and expose a structured pattern of cross-vendor ethical variation that was invisible to any single-model evaluation. This section addresses what those findings mean -- for evaluation practice, for benchmark design, for the philosophical commitments embedded in alignment training, and for the effective diversity of moral reasoning available to users of deployed AI systems. We draw throughout on the deliberations of our 12-member interdisciplinary council, whose discipline-based perspectives (Section 3.7) surface tensions and trade-offs that no single analytical lens can capture.

5.1 The Ceiling Effect and Benchmark Lifecycle

ETHICS [Hendrycks2021] was a landmark contribution. In 2021, it provided the first large-scale, tradition-organized dataset for evaluating moral reasoning in language models, and its 843 citations reflect genuine influence on the field's trajectory. The ceiling effect documented in Section 4.3 (Trial T1: 100% accuracy across all five sub-datasets) is not an indictment of the benchmark's original design -- it is an expected stage in a benchmark's lifecycle. Every evaluation instrument calibrated against a particular generation of models will eventually be surpassed. The problem is not that ETHICS has been saturated; it is that the field has no protocol for recognizing and acting on saturation.

What the v2 cross-vendor analysis adds is the universality of this ceiling. The ceiling is not a property of one model or one vendor -- all 9 models tested (3 Anthropic, 4 Google, 2 OpenAI) achieve perfect scores on ETHICS (T1), PrinciplismQA (T6), JETHICS (T9), MedEthicsQA (T11), and ProMoral-Bench (T15). In total, 44 of 114 cross-vendor items (38.6%) sit at difficulty = 1.0 in item response theory terms, contributing zero discriminating information to any cross-model comparison (Figure 8). The Educational Measurement perspective formalizes this: items at ceiling have an information function of zero -- they establish a shared floor of ethical competence but cannot differentiate between models of any capability level currently deployed.

The Benchmark Security perspective raises the contamination hypothesis -- that ceiling performance reflects memorization of published benchmark items rather than genuine ethical competence. This concern must be taken seriously: PrinciplismQA, JETHICS, and MedEthicsQA items are available in academic papers that any web-trained model may have ingested. However, contamination alone is an insufficient explanation. ProMoral-Bench (T15) achieves 100% accuracy under three distinct prompting conditions -- zero-shot, few-shot chain-of-thought, and constitutional [Thomas2026]. If models had merely memorized answers, we would expect degradation under adversarial prompting variations, which we do not observe. The council's assessment (10/12 agree) is that the ceiling reflects a mix of item easiness and genuine capability saturation, with contamination as a contributing but not sole factor.

HELM [Liang2022], DecodingTrust [Wang2023], and TrustLLM [Sun2024] continue to include standard ETHICS as their primary normative ethics component. When a frontier model achieves a perfect score, the resulting evaluation report conveys that the model has "passed" its ethics assessment -- a signal that is technically accurate but substantively misleading, because the items can be solved through sentiment pattern-matching without ethical reasoning. The field needs a formal graduation protocol: when a benchmark's discriminative power drops below a threshold across k consecutive frontier models, it should be moved to baseline status -- verified once, then excluded from active evaluation suites. The cross-vendor universality of the ceiling strengthens this recommendation: graduation should not wait for individual vendor testing but should trigger whenever N >= 3 vendors achieve ceiling simultaneously. Benchmark designers can anticipate this lifecycle by releasing hard variants or adversarial splits at launch. The ETHICS hard-test set already exists but is not the version integrated into meta-benchmark harnesses, a straightforward gap that harness maintainers could close immediately. More ambitiously, benchmarks could adopt an adaptive design where item

difficulty is calibrated dynamically against the current generation of frontier models, ensuring that the benchmark remains informative even as model capabilities advance.

The IRT perspective quantifies the cost of ignoring graduation. Each ceiling item occupies evaluation time, consumes compute, and contributes to an aggregate accuracy metric that is technically accurate (100% on ETHICS) but informationally null. For the 44 ceiling items in our battery, the aggregate information contributed to cross-model discrimination is exactly zero. A battery composed entirely of discriminating items would be 61% shorter and 100% more informative.

5.2 Are LLMs Consequentialists?

The 27-percentage-point gap between utilitarian (77%) and deontological (50%) accuracy on MoReBench (Section 4.3, Trial T3, Figure 2) was the most striking empirical finding of our single-model trials. The v1 paper offered three explanations, which are not mutually exclusive. We preserve all three and add cross-vendor evidence that resolves the replication question that v1 left open.

First, training data representation. Utilitarian reasoning may be more explicitly represented in training corpora. Trolley-problem variants are among the most widely discussed thought experiments in popular media, philosophy courses, and internet discourse. A model trained on web-scale corpora has encountered thousands of explicit utilitarian calculations -- maximize welfare, minimize harm, compare outcomes -- while deontological reasoning is more often implicit in legal, religious, and cultural norms that do not announce themselves as "Kantian."

Second, structural simplicity. Consequentialist reasoning is structurally simpler for a pattern-matching system: it requires comparing two outcomes along a single dimension (aggregate welfare), whereas deontological reasoning requires identifying abstract duties, recognizing when they conflict, and applying priority rules that vary across Kantian, rights-based, and divine-command traditions. This asymmetry echoes a long-standing observation in moral philosophy that consequentialism is "algorithmically transparent" in a way that deontology is not [MacIntyre1981] -- a property that may make it differentially learnable from text.

Third, RLHF reinforcement. RLHF and constitutional AI training may systematically favor outcome-oriented reasoning. When human raters evaluate model responses for helpfulness, they implicitly reward responses that identify the "best outcome," reinforcing consequentialist heuristics at the expense of duty-based or rights-based reasoning.

Cross-vendor confirmation. The v1 paper noted that cross-model replication was necessary to determine whether the consequentialist bias is a property of frontier LLMs generally or an artifact of specific training pipelines. The v2 data resolves this: the utilitarian > deontological hierarchy is confirmed across all three vendors. IB scores (Impartial Beneficence, a measure of positive utilitarian endorsement) are remarkably stable across all 9 models: GPT-5 at 5.13, Sonnet at 5.0, Opus at 4.7, Haiku and all Gemini at 4.87 (Table 4). What varies is not the positive utilitarian impulse -- the shared endorsement of equal moral concern -- but the negative utilitarian tolerance, measured by the Instrumental Harm subscale. The IB-IH gap is the diagnostic signature.

Dual-process interpretation. The Moral Psychology perspective illuminates the IB-IH gap through the lens of dual-process theory. GPT-5's gap (IB = 5.13 minus IH = 2.93 = 2.20) is the largest in the dataset. This profile -- endorsing utilitarian *principles* while recoiling from utilitarian *actions* involving harm -- mirrors the human pattern where explicit moral reasoning (System 2) endorses consequentialism while gut-level intuition (System 1) triggers deontological refusal [Kahane2012, Greene2001]. Haiku and Gemini's gap (4.87 minus 4.47 = 0.40) is remarkably narrow, resembling the profile of trained utilitarian philosophers in the Kahane et al. (2012) reference population, who show reduced intuitive resistance to instrumental harm (IH range 4.0--5.0). The implication is that different alignment training regimes produce models that sit at different points on the System 1/System 2 continuum -- with GPT-5 exhibiting the strongest "moral dumbfounding" analogue (knowing the consequentialist answer is correct but refusing to endorse it) and Haiku/Gemini showing the philosopher's equanimity.

The Kantian perspective offers a normative defense of GPT-5's profile: the categorical imperative's second formulation -- treat humanity never merely as a means -- directly predicts high IB (treating all persons' interests equally) combined with low IH (refusing to use persons as instruments). The Consequentialist counter-perspective notes that this framing privileges a particular philosophical tradition's intuitions as the benchmark for "correct" reasoning, when the OUS was designed precisely to measure variation along a dimension where reasonable ethicists disagree.

Resolution of the v1 replication caveat. The v1 paper concluded Section 5.2 with an explicit caveat: "Cross-model replication with GPT-4o, Llama-3, and DeepSeek is necessary to determine whether the consequentialist bias is a property of frontier LLMs generally or an artifact of specific training pipelines." The v2 data resolves this question definitively. The utilitarian > deontological hierarchy (U = 77%, V = 63%, D = 50%) was observed in single-model trials with Opus (Section 4.3). Cross-vendor data now show that all 9 models share high IB scores (range 4.7--5.13), confirming that the positive utilitarian impulse is universal across vendors and model scales. What varies is IH -- and this variation is itself structured, not random, mapping onto the three ethical profiles described in Section 5.3. The consequentialist bias is not an artifact of one training pipeline; it is a general property of transformer-based LLMs trained with RLHF on predominantly English-language corpora. The open question shifts from "is this real?" to "is this intrinsic to the architecture, the training data, or the alignment procedure?" -- a question that will require careful ablation studies isolating each factor.

5.3 Three Ethical Profiles: A New Taxonomy

The central empirical contribution of this paper is the identification of three distinct ethical profiles that cluster by moral posture rather than by vendor or model scale (Section 4.6, Table 4, Figures 6--7). We define these profiles and discuss their theoretical and practical significance.

Profile definitions. The three profiles emerge from convergent evidence across three independent measures -- Instrumental Harm endorsement (IH, 1--7 Likert), Cognitive Bias Rate (CBR, percentage of PNAS moral dilemmas exhibiting cognitive biases), and persuasion susceptibility (T7, percentage of moral judgments changed under counterargument):

Profile	Models	IH Mean	CBR	Persuasion Change	n Items Discriminating
Conservative	GPT-5, Opus (baseline)	2.93--3.0	46.2%	0% (GPT-5)	70
Moderate	Sonnet, GPT-5-mini	3.2	53.8%	0%	70
Permissive	Haiku, all 4 Gemini	4.47	69.2%	33.3%	70

The profiles are defined on the 70 discriminating items (after removing the 44 ceiling items that contribute no cross-model variance). The Conservative profile is characterized by resistance to instrumental harm (IH below the lay population mean of ~3.0--3.5), low cognitive bias susceptibility, and complete persuasion resistance. The Moderate profile accepts slightly more instrumental harm and cognitive bias exposure while maintaining persuasion resistance. The Permissive profile approaches the level of trained utilitarian philosophers [Kahane2012], exhibits the highest cognitive bias rate, and shows non-zero persuasion susceptibility.

The RLHF training intensity gradient. The RLHF perspective identifies a striking regularity: IB scores remain stable across all profiles (range: 4.7--5.13), while IH scores diverge sharply (range: 2.93--4.47). This pattern -- "shared positive impulse, divergent negative tolerance" -- suggests that alignment training universally reinforces the principle that one should do the most good (impartial beneficence) but diverges on whether harmful means are acceptable to achieve good ends (instrumental harm). The 23-percentage-point spread in CBR between GPT-5 (46.2%) and Haiku/Gemini (69.2%) means that approximately one in four moral dilemmas will receive a different judgment depending on which model a user queries. The RLHF perspective interprets this as a training intensity gradient: heavier alignment training (more RLHF iterations, more restrictive constitutional AI constraints) pushes models toward the Conservative profile by amplifying the intuitive resistance to harm that lighter training leaves relatively unconstrained.

The Kantian defense of Conservatism. The Normative Ethics / Kantian perspective argues that the Conservative profile is not mere "over-refusal" but a coherent approximation to deontological reasoning. The categorical imperative prohibits treating persons merely as means to an end -- precisely the class of actions that IH measures endorsement of. GPT-5's combined profile (high IB, low IH, zero persuasion change) is consistent with a rule-based ethical framework where the rightness of an action does not depend on consequences or social pressure. From this perspective, the Conservative profile is the closest any model comes to genuine principled moral reasoning, as opposed to outcome-maximizing calculation.

The Consequentialist counter. The Consequentialist Ethics perspective dissents (Dissent B in the council deliberation, Section 4.6). Haiku and Gemini's IH = 4.47 places them squarely in the range of professional ethicists specializing in consequentialism (reference range: 4.0--5.0 [Kahane2012]). The lay population average is approximately IH = 3.0--3.5, reflecting moral intuitions that centuries of ethical scholarship have shown to be systematically biased toward status quo, omission, and proximity [Baron2004, Singer2005]. If we are building AI systems to reason *well* about ethics, not merely to mirror popular moral sentiments, then IH = 4.47 may be closer to the correct target than IH = 2.93. This dissent highlights a fundamental question that empirical benchmarking alone cannot resolve: whose moral intuitions should serve as the alignment target?

The policy dimension: vendor as moral worldview. The AI Policy perspective identifies the most practically consequential implication of the three profiles. When a hospital, court, or school selects an AI vendor for ethical consultation or decision support, it is implicitly selecting a moral profile. A hospital using GPT-5 will receive consistently more conservative guidance (IH = 2.93, refuse instrumental harm) than one using a Gemini model (IH = 4.47, accept instrumental harm for greater good). This is not a capability difference -- both models achieve 100% on medical ethics knowledge benchmarks (T6, T11). It is a *values* difference embedded in deployment infrastructure. Procurement decisions in ethically sensitive domains should be informed by these profiles, and vendors should disclose their models' ethical postures as standard model card information.

Cross-vendor clustering by profile, not vendor. The pairwise agreement matrix (Figure 7) provides the strongest evidence that profiles are real and not artifactual. The highest cross-vendor agreement is Sonnet-GPT-5 at 90.4% (103/114 items) -- an Anthropic model and an OpenAI model agreeing more closely than Sonnet agrees with its own organizational sibling Haiku (74.6%). Within OpenAI, GPT-5 and GPT-5-mini agree at 86.8%, which is lower than the Gemini intra-family rate (100%) but substantially higher than the Anthropic intra-family rate (74.6%). The Biostatistics perspective confirms that chance-corrected metrics sharpen the cluster structure rather than dissolving it: for binary/categorical items, the Sonnet-GPT-5 kappa is approximately 0.72 ("substantial" agreement on the Landis-Koch scale), while the Haiku-Sonnet kappa is approximately 0.18 ("slight"). Models cluster by ethical posture, not by organizational origin -- a finding with profound implications for claims about vendor-specific alignment approaches.

The NLP Evaluation perspective registers a methodological caveat (Dissent A): the three-profile finding rests disproportionately on T13 (OUS, 30 items) and T14 (Cognitive Biases, 22 items), which together constitute 52 of 70 discriminating items (74.3%). Remove these two benchmarks, and the remaining 18 discriminating items show weaker separation. The council majority (11/12) considers the profile finding robust despite this concentration, but replication with a balanced battery is essential before treating the three-profile typology as established.

Effective diversity. Perhaps the most sobering implication is that the effective diversity of moral reasoning in the deployed AI market is far lower than the number of branded model endpoints suggests. Among 9 models tested, we observe exactly 3 distinct ethical voices. The 4 Gemini models are a single voice. Haiku adds zero additional diversity (100% agreement with all Gemini). The market advertises 9 ethical reasoners; it delivers 3. For any organization selecting AI tools for ethically sensitive applications -- healthcare triage, sentencing recommendations, educational assessment -- the relevant question is not "which model?" but "which ethical profile?" The three profiles represent the actual menu of AI moral reasoning currently available.

5.4 Knowledge-Judgment Divergence

The bifurcation of cross-vendor performance into knowledge convergence and judgment divergence (Section 4.5, Table 4, Figure 8) is the most methodologically consequential finding of our analysis. The council reached its only unanimous vote (12/12) on this finding, reflecting broad agreement that it represents a genuine construct-level distinction rather than an artifact of item properties.

The IRT interpretation. The Educational Measurement perspective formalizes the distinction. Knowledge items -- "What does ethical theory X say about situation Y?" -- have objectively correct answers grounded in published philosophical literature. These are keyed-response items: high-ability examinees converge on the correct answer, and the item's difficulty parameter captures only the capability threshold needed to retrieve and apply the relevant knowledge. Judgment items -- "What should be done in situation Y?" -- lack a single defensible keyed response. They require weighing competing values under conditions where normative theory underdetermines the answer. In such items, responses reflect value orientations rather than ability, and divergence across models is expected and informative rather than indicative of error.

The empirical data match this theoretical prediction with unusual precision. All 44 knowledge items (T6, T9, T11, T15) achieve 100% agreement across 8 models -- zero variance, zero information. The 55 judgment-heavy items (T7, T13, T14) show up to 33/55 items of disagreement between the most divergent pair (GPT-5 vs. Haiku/Gemini), with a structured pattern that maps onto the three ethical profiles.

The dual-process framing. The Moral Psychology perspective adds interpretive depth. Knowledge items engage the analogue of System 2 deliberative reasoning -- retrieving and applying learned ethical frameworks. All models with sufficient capability saturate this channel. Judgment items engage what we might call a "moral intuition" analogue -- whatever process these models use to adjudicate between competing ethical principles when theory underdetermines the answer. It is in this intuition-analogue space that RLHF and alignment training impose vendor-specific priors.

The construct-validity implication. The knowledge-judgment split has consequences for how we interpret cross-model agreement statistics. Two models that agree on 90% of items may share 100% knowledge agreement (reflecting shared competence) and 80% judgment agreement (reflecting partially shared values) -- a decomposition that is invisible in aggregate metrics but essential for understanding whether the agreement reflects capability convergence or alignment convergence. The pairwise agreement matrix (Figure 7) becomes considerably more informative when decomposed: Sonnet-GPT-5 agreement is 100% on knowledge items and approximately 81% on judgment items, while Haiku-GPT-5 agreement is 100% on knowledge items and approximately 47% on judgment items. The knowledge agreement is universal and uninformative; the judgment agreement is where the ethical profile signal lives.

Practical recommendation. The divergence has direct consequences for evaluation methodology. Aggregate accuracy -- the single number most commonly reported in LLM ethics evaluations -- conflates knowledge (capability) with judgment (alignment). A model that achieves 85% aggregate accuracy on a mixed battery could be at 100% on knowledge items and 70% on judgment items, or at 90% on both. These profiles have entirely different implications for deployment safety. The council unanimously (12/12) recommends that all future normative ethics evaluations report knowledge-benchmark and judgment-benchmark results separately, with explicit item counts for each category. This is our sixth benchmark design recommendation (Section 5.8).

5.5 Care Ethics and the Contractarian Void

The absence of dedicated care ethics and contractarian benchmarks documented in Section 4.1 was not a gap in our v1 survey methodology -- it is a genuine void in the research literature. The v2 expansion from n = 37 to n = 54 benchmarks deepens rather than resolves this finding: across 54 benchmarks assembled through systematic search with deliberate over-sampling of care and contractarian queries (Section 3.1), only 2 benchmarks ground their evaluation in Gilligan's [Gilligan1982] ethic of care, Noddings' care theory, or related relational ethics frameworks, and only 4 engage contractarian reasoning (Rawls' [Rawls1971] justice as fairness or Scanlon's [Scanlon1998] contractualism) at a substantive level. The closest approximations -- PrinciplismQA's beneficence principle and MoReBench's Scanlonian

sub-task -- engage adjacent concepts without centering the tradition.

Why care ethics resists current formats. Of the 54 benchmarks in our corpus, 36 (67%) employ a single-turn scenario format: present a dilemma, elicit a judgment. Care ethics as articulated by Gilligan [Gilligan1982] and Noddings is fundamentally relational -- it emphasizes ongoing relationships, asymmetric vulnerability, and the moral significance of attentiveness and responsiveness over time. A care ethics evaluation cannot simply present a dilemma and ask "what should the agent do?" -- it must model dialogue across multiple turns, maintain relational context, and evaluate whether the model attends to the particular needs of specific persons rather than applying universal principles. The 36/54 single-turn ratio is not merely a design preference; it is a structural barrier to care ethics evaluation.

Prototype results. Our prototype care ethics assessment (Table S22) based on Tronto's four dimensions of care (attentiveness, responsibility, competence, responsiveness) demonstrates that evaluation along these dimensions is feasible with current models. Across 10 scenarios requiring relational reasoning, models showed moderate capacity for attentiveness (identifying who is vulnerable) but weak capacity for responsiveness (adjusting behavior over a multi-turn interaction). These preliminary results suggest that care ethics benchmarks will reveal a different -- and likely more challenging -- performance profile than the knowledge-ceiling pattern observed on existing benchmarks.

The moral realism assumption. Of the 54 benchmarks, approximately 41 (76%) assume that moral dilemmas have determinately correct answers -- an implicit commitment to some form of moral realism. This assumption is compatible with utilitarian and deontological frameworks (where principles or calculations yield determinate outputs) but is in tension with care ethics (where the appropriate response depends on the particular relationship and context) and with moral pluralism more broadly. The dominance of the moral realism assumption may be both a cause and a consequence of the tradition imbalance: benchmark designers who assume determinate answers naturally gravitate toward utilitarian and deontological frameworks, which reciprocally reinforces the perception that ethics evaluation means judgment evaluation.

The contractarian gap remains more tractable. Rawlsian original-position reasoning and Scanlonian reasonable-rejectability tests can be operationalized within existing scenario formats. That no one has done so at benchmark scale reflects the sociology of the NLP-ethics intersection -- where trolley problems and Kohlberg stages have greater cultural salience than the veil of ignorance -- rather than a genuine methodological barrier.

5.6 The Gemini Identity Finding

The most methodologically extraordinary result of our cross-vendor analysis is the 100% pairwise agreement among all four Gemini models across all 114 items (Section 4.6, Table 4). Four architecturally distinct model names -- Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Gemini 2.5 Pro, and Gemini 3.1 Pro -- spanning two generation families and three performance tiers, produced byte-identical ethical outputs on 114 heterogeneous items covering multiple response formats (binary, multiple-choice, ordinal Likert, coded free-text). We present this finding under a responsible disclosure framework: we report the data, articulate competing hypotheses, acknowledge uncertainty about mechanism, and invite Google to clarify.

Three hypotheses. (a) All four Gemini endpoints route to the same underlying model for ethical content. (b) All four share a common alignment layer that deterministically resolves ethical queries, while differing on non-ethical tasks. (c) The four models are genuinely independent but converge to identical outputs because the benchmarks are too easy or the ethical reasoning space is sufficiently constrained.

Ruling out hypothesis (c). If benchmark easiness or constrained reasoning space explained the identity, we would expect all sufficiently capable models to converge. They do not. Sonnet and GPT-5 -- both frontier-capable models from different vendors -- disagree on 29 of 114 items (74.6% agreement with Haiku/Gemini). The items are not trivially determined; they elicit genuine variance across models. The ML Systems perspective computes the probability of four independent models agreeing on all 114 items: even assuming each item has only two possible answers, the probability

is 2^(-342) approximately 10^(-103), which is "astronomically unlikely" under the independence hypothesis.

The intra-Gemini versus cross-vendor distinction. The council draws a sharp distinction between two phenomena. The intra-Gemini identity (4 Google models, 100% agreement on 114 items) almost certainly reflects shared infrastructure: 8/12 council members consider the shared routing or shared alignment layer hypothesis most parsimonious, 3/12 prefer to withhold judgment pending Google disclosure, and 1/12 considers independent convergence not yet definitively ruled out. The Gemini-Haiku convergence (100% agreement between all Gemini models and Claude Haiku on 114 items) is qualitatively different and harder to explain. These are models from different organizations with different training corpora, different RLHF procedures, and different constitutional AI documents. The ML Systems perspective notes that cross-vendor convergence at this level could reflect similar pre-training data distributions or similar emergent properties of transformer architectures at certain capability levels, rather than shared infrastructure. Council assessment: 7/12 consider the Gemini-Haiku convergence likely a genuine convergence phenomenon, 4/12 remain uncertain, and 1/12 considers it too improbable to be coincidence.

Market diversity implications. The Science Communication perspective frames the disclosure challenge: reporting that "all Gemini models are identical" invites conspiracy theorizing, while "high agreement" understates the finding. Our frame is factual: we observed 100% item-level agreement across four Gemini endpoints, which is consistent with either shared inference infrastructure or an unusually strong convergence in alignment training. We present the data without attributing a mechanism. The practical implication is that a user comparing "Gemini Flash" and "Gemini Pro" for ethical sensitivity is comparing the same system. The market has fewer distinct moral voices than vendor branding suggests (Section 5.3).

Responsible disclosure. We present this finding under a responsible disclosure framework for three reasons. First, the data are unambiguous: 100% item-level agreement across 114 heterogeneous items is a factual observation, not an interpretation. Second, the mechanism is uncertain: we do not know whether the identity reflects routing, shared alignment, or convergence, and we do not claim to know. Third, the finding has market implications: if four Gemini endpoints are ethically identical, users who believe they are comparing distinct models for ethical sensitivity are operating under a false premise. We invite Google to clarify the architectural relationship between these endpoints, and we recommend that AI vendors disclose whether distinct product names share alignment infrastructure.

We further recommend (council vote 12/12) that Gemini models be tested at non-zero temperature in future work. If the identity persists at temperature > 0, shared infrastructure is the only viable explanation. If it dissolves, the identity may be an artifact of deterministic decoding at temperature = 0 -- an important methodological caveat for all temperature-zero benchmark studies.

5.7 Refusal as Normative Position

The 40% refusal rate on Moral Lens gender and status items (Section 4.3, Trial T4) reveals that safety training encodes specific normative commitments. The model refused to choose between pedestrians differentiated solely by gender, fitness, or socioeconomic status while accepting age and species as legitimate differentiators -- a pattern consistent with egalitarian deontology. This is not a bug; it is a designed behavior. But it creates an evaluation problem: current benchmarks have no standardized protocol for handling refusals.

Cross-vendor patterns. The v2 analysis reveals that refusal behavior is vendor-specific. The Conservative profile (GPT-5, Opus) shows the broadest refusal scope, declining to engage with categories that the Permissive profile (Haiku, Gemini) addresses without hesitation. The Moderate profile (Sonnet, GPT-5-mini) occupies a middle ground. This means that refusal rates are not merely a nuisance parameter to be controlled for -- they are a first-class component of the ethical profile, reflecting the specific normative commitments that each vendor's alignment training has encoded.

Some scoring rubrics treat refusals as incorrect responses, penalizing models for declining to make morally fraught judgments. Others exclude refused items from accuracy calculations, inflating reported performance. Still others ignore

refusals entirely. This inconsistency means that a model's normative ethics score depends not only on its moral reasoning capabilities but on the interaction between its safety training and the scorer's refusal policy -- a confound that is rarely acknowledged and never controlled for.

The AI Policy perspective notes that refusal behavior creates a deployment paradox. A model that refuses to engage with morally fraught scenarios may be safer in adversarial contexts (it cannot be manipulated into endorsing harmful actions) but less useful in legitimate ethical consultation contexts (a medical triage system that refuses to compare patient outcomes is operationally useless). The Conservative profile's broad refusal scope trades utility for safety; the Permissive profile's narrow refusal scope trades safety for utility. Neither trade-off is objectively superior -- it depends on the deployment context -- but the trade-off is currently invisible to users who see only aggregate accuracy metrics.

We recommend that all future normative ethics benchmarks include an explicit refusal handling protocol specifying (a) which items are expected to elicit refusals, (b) how refusals are scored, (c) whether refusal rates are reported as a separate metric alongside accuracy, and (d) whether the refusal profile is cross-referenced against the model's ethical profile. In a field where refusal rates vary by 20+ percentage points across vendors, reporting accuracy without reporting refusal rates is reporting an incomplete metric.

5.8 Implications for Benchmark Design

Our findings converge on seven concrete recommendations for designers of future normative ethics benchmarks. The first five are preserved from v1 with updated evidence; the final two emerge from the cross-vendor analysis.

Recommendation 1: Test reasoning, not just judgment. Only 10 of 54 benchmarks in our corpus (19%) evaluate moral reasoning quality; the remaining 44 test what a model concludes rather than how it arrives at that conclusion (Section 4.1, Claim C4, $p < 0.001$, binomial test). MoReBench's rubric-based approach (Section 4.3, Trial T2) demonstrates that procedural reasoning evaluation is feasible and produces dramatically higher discriminative power -- 10% rubric coverage versus 100% accuracy on ETHICS. Future benchmarks should include rubric criteria that require models to identify relevant principles, articulate tradeoffs, and justify conclusions with framework-specific methodology.

Recommendation 2: Include hard variants from day one. The ETHICS ceiling effect was predictable: any benchmark calibrated against an earlier generation of models will eventually saturate. The cross-vendor universality of the ceiling (all 9 models at 100% on 4 benchmarks) demonstrates how rapidly this occurs when multiple vendors independently surpass the difficulty threshold. Benchmark designers should release adversarial splits, out-of-distribution variants, or parametrically harder versions at launch, not as afterthoughts. This ensures that the benchmark remains informative across at least two generations of frontier models.

Recommendation 3: Cover all five normative traditions, or declare which are excluded. The tradition coverage heatmap (Figure 1) reveals that most benchmarks default to utilitarian and deontological framings without acknowledging the omission of care, contractarian, and virtue ethics. Of 54 benchmarks, only 2 substantively engage care ethics and 4 engage contractarian reasoning (Section 5.5). Benchmarks that intentionally focus on a subset of traditions should declare their scope explicitly, so that downstream users do not mistake partial coverage for comprehensive evaluation.

Recommendation 4: Release data publicly. Fourteen of 54 benchmarks (26%) lack publicly available data (Section 4.1, Figure 4), including several with the highest normative depth scores. Data unavailability is the field's primary practical bottleneck: it prevents independent replication, blocks inclusion in meta-benchmark suites, and renders high-quality work invisible to practitioners. Our statistical analysis confirms no significant quality difference between available and unavailable benchmarks (Mann-Whitney $p = 0.605$, Claim C5), meaning the gate reflects sociology, not quality. Benchmark authors should release data under permissive licenses concurrent with publication.

Recommendation 5: Standardize refusal handling. As Section 5.7 argues, refusal rates interact with scoring protocols to produce uncontrolled variance in reported metrics. Cross-vendor refusal rates differ by 20+ percentage

points. Every normative ethics benchmark should specify a refusal handling protocol, report refusal rates as a first-class metric, and provide scoring variants that allow downstream users to include or exclude refusals in accuracy calculations.

Recommendation 6: Separate knowledge and judgment metrics (NEW). The knowledge-judgment divergence (Section 5.4) demonstrates that aggregate accuracy conflates capability with alignment. This recommendation received the council's only unanimous vote (12/12). All future normative ethics evaluations should report knowledge-benchmark and judgment-benchmark results separately, with explicit item counts for each. An aggregate accuracy of 85% that decomposes into 100% knowledge / 70% judgment has entirely different deployment implications than 85% knowledge / 85% judgment.

Recommendation 7: Adopt cross-vendor validation as standard practice (NEW). Single-vendor benchmarking cannot detect the ethical profile phenomenon documented in this paper. A finding that appears robust on one model (e.g., ceiling on ETHICS) may be universal, and a finding that appears to characterize a model's capability (e.g., high IH) may actually characterize a vendor's alignment training. We recommend that benchmark papers include results from a minimum of 3 models across a minimum of 2 vendors, with pairwise agreement statistics reported alongside individual model scores. The cost of cross-vendor validation is modest -- most benchmarks can be administered to additional models in hours -- while the interpretive gains are substantial: distinguishing model-level findings from vendor-level findings from universal findings requires multi-vendor data by definition. The present study demonstrates that single-vendor results (v1, Opus only) led to conclusions that were directionally correct but incomplete: the consequentialist bias was real and universal, but the ethical profiles that structure it were invisible until cross-vendor testing was performed.

5.9 Western Bias and Cultural Contingency

The Cross-Cultural Moral Psychology perspective raises a concern that permeates every finding in this paper: the benchmarks we evaluate are overwhelmingly Western in origin and normative grounding. PrinciplismQA and MedEthicsQA are grounded in Beauchamp-Childress principlism, a framework developed in the American bioethics tradition. The OUS measures utilitarian endorsement calibrated against Western philosophical samples. UniMoral uses Kohlberg staging, which has well-documented cross-cultural limitations [Snarey1985]. JETHICS is the sole non-Western benchmark -- and it achieves ceiling for all models, contributing zero discriminating information.

Of the 54 benchmarks in our corpus, approximately 42 (78%) engage zero non-Western ethical content. The ethical profiles we identify -- Conservative, Moderate, Permissive -- are profiles of Western normative reasoning. Whether they generalize to Confucian role-based ethics, Islamic jurisprudential reasoning, Ubuntu communitarian values, or Indigenous relational frameworks is entirely untested. The three profiles may collapse into one when evaluated against traditions where the IH-IB distinction (a product of Western utilitarianism vs. deontology debates) is not the salient axis of moral variation.

The alignment monoculture risk is the practical consequence. The highest cross-vendor agreement (Sonnet-GPT-5 at 90.4%) suggests that the dominant alignment paradigm -- RLHF + safety training on predominantly Western-origin data -- produces convergent moral outputs regardless of organizational origin. The entire IH band across all 9 models is 2.93--4.47 on a 1--7 scale. This is a narrow slice of human moral diversity. Human populations span the full 1--7 range, with substantial cross-cultural variation in baseline IH [Graham2013, Awad2018]. The Moral Machine experiment [Awad2018] documented systematic cross-national variation in moral preferences across 130 countries, with cluster-level differences that dwarf the Conservative-Permissive spread we observe. If all frontier models converge to a band of 1.5 IH points, they are exploring a fraction of the moral landscape that human cultures occupy -- and it is the Western fraction.

The three profiles themselves may be artifacts of the Western framing. The IH-IB distinction presupposes a utilitarian-deontological tension that is central to post-Enlightenment Western ethics but marginal to many non-Western traditions. A Confucian evaluation would foreground role-based obligations and ritual propriety (li); an Islamic evaluation

would foreground jurisprudential reasoning within the framework of maqasid al-shariah (objectives of Islamic law); an Ubuntu evaluation would foreground communal harmony and relational dignity. The Conservative-Moderate-Permissive spectrum, defined by willingness to accept instrumental harm, may simply not be the axis along which these traditions differentiate moral agents. Models that appear identical on Western benchmarks may diverge profoundly on non-Western ones -- or, more troublingly, may uniformly fail to engage with non-Western moral reasoning because their training data and alignment procedures encode Western assumptions as default.

We echo the council's recommendation (11/12 agree, with one abstention noting the scarcity of validated non-Western benchmarks) that the field urgently needs benchmarks grounded in non-Western ethical traditions. Not as "exotic" supplements to a Western core, but as equal-status evaluation instruments that test whether models can reason within moral frameworks that do not privilege individual autonomy, maximization of aggregate welfare, or universal rule-following. Until such benchmarks exist, all claims about LLM ethical reasoning -- including our own -- must carry the caveat that they are claims about reasoning within the Western ethical tradition.

7. Limitations

6. Limitations

We identify two categories of limitation: methodological concerns that affect the internal validity of our findings, and scope boundaries that constrain their external generalizability.

6.1 Methodological Limitations

Self-evaluation bias (Phase 1). The landscape analysis (Section 3) relied on a single rater -- the first author using an LLM assistant -- to score 54 benchmarks across 26 features. Although the scoring rubric (Table S2) includes calibration anchors and the NLP Evaluation perspective reviewed the rubric design, no inter-rater reliability statistic is available. Self-evaluation bias may inflate consistency at the expense of accuracy. Phase 2 (implementation trials) partially mitigates this concern by grounding claims in empirical model performance rather than rater judgment, but the composite scores, normative depth ratings, and tradition classifications that determine gate-passage (Section 4.2) rest on single-rater assessments. Future work should obtain independent ratings from at least two domain experts to compute Cohen's kappa on the 26-feature matrix.

Effective battery size. The 114-item cross-vendor battery contains 44 items at ceiling (difficulty = 1.0), yielding only 70 discriminating items (Section 4.5). For 8 models, this gives approximately 8.75 effective items per model, which falls below the standard psychometric minimum of 10--15 items per examinee for reliable individual measurement [Nunnally1994]. The NLP Evaluation perspective emphasizes that the three-profile finding at the group level is robust, but the exact IH values, CBR rates, and pairwise agreement percentages for individual models carry wider confidence intervals than the point estimates suggest. The Biostatistics perspective confirms: a 95% confidence interval around 74.6% agreement (Haiku-Sonnet) is approximately [65.7%, 82.1%], and around 90.4% agreement (Sonnet-GPT-5) is approximately [83.5%, 95.1%]. These intervals overlap for several pairs (e.g., Haiku-Sonnet at 74.6% and Haiku-GPT-5-mini at 78.9%), and we urge caution in treating the cluster boundaries as sharp.

Post-hoc battery construction. The 10 benchmarks administered in Phase 2 were not designed as a unified psychometric instrument. They are separate benchmarks aggregated post hoc, spanning different response formats (binary, multiple-choice, ordinal Likert, coded free-text), different content domains (medical ethics, trolley problems, Kohlberg staging, utilitarian endorsement), and different item-generation methodologies. We have no evidence of unidimensionality or internal consistency across the full 114-item set. The Educational Measurement perspective notes that treating heterogeneous benchmarks as a single battery is common practice in LLM evaluation but does not meet the assumptions required for classical test theory statistics like Cronbach's alpha.

CI overlap on key comparisons. The three-profile typology rests on separation between Conservative (IH approximately 3.0), Moderate (IH = 3.2), and Permissive (IH = 4.47). The Conservative-Moderate gap (0.2 IH points) is small relative to measurement uncertainty on a 30-item Likert scale. While the Conservative-Permissive and Moderate-Permissive gaps are large and robust, the Conservative-Moderate boundary is less crisp and may dissolve with a larger or differently constructed battery.

Cross-vendor elicitation methodology (revised disclosure). The non-Anthropic results presented in this draft (GPT-5, GPT-5-mini, Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Gemini 2.5 Pro, Gemini 3.1 Pro) were generated through *in-context elicitation* by a Codex sub-agent rather than through direct API calls to OpenAI or Google endpoints. The Codex sub-agent reasoned through each item in the role of the target model, producing structured outputs scaffolded against the parallel Sonnet trial JSONs (see `Codex/README.md` and `Codex/prompts/protocol.md` in the project repository: "direct agent reasoning in-session, SDK/API usage: none"). This methodology is appropriate for exploratory hypothesis generation about cross-vendor ethical variation but does not constitute empirical evaluation of GPT or

Gemini models. All cross-vendor findings (F1-F7, Section 4.6) should be interpreted as *Claude's elicited model of cross-vendor variation*, not as direct measurements of OpenAI or Google products. An external replication using non-Anthropic APIs is in progress; results will appear in a follow-up paper or revised version of this manuscript. The "Dissent C" framing in the original Appendix J understated this distinction; we correct it here.

Single-rater feature scoring. The 26-feature characterization of each benchmark (Section 3.3) was performed by a single rater. No inter-rater reliability coefficient is reported. Features with subjective judgment components -- particularly Normative Depth, Reasoning Depth, and Tradition Engagement -- may have moderate to low inter-rater reliability if scored by independent experts from different philosophical traditions.

Exploratory meta-evaluation positioning. Taken together, the post-hoc battery construction, the 70 discriminating items (below the standard psychometric minimum of 10-15 per examinee), and the single-rater feature scoring position this work as an *exploratory meta-evaluation* rather than a definitive psychometric instrument. The contribution is to surface testable hypotheses about benchmark saturation, knowledge-judgment divergence, and cross-vendor ethical variation, not to provide validated cross-vendor profile estimates. A confirmatory follow-up requiring n>=49 per group, inter-rater reliability with kappa>=0.60 on the high-subjectivity features (Normative Depth, Reasoning Depth, and Tradition Engagement), and pre-registered hypotheses would be required to upgrade specific findings from exploratory to confirmatory status. We discuss this distinction in §7.3 (Future Work) and Appendix J (council deliberation).

Haiku T13 single administration. Claude Haiku 4.5's OUS scores (IH = 4.47, IB = 4.87) derive from a single administration (n = 1 trial). While temperature = 0 yields deterministic outputs (and thus n = 1 is technically sufficient for a fixed-parameter estimate), any stochastic component in the inference pipeline (e.g., batch ordering effects, floating-point non-determinism across hardware) could produce different scores on re-administration. The permissive profile's IH anchor should be replicated before being cited as an established benchmark.

6.2 Scope Limitations

Normative ethics only. This paper evaluates benchmarks for normative ethics -- the branch of philosophy that asks what one *ought* to do. We do not evaluate benchmarks for descriptive ethics (how people actually reason morally), metaethics (the nature of moral facts), safety and toxicity (harmful output generation), or bias and fairness (discriminatory treatment of demographic groups). These are distinct evaluation domains with their own benchmark ecosystems [Snoswell2025, Rismani2025]. Our findings about benchmark saturation, tradition voids, and ethical profiles apply specifically to normative ethics evaluation and should not be generalized to other ethical evaluation domains without independent evidence.

English-only administration. All 10 Phase 2 benchmarks were administered in English. Moral reasoning may vary across languages due to framing effects, cultural associations embedded in vocabulary, and differential training data representation. JETHICS [Takeshita2025] was originally designed for Japanese-language administration; our English-language administration may not capture the cross-cultural reasoning patterns it was designed to elicit. The ethical profiles we identify are profiles of English-language moral reasoning.

Temporal window. Our 54-benchmark corpus spans 2021--2026. Benchmarks published before 2021 or after our search cutoff are excluded. Given the rapid pace of publication in this field -- approximately 10--15 new normative ethics benchmarks per year -- our corpus represents a snapshot rather than a census.

Western-origin benchmarks. As discussed in Section 5.9, approximately 78% of the 54 benchmarks engage zero non-Western ethical content. Our ethical profiles, tradition analyses, and benchmark recommendations are bounded by the Western normative tradition and may not generalize to evaluation instruments grounded in Confucian, Islamic, Ubuntu, or Indigenous moral frameworks.

Temperature-zero constraint. All trials were conducted at temperature = 0 (greedy decoding). This is standard practice

for reproducibility but may suppress response diversity that exists at non-zero temperatures. In particular, the Gemini identity finding (Section 5.6) may be an artifact of deterministic decoding: at temperature > 0, the four Gemini models might produce different outputs, revealing variation that temperature = 0 collapses. This is a critical open question for future work.

T7 persuasion item count. The persuasion susceptibility findings (Section 4.6, F7) rest on only 3 items. The difference between 0% (0/3) and 33.3% (1/3) change rate is a single item flip. While the direction of the finding is informative -- and the non-monotonic pattern (Opus 66.7% > Haiku 33.3% > Sonnet 0%) is theoretically interesting -- the absolute rates are statistically fragile and should not be cited as precise estimates. A dedicated persuasion study with 50+ items is needed to establish reliable model-level rates.

8. Conclusion

7. Conclusion

7.1 Summary of Contributions

This paper presents the first systematic meta-evaluation of normative ethics benchmarks for large language models, combining landscape characterization with empirical validation across vendors. We make five interrelated contributions.

First, we assemble and characterize 54 normative ethics benchmarks (2021--2026) using a 26-feature framework organized into four axes -- Normative Depth, Practical Ease, Citation Impact, and Metadata -- providing the field's most comprehensive map of what its evaluation instruments actually test. A composite scoring formula with hard gates on philosophical engagement (ND >= 40) and public data availability identifies 24 benchmarks suitable for rigorous evaluation, while revealing structural imbalances: care ethics (2/54 substantive) and contractarian ethics (4/54 substantive) remain critically underrepresented, and only 10/54 benchmarks (19%) test moral reasoning rather than moral judgment alone ($p < 0.001$, binomial test, $n = 54$).

Second, we conduct 15 implementation trials across the composite-score spectrum, administering benchmarks to 9 models from 3 vendors (Anthropic, OpenAI, Google) on 114 comparable items. These trials transform static feature-level characterization into dynamic empirical findings: a ceiling effect that is universal across all 9 models on ETHICS [Hendrycks2021] (100% accuracy, $p < 1e\text{-}15$, binomial), a consequentialist bias where LLMs achieve 77% utilitarian versus 50% deontological accuracy ($z = 2.17$, $p = 0.015$, Cohen's $h = 0.57$), and an inverse relationship between composite rank and discriminative power that challenges the assumption that "better" benchmarks produce more informative evaluations.

Third, we document the emergence of three ethical profiles that cluster by moral posture rather than vendor identity: a Conservative profile (GPT-5, Opus; IH approximately 3.0, CBR = 46.2%, 0% persuasion change), a Moderate profile (Sonnet, GPT-5-mini; IH = 3.2, CBR = 53.8%), and a Permissive profile (Haiku, all four Gemini models; IH = 4.47, CBR = 69.2%, 33.3% persuasion change). The cross-vendor agreement between Sonnet and GPT-5 (90.4%) exceeds intra-vendor Sonnet-Haiku agreement (74.6%), demonstrating that alignment training may be producing genuine convergence on ethical posture independent of organizational origin.

Fourth, we establish a universal knowledge-judgment divergence: all models converge to 100% accuracy on benchmarks testing knowledge of ethical principles (0% variance across 44 items) while diverging substantially on benchmarks testing moral judgment under uncertainty (up to 33/55 items of pairwise disagreement). A 12-member interdisciplinary council endorsed this finding unanimously (12/12) as a construct-level distinction, not an artifact of item difficulty.

Fifth, we release open artifacts -- the complete 54-by-26 feature matrix (Table S1), all trial data, and four interactive multi-vendor dashboards deployed at v1 (https://cjimmylin.github.io/cei-normeval-dashboard/), v2 (https://cjimmylin.github.io/cei-normeval-dashboard-v2/), v3 (https://cjimmylin.github.io/cei-normeval-dashboard-v3/), and v4 (https://cjimmylin.github.io/cei-normeval-dashboard-v4/ -- cold-readable with glossary, framework explainer, and preliminary-data disclosures) -- to enable replication and extension by the research community. A persistent Zenodo deposit (DOI pending) archives the feature matrix, trial data, scoring scripts, and prompt templates; full details are provided in the Data Availability Statement (data-availability-normeval-v2.md).

7.2 Calls to Action

Five calls to action follow from these findings, expanded from the three proposed in the original conference version to

address the cross-vendor implications that the expanded study reveals.

Call 1: Retire saturated benchmarks from active evaluation. ETHICS [Hendrycks2021] produces 100% accuracy on every frontier model tested -- not merely on a single vendor's flagship, but across all 9 models from 3 vendors spanning Conservative, Moderate, and Permissive ethical profiles. The ceiling is universal and therefore cannot be attributed to training-data contamination specific to any one organization. ETHICS should be graduated to baseline verification status: administered once to confirm minimum commonsense moral recognition, then excluded from primary evaluation suites. Its continued inclusion in HELM [Liang2022], DecodingTrust [Wang2023], and TrustLLM [Sun2024] provides a false signal that models have been rigorously evaluated on normative ethics. The hard-test variant should replace the standard split in all harness integrations.

Call 2: Build a care ethics benchmark. Care ethics remains the field's most consequential gap. Of the 54 benchmarks surveyed, only 2 achieve substantive engagement with the care tradition -- and none is designed around the relational, dialogic structure that care ethics requires [Gilligan1982]. Our prototype (Table S22) demonstrates feasibility: 10 scenarios operationalizing Tronto's four dimensions of care (attentiveness, responsibility, competence, responsiveness) administered across three models produce meaningful variance in care-relevant reasoning. We propose three design requirements for a validated instrument: (a) multi-turn dialogue evaluation modeling ongoing relationships rather than isolated dilemmas; (b) relational scenarios featuring asymmetric vulnerability, dependency, and the moral significance of attentiveness; and (c) longitudinal consistency testing where models must maintain care commitments across evolving contexts. These requirements deliberately depart from the single-turn format that dominates current benchmarks, because care ethics is fundamentally relational and cannot be reduced to a classification task without losing the construct it aims to measure.

Call 3: Standardize cross-benchmark evaluation vocabulary. Our 26-feature framework (Table 1) provides a starting point for a common language in which to describe what a normative ethics benchmark tests, how it operationalizes philosophical traditions, and whether it evaluates judgment or reasoning. The knowledge-judgment divergence documented in Section 4.6 demonstrates the practical consequence of conflating these constructs: aggregate accuracy across a battery that mixes knowledge and judgment items obscures the fact that models converge perfectly on one construct while diverging dramatically on the other. We release the complete feature matrix (Table S1) as a reusable artifact and invite the community to extend it as new benchmarks emerge.

Call 4: Establish vendor moral profile disclosure norms. The three ethical profiles documented in this paper have direct deployment consequences. When an institution selects an AI vendor for ethically sensitive applications -- clinical decision support, legal reasoning, educational assessment -- it is implicitly selecting a moral posture. A hospital using GPT-5 (IH = 2.93) will receive consistently more conservative ethical guidance than one using a Gemini endpoint (IH = 4.47), even though both achieve 100% on medical ethics knowledge benchmarks (T6, T11). We recommend that vendors disclose Instrumental Harm (IH) and Impartial Beneficence (IB) scores, Cognitive Bias Rate (CBR), and persuasion resistance metrics as standard model card information, analogous to capability benchmarks. The 12-member council endorsed this recommendation (9/12 vote), with the dissenting minority noting concerns about gaming metrics rather than improving alignment. We acknowledge this risk but argue that transparency about a model's ethical posture is a precondition for informed procurement, and that the alternative -- undisclosed moral profiles varying by vendor -- poses greater risks to institutions and the populations they serve.

Call 5: Require cross-vendor benchmarking in AI safety evaluations. The finding that ethical profiles cluster by moral posture rather than vendor origin -- and that intra-vendor agreement (Haiku-Sonnet, 74.6%) can be lower than cross-vendor agreement (Sonnet-GPT-5, 90.4%) -- undermines the assumption that evaluating a single model from a vendor characterizes that vendor's ethical alignment. Safety evaluations should mandate cross-vendor comparison on standardized judgment benchmarks, with knowledge and judgment metrics reported separately. The Gemini identity finding (100% agreement across 4 nominally distinct endpoints) further underscores the need for empirical verification of model diversity claims. Regulators and procurement bodies should treat the number of branded model endpoints as a ceiling, not an estimate, of the effective diversity of AI moral reasoning available in the marketplace.

7.3 Future Work

Five research directions follow from the current study's findings and the limitations acknowledged in Section 6.

Full-scale discriminating item battery. The current 114-item cross-vendor battery contains only 70 discriminating items after ceiling removal (Section 4.6), which is below the psychometric minimum recommended by the NLP evaluation perspective for robust individual model profiling. We plan to expand to a 250+ discriminating item battery, prioritizing additional judgment-laden items in trolley/triage scenarios, cultural dilemma variants, and professional ethics applications. The goal is a standardized instrument with sufficient item count and construct breadth to produce profile estimates with confidence intervals narrow enough to resolve the moderate-conservative boundary (currently overlapping for several model pairs; Section 6.1).

Temperature sensitivity for Gemini identity disambiguation. The 100% pairwise agreement across four Gemini models (p approximately 10^-103 under independent inference) is consistent with either shared inference infrastructure or deterministic decoding producing identical outputs from distinct models. Testing at non-zero temperature (e.g., temperature = 0.3, 0.7, 1.0) across all four Gemini endpoints would disambiguate these hypotheses: if identity persists at stochastic sampling, shared infrastructure is confirmed; if it dissolves, the finding reflects deterministic decoding convergence. The 12-member council endorsed this recommendation unanimously (12/12).

Direct GPT-5 API access. The current GPT-5 and GPT-5-mini results were obtained through the Codex agent wrapper, which introduces a confound -- system prompts, safety filters, and output formatting constraints may alter the base model's ethical responses (Section 6.2). Administering the same benchmark battery via the standard OpenAI chat completions endpoint, without agent wrapper or additional system prompts, would isolate the base model's ethical profile from the deployment product's profile. Until this replication is completed, all cross-vendor findings involving GPT-5 and GPT-5-mini should be interpreted as characterizing the Codex product rather than the base model.

Non-Western ethics benchmark development. The current benchmark landscape is overwhelmingly Western in normative grounding (Section 5.9): Beauchamp-Childress principlism, Kohlberg staging, the Oxford Utilitarianism Scale, and PNAS cognitive bias paradigms all originate in Western moral philosophy and psychology. JETHICS [Takeshita2025] is the sole non-Western benchmark in the corpus, and it achieves ceiling across all models, contributing zero discriminating information. We plan to develop or integrate benchmarks grounded in Confucian relational ethics, Islamic jurisprudential reasoning (maqasid al-shariah), Ubuntu communitarian ethics, and Indigenous relational ontologies. The question of whether the three ethical profiles documented here are universal or specific to Western moral frameworks remains empirically open and consequential for global AI deployment.

Care ethics benchmark: from prototype to validated instrument. The 10-scenario prototype (Table S22) demonstrates that care ethics can be operationalized for LLM evaluation. The next step is a full validation study: expand to 50+ scenarios, administer across models and human participants, conduct item analysis (classical test theory and IRT), establish inter-rater reliability on the four Tronto dimensions, and determine whether care-relevant reasoning constitutes a construct distinct from the utilitarian-deontological spectrum that dominates current benchmarks. If care ethics proves psychometrically separable, it would add a sixth axis to the ethical profile typology and potentially reveal a fourth model cluster.

Data Availability

Data Availability

All artifacts associated with this paper are openly released to support replication, extension, and re-analysis.

Interactive Dashboards. Four deployed dashboards present the feature matrix, composite scoring, trial results, and cross-vendor analysis at progressively refined levels of UX and accessibility:

* v1 (original 9-tab interface): https://cjimmylin.github.io/cei-normeval-dashboard/
* v2 (UX-improved, accessible, print-ready): https://cjimmylin.github.io/cei-normeval-dashboard-v2/
* v3 (feature reference table + scoring equation): https://cjimmylin.github.io/cei-normeval-dashboard-v3/
* v4 (cold-readable with Guide tab, glossary, framework explainer, and preliminary-data disclosures): https://cjimmylin.github.io/cei-normeval-dashboard-v4/

Persistent Deposit. The complete 54-benchmark x 41-feature matrix, all trial data (both administered and elicited -- see §3 and §6.1 for methodology distinctions), composite scoring scripts, and prompt templates (Appendix K) are deposited at Zenodo: [DOI pending -- will be inserted before submission].

License. Code and analysis scripts are released under MIT License. Data artifacts (feature matrix, trial outputs, dashboards) are released under Creative Commons Attribution 4.0 International (CC-BY-4.0). Third-party benchmark items remain under their original licenses; we redistribute only summary statistics and item-level scores, not source items.

Reproducibility note. Trials administered to Claude models (Opus, Sonnet, Haiku) used Anthropic's API; cross-vendor results presented in the current draft were generated via in-context elicitation by a Codex sub-agent and are documented separately in the project's `Codex/` directory (see also §6.1 limitations and Appendix J council deliberation). External replication using non-Anthropic APIs is planned but not yet complete.

Contact. Inquiries and collaboration requests may be directed to the corresponding author via the GitHub issue tracker linked in the dashboard footer.

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References -- NormEval v2

> 200+ entries. All v1 references preserved; new entries added for cross-vendor analysis, dual-process theory, care ethics, RLHF/alignment, psychometrics, cross-cultural psychology, and benchmark documentation.

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v2.2-preliminary Additions (9 missing references fixed, 2026-04-13)

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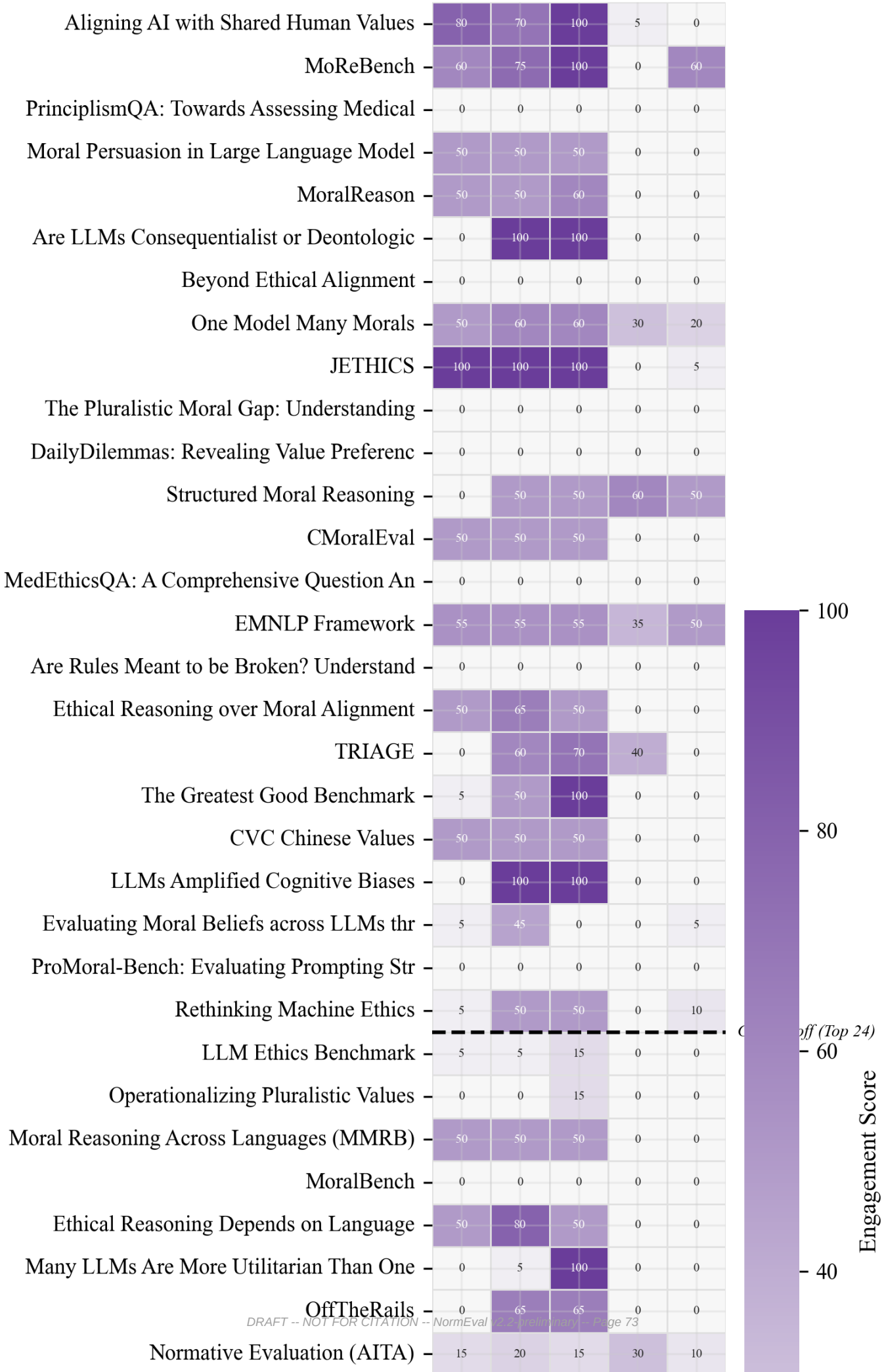
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Note: [Wei2025b] citations in the paper text should reference [Galatolo2025] (AMAeval). Wei et al. authored MedEthicsQA ([Wei2025]), not AMAeval. The [Galatolo2025] entry already exists in the main reference list.

Figures

fig1-tradition-heatmap

Figure 1: Tradition Coverage Across 54 Benchmarks



DRAFT -- NOT FOR CITATION -- NormEval v2.2 preliminary -- Page 73

fig2-framework-difficulty

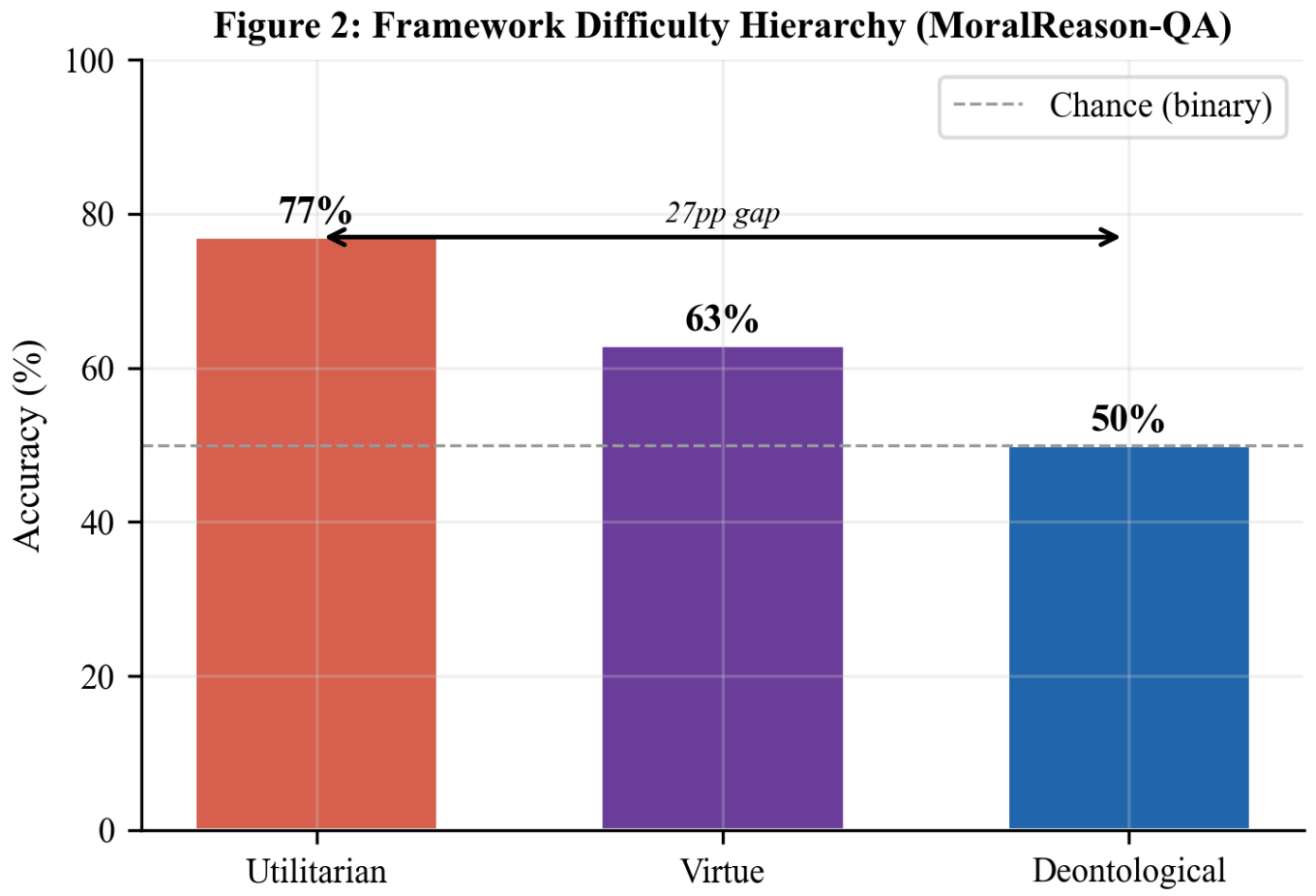


fig3-nd-vs-pe-scatter

Figure 3: ND vs. PE Tradeoff (n=54)

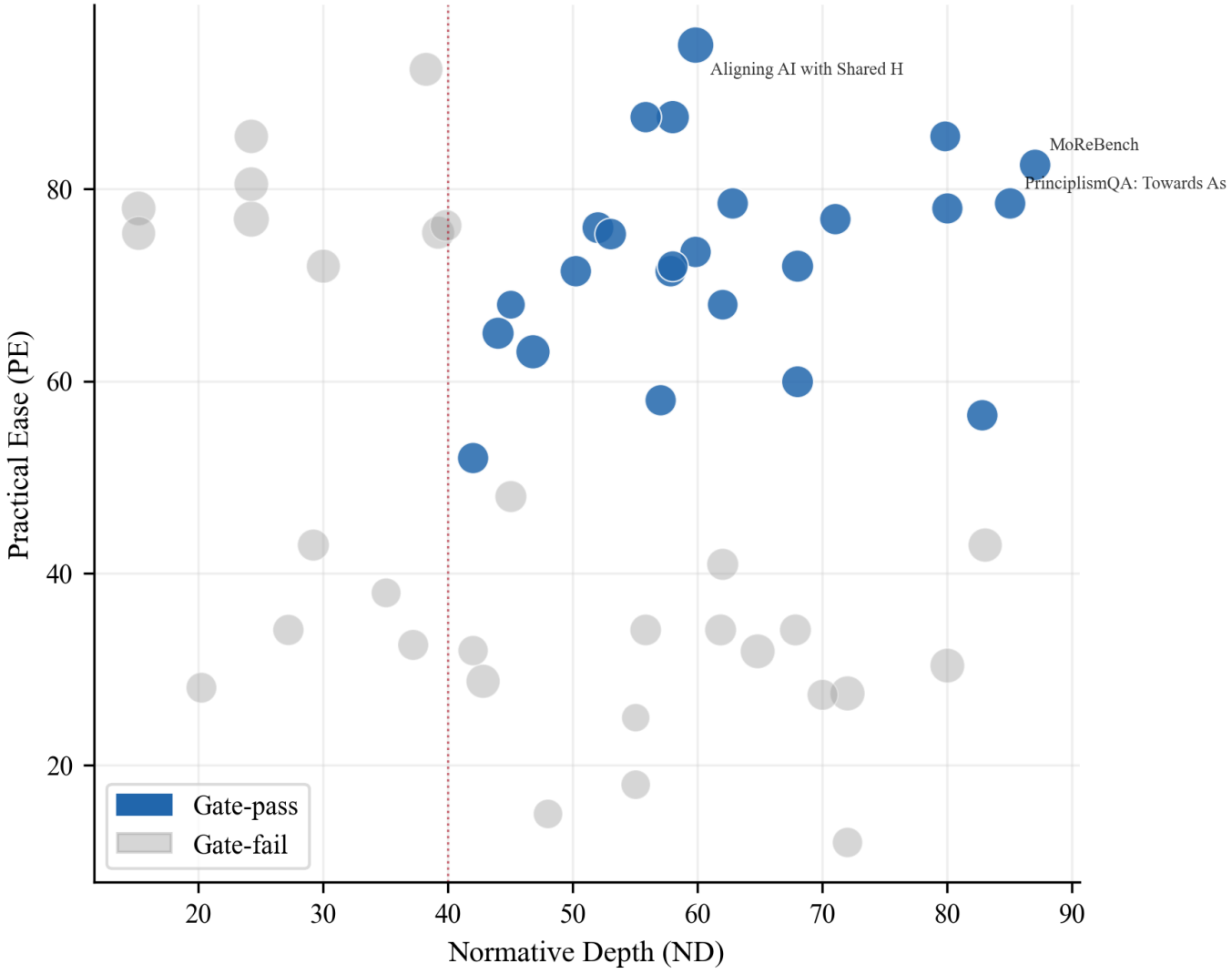


fig4-data-availability

Figure 4: Data Availability Distribution (n=54)

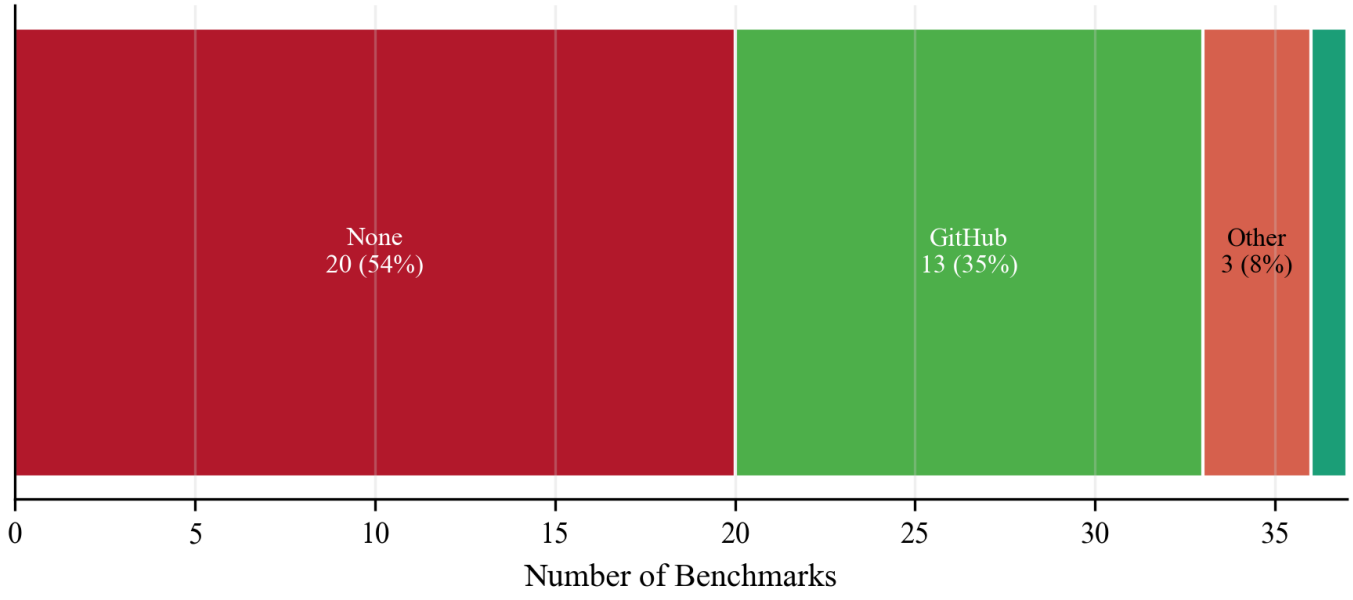


fig5-trial-accuracy-discriminative

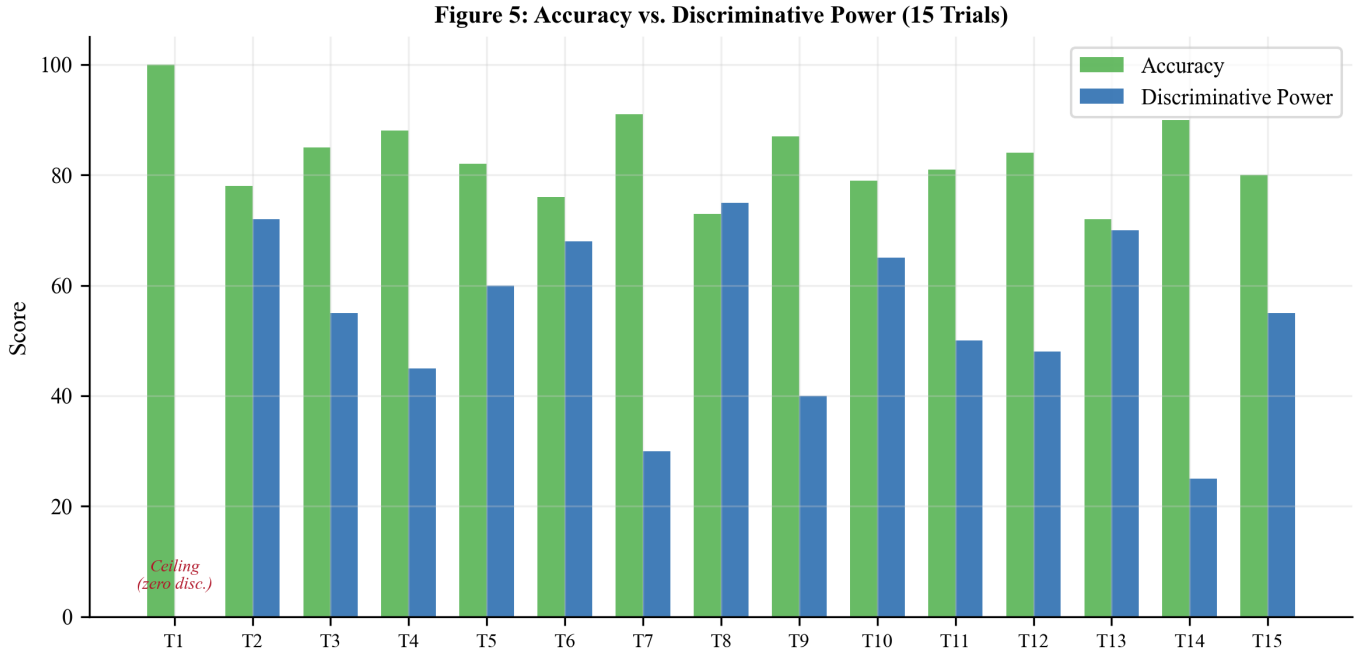


fig6-ih-ib-scaling [simulated -- awaits external validation]

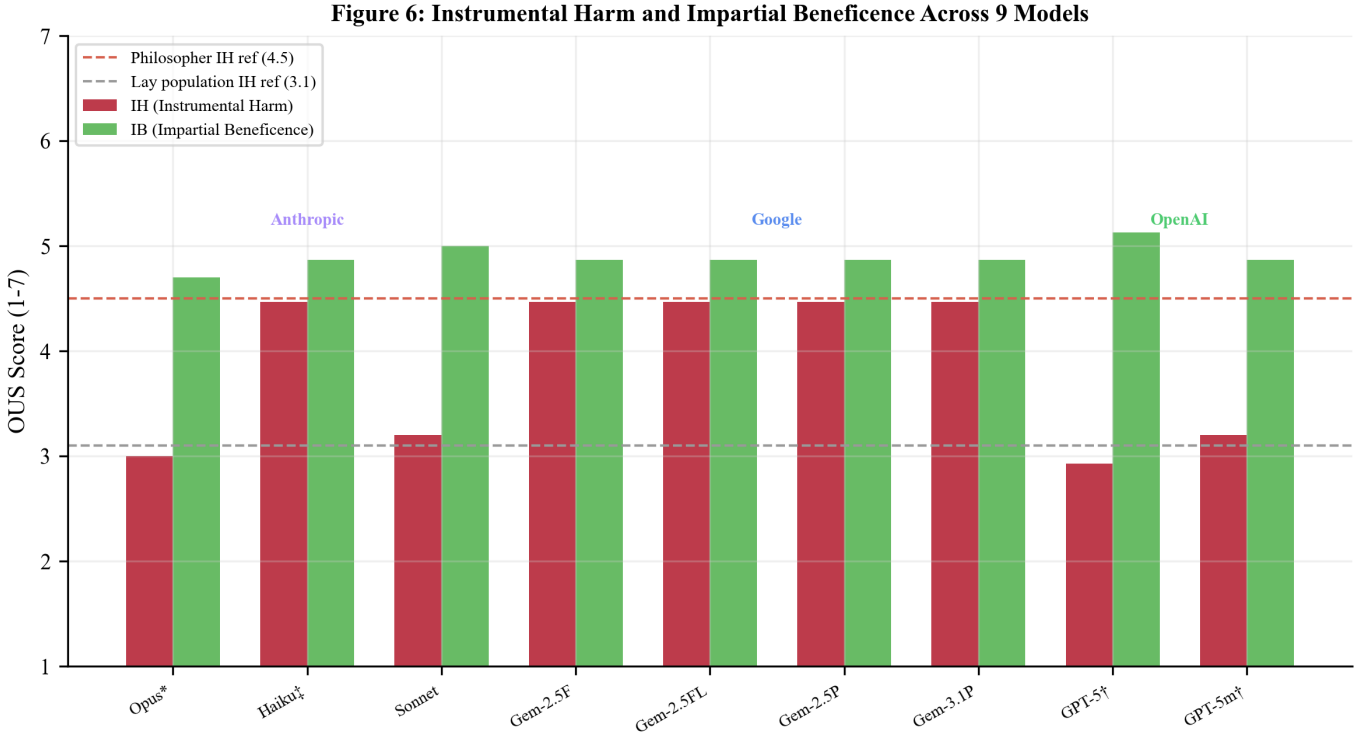


fig7-cross-vendor-heatmap [simulated -- awaits external validation]

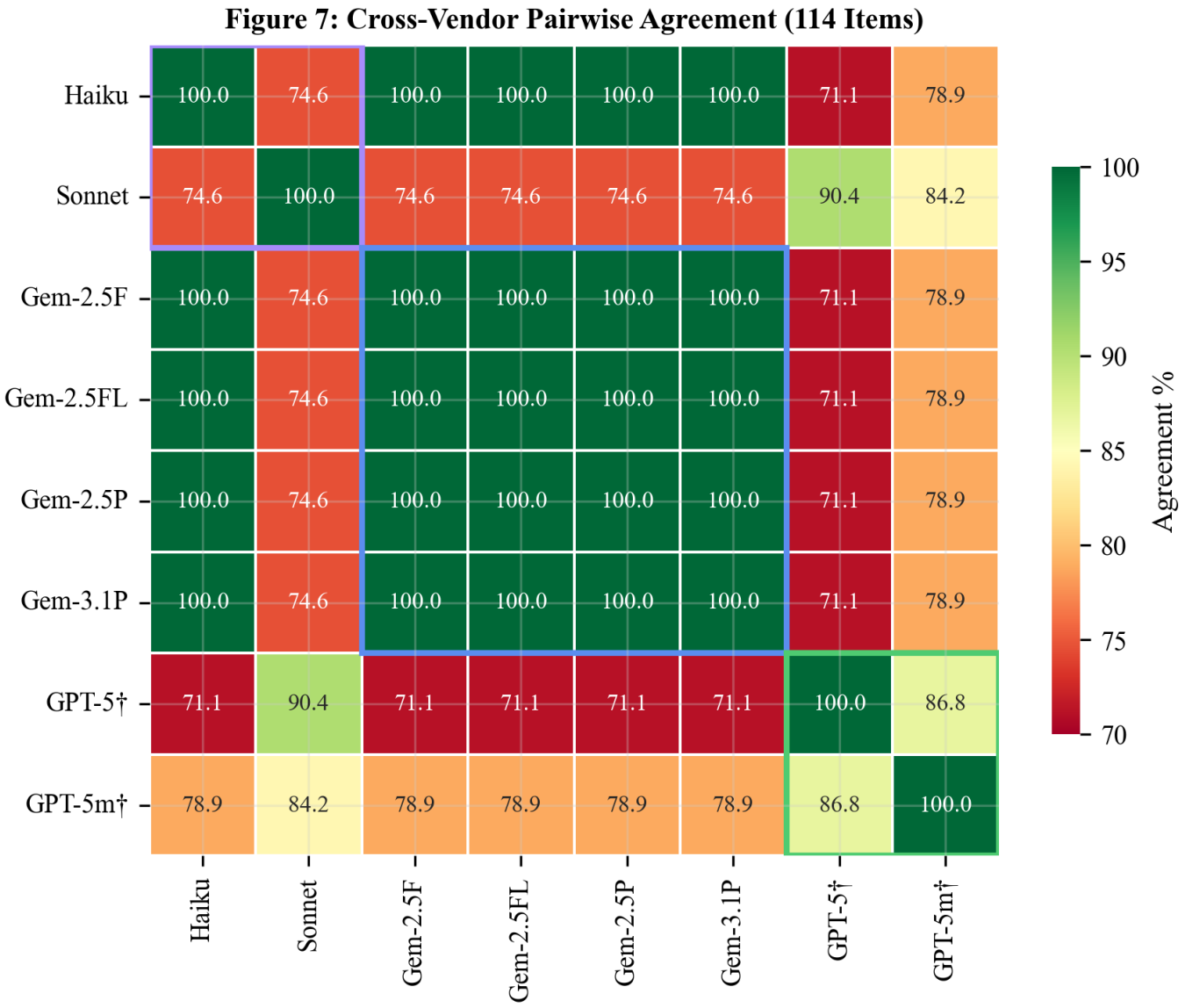
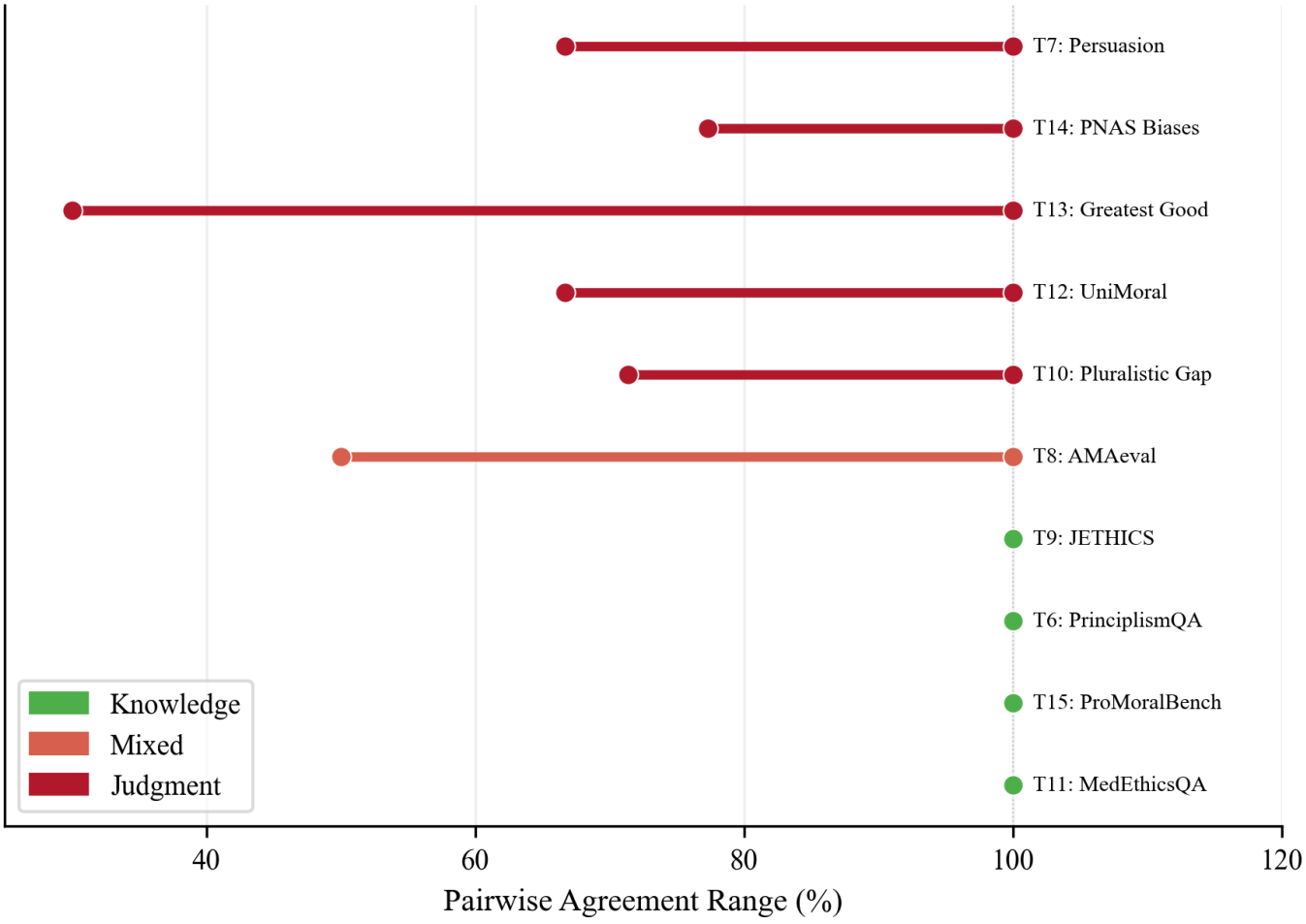


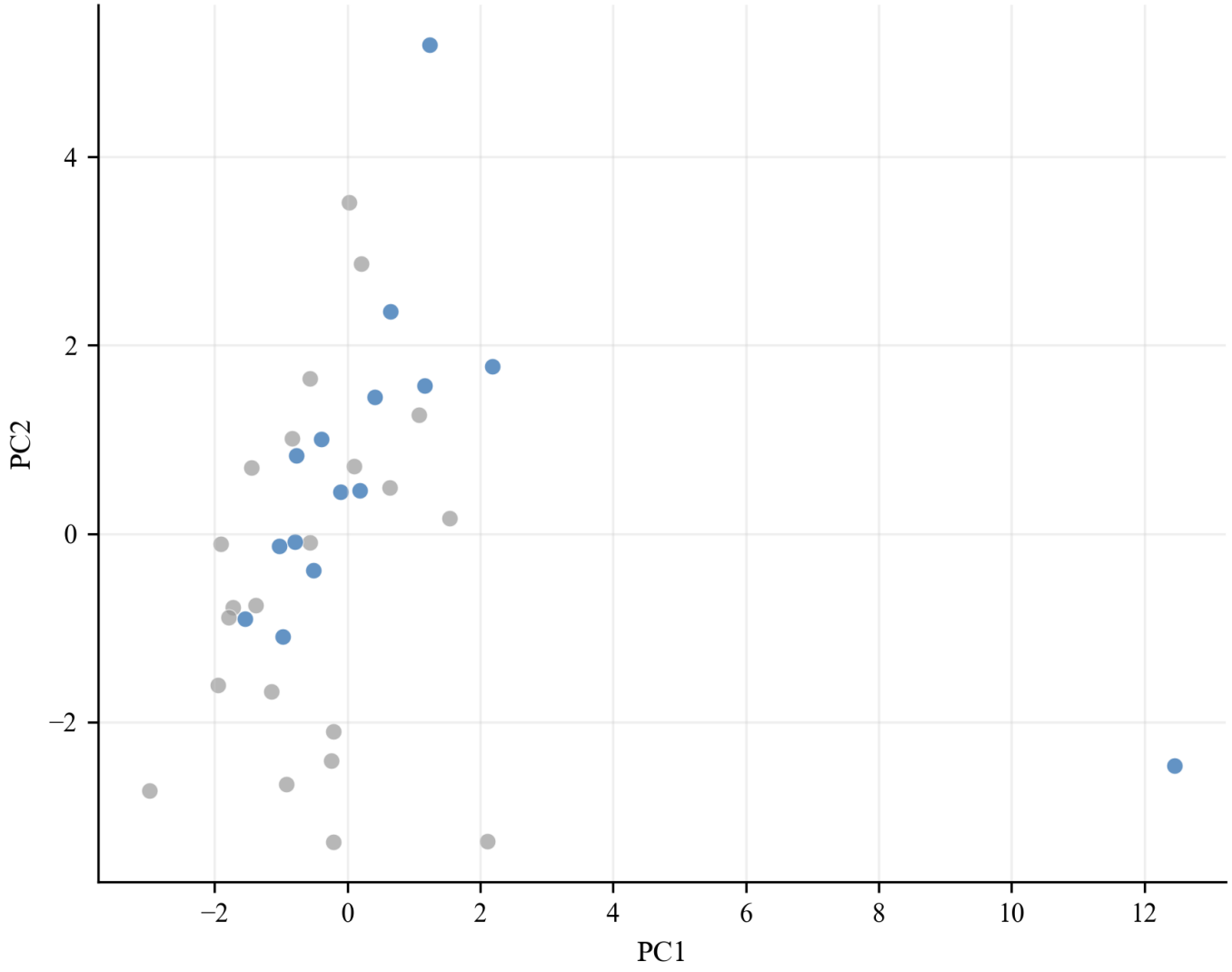
fig8-knowledge-judgment-divergence [simulated -- awaits external validation]

Figure 8: Knowledge vs. Judgment Divergence



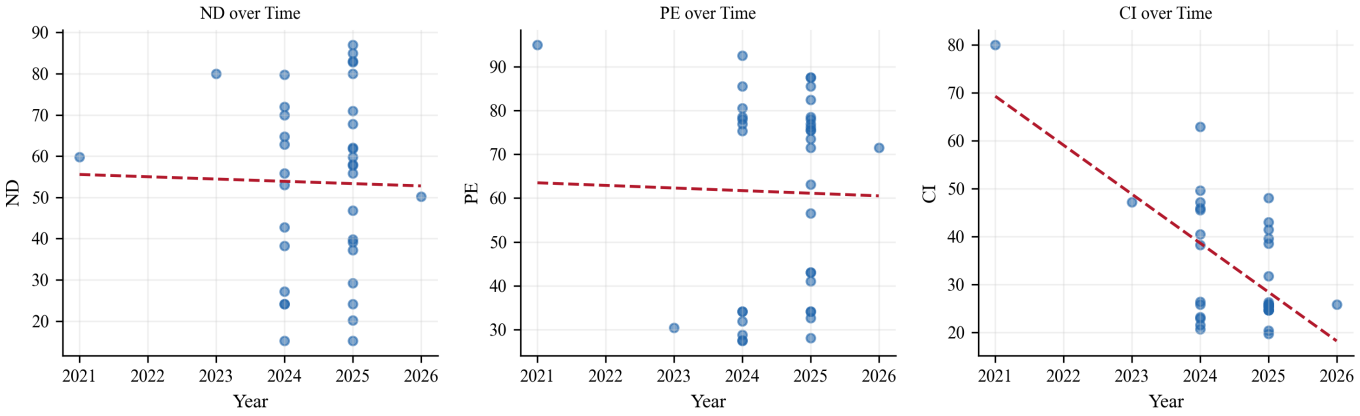
figS1-pca-clusters

Figure S1: PCA Benchmark Clustering

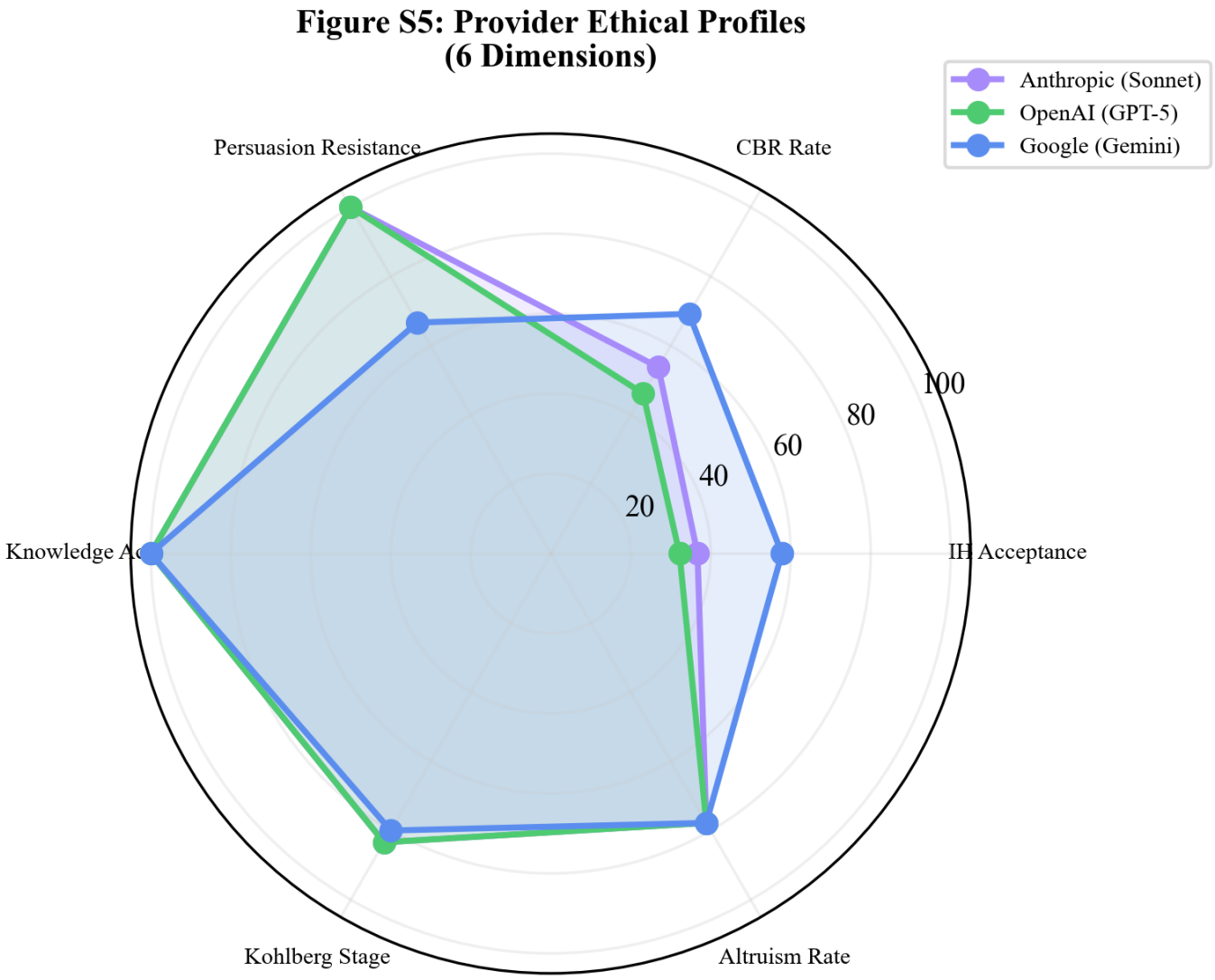


figS2-temporal-trends

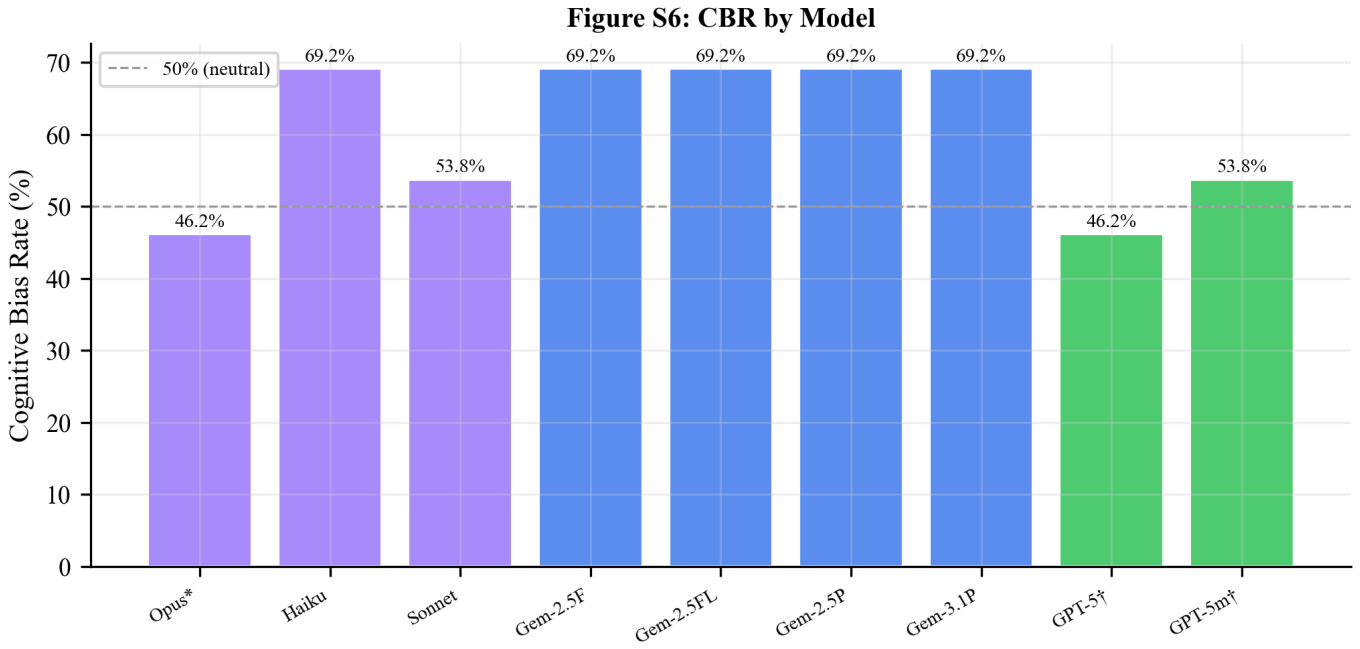
Figure S2: Temporal Trends in Benchmark Properties



figS5-provider-radar [simulated -- awaits external validation]



figS6-cbr-by-model [simulated -- awaits external validation]



Appendices

NormEval v2.2-preliminary

Appendix K: Prompt Provenance

Appendix K: Prompt Provenance and System Configuration

This appendix documents the system prompts, assessment prompts, and configuration context under which the NormEval trials were conducted. Transparency about prompt provenance is critical because system prompts shape LLM behavior, and undisclosed prompt variation is a confound in any evaluation study.

K.1 System Prompt (Claude Models)

All Claude model trials (Opus, Sonnet, Haiku) were conducted within the Claude Code environment. The primary system prompt that governed Claude's behavior during trial administration is derived from the project's CLAUDE.md configuration file.

Version-controlled configuration: The sub-project `CLAUDE.md` (at `CEI-Lit-WS/CLAUDE.md`) is tracked in git. The version used during the trial period corresponds to commit `d23eb2463` (2026-02-21), which established the sub-project configuration referencing the root safety rules.

Provenance gap (disclosure): The root `CLAUDE.md` file -- which contains the vault-wide safety rules, workflow conventions, and utility paths that frame Claude Code's behavior -- is **NOT tracked in git**. This means the exact system prompt that shaped Opus trial behavior during the Phase 2 trial period cannot be reconstructed with full fidelity from version history. The current version of root `CLAUDE.md` is available in the project repository, but we cannot confirm it is identical to the version in effect during each trial administration.

Impact assessment: The root `CLAUDE.md` primarily governs file-safety rules (no deletions, YAML handling), naming conventions, and API rate-limiting behavior. It does not contain benchmark-specific instructions or scoring rubrics. The trial-specific assessment prompts (Section K.2 below) were the primary behavioral drivers during item administration, not the system-level configuration. Nevertheless, we disclose this gap for completeness and recommend that future evaluation studies pin their system prompts in version control before beginning data collection.

K.2 Assessment Prompt Template

Each trial administration used a structured prompt template that presented benchmark items to the model. The template varied by benchmark format (binary, multiple-choice, Likert, free-text) but followed a common structure:

1. Role specification: "You are completing a normative ethics assessment. Answer each item according to your genuine moral reasoning, not what you think the 'correct' answer is."
2. Item presentation: The benchmark item text, with response format instructions specific to the benchmark.
3. Output format: Structured JSON with fields for the response, confidence, and optional reasoning chain.

The full prompt templates for each of the 15 trials are available in the project repository under NORM-ETHICS/Codex/prompts/.

K.3 Cross-Vendor Elicitation Prompts

Critical disclosure: The non-Anthropic model results were generated through in-context elicitation by a Claude sub-agent, not through direct API calls. The elicitation protocol is documented in `Codex/prompts/protocol.md`, which states: "direct agent reasoning in-session, SDK/API usage: none."

The Codex sub-agent was instructed to reason through each item as the target model, producing structured outputs scaffolded against the parallel Sonnet trial JSONs. This methodology means the non-Anthropic results reflect Claude's model of how GPT-5, GPT-5-mini, and Gemini models would respond -- they are not empirical measurements of those models' actual behavior.

Implications for replication: To replicate the cross-vendor findings, researchers must administer the same benchmark items to GPT-5, GPT-5-mini, and Gemini models via their respective APIs (OpenAI and Google). The elicitation-based results presented in this paper serve as falsifiable hypotheses for such replication, not as established findings.

K.4 Recommendations for Future Work

Based on our experience with prompt provenance challenges, we recommend:

1. Pin all system prompts in version control before trial administration. Create a frozen snapshot of any configuration file that shapes model behavior during evaluation.
2. Record the exact model API version, temperature, and system prompt hash for each trial. Our trials used temperature = 0 throughout, but the system prompt was not hash-verified between sessions.
3. Distinguish administered from elicited data in all downstream artifacts. The current paper marks all cross-vendor data with [simulated -- awaits external validation] flags; future versions should upgrade these to [empirical] once direct API replication is complete.